

Polylogarithms and their Arithmetic Bridges
Special Values, Zeta Jets, Lattices and Experimental Discovery

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Preface

The purpose of this manuscript is to develop a common mathematical language for special values that first appear in rather different guises: nested sums, arctangent integrals, gamma products, lattice row sums and coefficients of Hurwitz zeta. Its central movement is from experimental discovery to exact structure. A well-chosen computation can suggest a transformation, reveal an unexpected character component or expose a missing generator. A proof then explains the phenomenon and determines its range of validity.

The material originates in thirty-nine articles, reports and literature surveys from the PolyLog programme in the Smithereens repository. The present text merges overlapping treatments, uses later corrected formulas, and revises claims whose original wording exceeded the evidence. It is organized by mathematical dependency rather than by the order of research sessions. The source inventory and editorial ledger accompany the manuscript.

There are three distinct meanings of “exact” in this work. An analytic identity may have a proof in ordinary mathematics. A row combination or a shuffle calculation may be checked by finite exact arithmetic, relative to its stated law vocabulary. A numerical identity may instead have a very small independently recomputed residual. These are valuable and different forms of evidence. In particular, an unsuccessful integer-relation search proves neither independence nor minimal depth, and a zero canary coefficient does not prove that a proposed relation is exact. No result in this project has yet been formalized in a proof assistant.

The chapter order follows this distinction. Nested sums and iterated integrals establish the algebra; cyclotomic Fourier transforms organize its arithmetic coordinates; algebraic-argument transformations and depth experiments identify candidate special values. Gamma certificates and CM lattices connect these values to periods. Hurwitz-zeta jets unify integration and differentiation, and the Herglotz function carries these bridges into real-quadratic arithmetic. The final chapter returns to experimental design and the proof problems that this progression leaves open.

Historical residuals, performance measurements and search counts refer to the source snapshots, whose scripts and identity stores were not all imported into ProveIt. Fresh checks in the accompanying verification directory are labeled separately. Attribution is retained for classical identities and published theorems; an experimental rediscovery is not a claim of priority. The original drafts were prepared with agentic assistance from Anthropic Claude, with an external OpenAI review of the Stieltjes integration article; the present consolidation and additional audit were assisted by OpenAI Codex.

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Chapter 1

From one polylogarithm to many

An ordinary polylogarithm has two descriptions: a series and an iterated integral. Nesting the first and extending the alphabet of the second produce the same larger class of functions. Their two multiplication laws then provide exact relations. This chapter fixes conventions before those relations are used to organize special values.

1.1 Multiple polylogarithms: two faces

The classical polylogarithm $\text{Li}_s(z) = \sum_{n \geq 1} z^n / n^s$ has, for the reader of Lewin, two complementary descriptions: a power series and an iterated integral $\text{Li}_s(z) = \int_0^z \text{Li}_{s-1}(t) dt/t$. Each survives the passage to many indices, and the resulting two descriptions of one and the same function are the source of essentially every nontrivial identity in this circle of ideas. We set up both here; the products they generate occupy the next section.

1.1.1 The nested sum

Definition 1.1. For $s_1, \dots, s_k \in \mathbb{C}$ and $z_1, \dots, z_k \in \mathbb{C}$ the multiple polylogarithm is the strictly nested sum

$$\text{Li}_{s_1, \dots, s_k}(z_1, \dots, z_k) = \sum_{n_1 > n_2 > \dots > n_k \geq 1} \prod_{i=1}^k \frac{z_i^{n_i}}{n_i^{s_i}}.$$

Its weight is $w = s_1 + \dots + s_k$ and its depth is k . The single-index case $k = 1$ recovers $\text{Li}_s(z)$.

Two features of this definition demand comment before anything is proved.

Remark 1.2 (Conventions differ; ours is fixed). The inequality is *strict*, $n_1 > n_2 > \dots > n_k$. A sizable part of the literature (and several computer-algebra systems) instead uses a non-strict summation $n_1 \geq n_2 \geq \dots \geq n_k$, or reverses the role of the indices so that the *innermost* summation variable carries the first parameter. These choices are genuinely different functions (they differ by lower-depth terms, exactly the diagonal contributions $n_i = n_{i+1}$), and the discrepancy is the single most common source of sign and indexing confusion when comparing sources. We adhere throughout to the strict, “outermost variable carries s_1 ” convention above. The arguments are likewise ordered with z_1 paired to the largest index n_1 . When checking against `MultiplePolyLog`, confirm the convention numerically on a small case rather than assuming agreement.

Remark 1.3 (Multiple zeta values). Setting every $z_i = 1$ gives the *multiple zeta value*

$$\zeta(s_1, \dots, s_k) = \text{Li}_{s_1, \dots, s_k}(1, \dots, 1) = \sum_{n_1 > \dots > n_k \geq 1} \prod_i n_i^{-s_i},$$

the central special values of the theory (Zagier 1994; Hoffman 1992). The strict nesting is what makes $\zeta(s_1, \dots, s_k)$ finite for $s_1 \geq 2$.

Convergence. The single-variable series $\text{Li}_s(z)$ converges absolutely for $|z| < 1$ (any s), on $|z| = 1$ for $\Re s > 1$, and conditionally on $|z| = 1$, $z \neq 1$, for $\Re s > 0$. For the nested sum the corresponding statement is clean.

Proposition 1.4 (A sufficient absolute-convergence domain). *For positive integer indices, the nested series converges absolutely when $|z_i| \leq 1$ for all i and $s_1 \geq 2$. It also converges absolutely for $|z_1| < 1$, $|z_i| \leq 1$ for $i > 1$, with $s_1 \geq 1$.*

Proof. The absolute value of the inner nested sum is at most $H_{n_1-1}^{k-1} = O((1 + \log n_1)^{k-1})$. The outer series is therefore bounded by $\sum |z_1|^{n_1} (1 + \log n_1)^{k-1} / n_1^{s_1}$, which converges in either stated case. $\square$

These are sufficient conditions, not a complete classification of every boundary point. Oscillation can give conditional convergence: $\text{Li}_1(-1)$ converges, but $\sum |(-1)^n/n|$ diverges. More generally, away from the polydisc an interior domain is given by $|z_1 \cdots z_j| < 1$ for each $1 \leq j \leq k$. At unity, for positive integer indices, admissibility is exactly $s_1 \geq 2$.

1.1.2 The iterated integral

The second face is Chen's iterated integral. Fix the two differential one-forms

$$\omega_0 = \frac{dt}{t}, \quad \omega_1 = \frac{dt}{1-t},$$

and, for a word $u = a_1 a_2 \cdots a_w$ in the letters $\{0, 1\}$, write the iterated integral over the standard simplex

$$\int_0^z u = \int_{0 < t_1 < t_2 < \cdots < t_w < z} \omega_{a_w}(t_1) \omega_{a_{w-1}}(t_2) \cdots \omega_{a_1}(t_w),$$

i.e. the leftmost letter a_1 is attached to the *outermost* (largest) variable t_w and one integrates from the top down. This is the orientation under which d/dz strips the outermost form. With this normalization the one-variable multiple polylogarithm is an iterated integral over a single explicit word.

Theorem 1.5 (Iterated-integral representation). *For $s_1 \geq 2$ and the remaining $s_i \geq 1$,*

$$\text{Li}_{s_1, \dots, s_k}(z, 1, \dots, 1) = \int_0^z \underbrace{0^{s_1-1} 1}_{\text{block 1}} \underbrace{0^{s_2-1} 1}_{\text{block 2}} \cdots \underbrace{0^{s_k-1} 1}_{\text{block k}},$$

the iterated integral of the binary word

$$u(\mathbf{s}) = 0^{s_1-1} 1 0^{s_2-1} 1 \cdots 0^{s_k-1} 1$$

of length w . Equivalently, with the alternative kernel $\tilde{\omega}_1 = dt/(t-1) = -\omega_1$ one carries one sign per 1-letter, giving the textbook normalization

$$\text{Li}_{s_1, \dots, s_k}(z, 1, \dots, 1) = (-1)^k \int_{0 < t_1 < \cdots < t_w < z} \prod_{j=1}^w \eta_{a_{w+1-j}}(t_j), \quad \eta_0 = \frac{dt}{t}, \quad \eta_1 = \frac{dt}{t-1},$$

the factor $(-1)^k$ being exactly the number of 1-letters in $u(\mathbf{s})$, one per block.

The composition $\leftrightarrow$ word dictionary. The map $\mathbf{s} = (s_1, \dots, s_k) \mapsto u(\mathbf{s})$ is a bijection between compositions (ordered tuples with each $s_i \geq 1$) and binary words that *end in a 1*. Reading u from left to right: a 1 *closes a block* and contributes a new summation index; the run of 0's immediately preceding that 1 sets the corresponding exponent, $s_i - 1$ zeros for exponent s_i . Thus depth $k =$ (number of 1's) and weight $w =$ (length of the word). Admissibility $s_1 \geq 2$ is the requirement that u *begin with a 0*: a leading 1 would attach ω_1 to the top variable and produce the logarithmic divergence at $t_w \rightarrow 1$ when $z = 1$. The endpoints play dual roles — ω_0 is singular at 0, ω_1 at 1 — so a convergent word neither starts with 1 nor ends with 0.

Origin. Expand the innermost form geometrically. The depth-one case is Lewin's $\text{Li}_s(z) = \int_0^z 0^{s-1} 1$, obtained by iterating $\text{Li}_s(z) = \int_0^z \text{Li}_{s-1}(t) \omega_0$ down to $\text{Li}_1(z) = -\log(1-z) = \int_0^z \omega_1$. For general depth, $\int_0^z \omega_1$ on the innermost block introduces $\sum_{n_k \geq 1} t^{n_k}/n_k$, the subsequent ω_0 's raise the power of n_k , and the next ω_1 couples a strictly larger index $n_{k-1} > n_k$; the strict inequality is forced by the nesting of the simplex $t_1 < \dots < t_w$. Carrying this out reproduces the nested sum, and Chen's theory (1977) guarantees convergence and homotopy invariance of the integral. The general framework is due to Goncharov; the systematic exploitation for these functions is Goncharov's and Borwein–Bradley–Broadhurst–Lisonek's. $\square$

1.1.3 General alphabets and a fixed integral orientation

For a path from 0 to z avoiding its letters, use the outer-letter-first recursion

$$G(a_1, \dots, a_w; z) = \int_0^z \frac{G(a_2, \dots, a_w; t)}{t - a_1} dt, \quad G(; z) = 1.$$

The all-zero word is regularized as $G(0^w; z) = \log^w z / w!$. For positive integer indices the dictionary in our decreasing-index convention is

$$\text{Li}_{s_1, \dots, s_k}(z_1, \dots, z_k) = (-1)^k G(0^{s_1-1}, z_1^{-1}, 0^{s_2-1}, (z_1 z_2)^{-1}, \dots, 0^{s_k-1}, (z_1 \dots z_k)^{-1}; 1).$$

Thus the letters are inverse *prefix* products. Reversing the integration order also reverses the letter word; mixing these conventions changes the function. The formula follows by expanding each nonzero kernel geometrically near the origin and integrating recursively, then continuing along a common path. Wolfram's generalized-polylogarithm interface must be calibrated against this orientation before numerical comparisons.

1.1.4 Two facts to carry forward

(1) **The differential strips the first letter.** In the integral orientation above,

$$z \frac{d}{dz} \text{Li}_{s_1, \dots}(z, 1, \dots) = \begin{cases} \text{Li}_{s_1-1, s_2, \dots}(z, 1, \dots), & s_1 \geq 2, \\ \frac{z}{1-z} (\text{remove the leading block}), & s_1 = 1, \end{cases}$$

i.e. differentiation removes one outer letter — a 0 lowers s_1 , the leading 1 deletes the first index. This recursion (equivalently the integral equation defining the word) drives every analytic continuation and every series-rearrangement proof. It is also the recursion behind the appearance of `HarmonicPolyLog` in `DSolve/AsymptoticDSolveValue` output for Fuchsian systems with rational coefficients (Remiddi–Vermaseren 1999 for the $\{0, 1\}$ -alphabet case).

- (2) **One function, two presentations.** The *same* $\text{Li}_{s_1, \dots, s_k}$ is simultaneously the nested sum and the iterated integral. Multiplying two nested sums and re-sorting the indices yields the *stuffle* (harmonic, quasi-shuffle) product $*$; multiplying two iterated integrals over a common path and shuffling the letters yields the *shuffle* product Π . Both express a product of polylogarithms as a $\mathbb{Q}$ -linear combination of polylogarithms, but through different combinatorics; equating them is the engine of the next section.

1.1.5 The smallest nontrivial example, in full

Take depth $k = 2$, $\mathbf{s} = (2, 1)$, the lightest admissible non-classical case. The nested sum is

$$\text{Li}_{2,1}(z, 1) = \sum_{m>n\geq 1} \frac{z^m}{m^2 n} = \sum_{m=2}^{\infty} \frac{z^m}{m^2} H_{m-1}, \quad H_{m-1} = \sum_{n=1}^{m-1} \frac{1}{n},$$

exhibiting the characteristic harmonic-number coefficient. The associated word is

$$u(2, 1) = \underbrace{0^{2-1}1}_{\text{block } (s_1=2)} \underbrace{0^{1-1}1}_{\text{block } (s_2=1)} = 011,$$

a word of length $w = 3$ (weight 3) with two 1's (depth 2), beginning in 0 (admissible) and ending in 1 (convergent). By Theorem 1.5,

$$\text{Li}_{2,1}(z, 1) = \int_0^z 011 = \int_{0 < t_1 < t_2 < t_3 < z} \frac{dt_3}{t_3} \frac{dt_2}{1-t_2} \frac{dt_1}{1-t_1},$$

the leading 0 sitting on the outermost variable t_3 and the two 1's on t_2, t_1 . (Equivalently $(-1)^2 = +1$ in the $\eta_1 = dt/(t-1)$ normalization, consistent with the unsigned $\omega_1 = dt/(1-t)$ form.) A numerical check at $z = 0.7$ gives both sides equal to $0.2384965\dots$, and at $z = 1$ the integral $\int_0^1 011$ recovers Euler's celebrated

$$\zeta(2, 1) = \sum_{m>n\geq 1} \frac{1}{m^2 n} = \zeta(3),$$

the depth-two value whose collapse to a depth-one zeta was Euler's first multiple-zeta identity — and, in Wolfram 15, exactly the symbolic output of `MultipleZeta` on $\{2, 1\}$.

1.2 Multiple zeta values and their relations

Specializing the multiple polylogarithm at $z_1 = \dots = z_k = 1$ produces the central objects of this chapter, the *multiple zeta values* (MZVs):

$$\zeta(s_1, \dots, s_k) = \text{Li}_{s_1, \dots, s_k}(1, \dots, 1) = \sum_{n_1 > n_2 > \dots > n_k \geq 1} \prod_{i=1}^k \frac{1}{n_i^{s_i}}. \quad (1.1)$$

The outermost summation index n_1 carries the exponent s_1 , and the strict inequalities make the sum a genuine iterated, rather than ordinary, multiple series. The series (1.1) converges if and only if $s_1 \geq 2$; such an index tuple is called *admissible*. (Any $s_i \geq 1$ for $i \geq 2$ is allowed; only the leading entry is constrained.) We call $w = s_1 + \dots + s_k$ the *weight* and k the *depth*. In Wolfram Language these are `MultipleZeta`, with $\zeta(2, 1)$ written `MultipleZeta[{2, 1}]`.

The depth-one values are the classical zeta values. Euler's formula is

$$\zeta(2n) = \frac{(-1)^{n+1} B_{2n} (2\pi)^{2n}}{2(2n)!} \in \pi^{2n} \mathbb{Q}.$$

The first genuinely new phenomenon is that distinct MZVs can collapse onto a single lower-depth value. The prototype is Euler's identity

$$\zeta(2, 1) = \sum_{n>m \geq 1} \frac{1}{n^2 m} = \zeta(3), \quad (1.2)$$

already known to Euler in the 1740s. The conjecture is that the $\mathbb{Q}$ -vector space spanned by all MZVs is *weight-graded*: products and relations never mix weights, so one studies each weight w separately.

The sum theorem

Euler's $\zeta(2, 1) = \zeta(3)$ is the depth-two case of the first structural theorem. Summing all admissible MZVs of a fixed weight w and fixed depth k recovers the single zeta value $\zeta(w)$.

Theorem 1.6 (Sum theorem; Euler for $k = 2$, Granville and Zagier in general, 1990s). *For all $w > k \geq 1$,*

$$\sum_{\substack{s_1 + \dots + s_k = w \\ s_1 \geq 2, s_i \geq 1}} \zeta(s_1, \dots, s_k) = \zeta(w). \quad (1.3)$$

At $w = 3, k = 2$ the only admissible tuple is $(2, 1)$, recovering (1.2). At $w = 4, k = 2$ the admissible tuples are $(3, 1)$ and $(2, 2)$, so

$$\zeta(3, 1) + \zeta(2, 2) = \zeta(4) = \frac{\pi^4}{90}. \quad (1.4)$$

This one equation does not separate $\zeta(3, 1)$ from $\zeta(2, 2)$; for that we need a *second* multiplication law, to which we now turn.

The two products

MZVs admit two structurally different multiplication rules, because $\zeta(s_1, \dots, s_k)$ has two faithful representations: as a nested sum (1.1), and as an iterated integral. Each representation makes the product of two MZVs expand into a sum of MZVs, but the two expansions differ — and that discrepancy is the engine of the theory.

(a) The stuffle (harmonic / quasi-shuffle) product. Multiply two nested sums directly. For depth one,

$$\zeta(a) \zeta(b) = \left(\sum_{m \geq 1} m^{-a} \right) \left(\sum_{n \geq 1} n^{-b} \right) = \sum_{m > n} + \sum_{n > m} + \sum_{m = n},$$

splitting the lattice $\{(m, n)\}$ by the trichotomy $m > n$, $m < n$, $m = n$. The three regions are exactly two depth-two MZVs and one depth-one MZV:

$$\zeta(a) \zeta(b) = \zeta(a, b) + \zeta(b, a) + \zeta(a + b). \quad (1.5)$$

The general rule is the same combinatorial device — the *quasi-shuffle* of Hoffman — applied to the index words $(s_1, \dots, s_k)$ and $(t_1, \dots, t_\ell)$: interleave the two words in all order-preserving ways, and

additionally allow any aligned pair s_i, t_j to *collide* into a single slot $s_i + t_j$. (Setting $a = b = 2$ in (1.5) gives $\zeta(2)^2 = 2\zeta(2, 2) + \zeta(4)$, used below.) A depth-two example of the general rule:

$$\zeta(a)\zeta(b, c) = \zeta(a, b, c) + \zeta(b, a, c) + \zeta(b, c, a) + \zeta(a + b, c) + \zeta(b, a + c),$$

the first three terms inserting a into the three gaps of (b, c) , the last two colliding a with b and with c .

(b) The shuffle product. The integral representation reads each MZV as a Chen iterated integral over the simplex $1 > t_1 > \cdots > t_w > 0$ against the two one-forms $\omega_0 = \frac{dt}{t}$ and $\omega_1 = \frac{dt}{1-t}$. Encode an admissible index as a binary word: each s_i becomes $\underbrace{0 \cdots 0}_{s_i-1} 1$, so that, e.g., $\zeta(2) \leftrightarrow 01$, $\zeta(3) \leftrightarrow 001$,

$\zeta(2, 1) \leftrightarrow 011$, $\zeta(3, 1) \leftrightarrow 0011$, $\zeta(2, 2) \leftrightarrow 0101$. Admissibility ($s_1 \geq 2$) is the condition that the word *begins with 0*; convergence at the inner end requires the word to *end with 1*. Multiplying two iterated integrals over the same simplex produces the integral over the product region, which decomposes into the $\binom{w_1+w_2}{w_1}$ order-preserving interleavings — the *shuffle product* Π — of the two letter words, *without* any collision term. Thus

$$\zeta(a)\zeta(b) = \sum_{\text{words } u \in (\text{word } a) \Pi (\text{word } b)} \zeta(u). \quad (1.6)$$

The double-shuffle relation: $\zeta(3, 1)$ in closed form

The canonical demonstration computes $\zeta(2)^2$ by both products and equates the results. Stuffle, from (1.5) with $a = b = 2$:

$$\zeta(2)^2 = 2\zeta(2, 2) + \zeta(4). \quad (1.7)$$

Shuffle: $\zeta(2)$ is the word 01, and we shuffle 01 Π 01. Writing the two factors as $0_a 1_a$ and $0_b 1_b$ and listing the $\binom{4}{2} = 6$ interleavings that preserve $0_a \rightarrow 1_a$ and $0_b \rightarrow 1_b$:

$$0_a 0_b 1_a 1_b, \quad 0_a 0_b 1_b 1_a, \quad 0_b 0_a 1_a 1_b, \quad 0_b 0_a 1_b 1_a, \quad 0_a 1_a 0_b 1_b, \quad 0_b 1_b 0_a 1_a.$$

Forgetting the colour labels, these are the words 0011 (four times) and 0101 (twice). Since $0011 \leftrightarrow \zeta(3, 1)$ and $0101 \leftrightarrow \zeta(2, 2)$,

$$\zeta(2)^2 = 4\zeta(3, 1) + 2\zeta(2, 2). \quad (1.8)$$

Now subtract (1.7) from (1.8): the $2\zeta(2, 2)$ terms cancel and we obtain the *double-shuffle relation*

$$4\zeta(3, 1) = \zeta(4) \quad \implies \quad \zeta(3, 1) = \frac{1}{4}\zeta(4) = \frac{\pi^4}{360}. \quad (1.9)$$

Feeding this back into the sum-theorem relation (1.4) also pins $\zeta(2, 2) = \zeta(4) - \zeta(3, 1) = \frac{3}{4}\zeta(4) = \pi^4/120$ — consistent with the all-2's family below.

Remark 1.7 (Regularized double shuffle). The naive double-shuffle relations above are produced only by admissible words. To capture the boundary case, one regularizes the divergent $\zeta(1)$ (the word 1). Comparing the two regularizations — shuffle versus stuffle — as polynomials in a formal variable T standing for the regularized $\zeta(1)$ yields the *extended* or *regularized double-shuffle relations* of Ihara, Kaneko, and Zagier (2006). These are conjectured to generate *all* $\mathbb{Q}$ -linear relations among MZVs.

Other relation families

The double-shuffle relations sit inside a larger, heavily overlapping web. We list the principal families, one line each.

Duality. The integral substitution $t \mapsto 1 - t$ swaps $\omega_0 \leftrightarrow \omega_1$ and reverses the order of integration, giving $\zeta(\mathbf{s}) = \zeta(\mathbf{s}^\dagger)$, where the *dual* word $\mathbf{s}^\dagger$ is obtained from the binary word by reading it backwards and exchanging $0 \leftrightarrow 1$. The simplest instance is $011 \mapsto 011$ read backwards is 110 , complemented is 001 : thus $\zeta(2, 1) = \zeta(3)$ (1.2) is precisely a self/dual coincidence. Likewise 0011 ($\zeta(3, 1)$) is *self*-dual—read backwards and complemented it is again 0011 —so duality says nothing new there; the cleanest nontrivial higher instance is the depth-three identity $\zeta(2, 1, 1) = \zeta(4)$, from $0111 \mapsto 0001$.

Hoffman’s relations. Relations obtained by comparing the regularized product with $\zeta(1)$ provide useful index identities. Their precise insertion and splitting terms matter; a schematic insertion of a single 1 in every position is not a valid general formula.

Derivation relations. An infinite family produced by certain Hopf-algebra derivations acting on the algebra of admissible words (Ihara–Kaneko–Zagier), equivalent over $\mathbb{Q}$ to large portions of the regularized double shuffle.

Ohno’s relation. A single generating identity (Ohno 1999) that simultaneously generalizes the sum theorem (1.3) *and* duality, obtained by summing MZVs over all ways of distributing a fixed total increment among the indices of $\mathbf{s}$ and of its dual.

Cyclic sum formula. Hoffman–Ohno (and Ohno–Wakabayashi): a cyclic sum of MZVs of weight w equals a weighted sum of weight- w values, refining the sum theorem with a cyclic symmetry on the index word.

Closed-form families

Beyond individual reductions, several infinite families of MZVs evaluate to powers of π in closed form. Throughout, $\{s\}_n$ denotes the index s repeated n times.

All 2’s. The generating function $\prod_{n \geq 1} (1 - x^2/n^2) = \sin(\pi x)/\pi x$ yields, after extracting the relevant power sums,

$$\zeta(\{2\}_n) = \zeta(\underbrace{2, 2, \dots, 2}_n) = \frac{\pi^{2n}}{(2n+1)!}. \quad (1.10)$$

For $n = 1$ this is $\zeta(2) = \pi^2/6$; for $n = 2$, $\zeta(2, 2) = \pi^4/120$, matching (1.9); for $n = 3$, $\zeta(2, 2, 2) = \pi^6/5040$.

The $\{3, 1\}$ family. A celebrated result of Borwein, Bradley, Broadhurst, and Lisoněk (1998), originally conjectured experimentally:

$$\zeta(\{3, 1\}_n) = \zeta(\underbrace{3, 1, 3, 1, \dots, 3, 1}_{n \text{ blocks}}) = \frac{2\pi^{4n}}{(4n+2)!}. \quad (1.11)$$

For $n = 1$ this reproduces $\zeta(3, 1) = 2\pi^4/6! = \pi^4/360$ of (1.9); for $n = 2$, $\zeta(3, 1, 3, 1) = 2\pi^8/10!$. The proof is a striking generating-function argument identifying $\sum_n \zeta(\{3, 1\}_n) x^{4n}$ with a ratio of cosh and cos.

The $\{2\}_a 3 \{2\}_b$ family. Zagier (2012) evaluated the two-parameter family $\zeta(\{2\}_a, 3, \{2\}_b)$ as an explicit $\mathbb{Q}$ -linear combination of products $\pi^{2m} \zeta(2n + 1)$, a result whose proof unexpectedly drew on the arithmetic of the modular-form-like generating series; it remains one of the deepest known closed-form families.

Dimensions and the weight grading

Let D_w be the $\mathbb{Q}$ -vector space spanned by all MZVs of weight w (with $D_0 = \mathbb{Q}$, $D_1 = 0$ since $\zeta(1)$ diverges), and $d_w = \dim_{\mathbb{Q}} D_w$. Zagier (1994) conjectured the remarkably simple recursion

$$d_w = d_{w-2} + d_{w-3}, \quad d_0 = 1, \quad d_1 = 0, \quad d_2 = 1, \quad (1.12)$$

giving the sequence

$$d_0, d_1, d_2, \dots = 1, 0, 1, 1, 1, 2, 2, 3, 4, 5, 7, 9, \dots$$

The *upper* bound $d_w \leq$ (the recursion value) is a theorem, proved independently by Goncharov and by Terasoma (around 2002), with the conceptual framework of mixed Tate motives completed by Deligne and Goncharov; it amounts to showing that the known relations (e.g. associator / motivic) suffice to cut the dimension down. Broadhurst and Kreimer (1997) refined the count by also grading by depth, conjecturing a two-variable generating function that brings in the dimensions of spaces of cusp forms (the first modular obstruction appearing at weight 12, depth 4). The matching *lower* bound — equivalently, that no further relations exist and the conjectured spanning sets are linearly independent — is wide open; even $\dim D_w \geq 2$ for some specific large w would already imply transcendence statements (such as the linear independence of π^6 and $\zeta(3)^2$ over $\mathbb{Q}$ at $w = 6$) that are presently out of reach.

A table of low-weight reductions

The following table records all admissible MZVs through weight 5 together with their reductions to the conjectural basis $\{1\}$ (weight ≤ 2), $\{\zeta(3)\}$, $\{\zeta(4)\} \sim \{\pi^4\}$, $\{\zeta(5), \zeta(2)\zeta(3)\}$. Every entry has been verified symbolically.

w	spanning count	MZVs of weight w and their reductions
2	1	$\zeta(2) = \frac{\pi^2}{6}$
3	1	$\zeta(3); \quad \zeta(2, 1) = \zeta(3)$
4	1	$\zeta(4) = \frac{\pi^4}{90}; \quad \zeta(3, 1) = \frac{\pi^4}{360}, \quad \zeta(2, 2) = \frac{\pi^4}{120}, \quad \zeta(2, 1, 1) = \zeta(4)$
5	2	$\zeta(5), \zeta(2)\zeta(3)$ span; $\zeta(4, 1) = 2\zeta(5) - \zeta(2)\zeta(3),$ $\zeta(3, 2) = -\frac{11}{2}\zeta(5) + 3\zeta(2)\zeta(3), \quad \zeta(2, 3) = \frac{9}{2}\zeta(5) - 2\zeta(2)\zeta(3),$ $\zeta(3, 1, 1) = 2\zeta(5) - \zeta(2)\zeta(3), \quad \zeta(2, 1, 1, 1) = \zeta(5)$ (duality of $\zeta(5)$)

The table gives spanning sets of sizes 1, 1, 1, 2 at weights 2, 3, 4, 5. The two-element weight-five set is not known to be independent. Every depth-two value in that row reduces to products of single zeta values. At weight eight, values such as $\zeta(6, 2)$ are candidates for a primitive direction beyond the polynomial algebra of single zetas.

1.3 Harmonic polylogarithms

The multiple polylogarithms of the previous sections are nested *sums*. For the practitioner who must actually manipulate weight- w transcendentals—most insistently the particle physicist computing Feynman integrals—it is the *iterated-integral* avatar that is decisive, because integration variables, differential equations, and analytic continuation all speak the language of integrals. The harmonic polylogarithms (HPLs) of Remiddi and Vermaseren [38] are precisely this avatar, specialised to the three integration kernels that arise when a rational integrand has singularities only at $t = 0, 1, -1$. They have become the de facto standard basis for one-scale problems, and Wolfram 15 exposes them directly as `HarmonicPolyLog`.

1.3.1 Definition: kernels and iterated integrals

Fix the alphabet $\{0, 1, -1\}$ and the three rational *kernels*

$$f_0(t) = \frac{1}{t}, \quad f_1(t) = \frac{1}{1-t}, \quad f_{-1}(t) = \frac{1}{1+t}.$$

A harmonic polylogarithm $H(a_1, \dots, a_w; z)$ is indexed by a word $\vec{a} = (a_1, \dots, a_w)$ over this alphabet; the length w is the *weight*. The definition is recursive in the leftmost letter. When the leading letter is nonzero,

$$H(a_1, a_2, \dots, a_w; z) = \int_0^z f_{a_1}(t) H(a_2, \dots, a_w; t) dt, \quad a_1 \neq 0, \quad (1.13)$$

with the depth-zero seed $H(; z) = 1$ (the empty word). The recursion also applies when $a_1 = 0$, provided the remaining word is not all zeros. Such an integrand is integrable at the origin. Only the all-zero words need the logarithmic regularization above; zeros at the end of a general word can be removed by shuffle identities. For example $H(0, 1; z) = \text{Li}_2(z)$ is a convergent integral with a leading zero. The leftmost letter always labels the outermost integration.

1.3.2 The weight-one and weight-two cases

At weight one (1.13)—the all-zero regularization give the three building blocks

$$H(0; z) = \ln z, \quad H(1; z) = -\ln(1-z), \quad H(-1; z) = \ln(1+z), \quad (1.14)$$

since $\int_0^z dt/(1-t) = -\ln(1-z)$ and $\int_0^z dt/(1+t) = \ln(1+z)$. At weight two, integrating these against another kernel reproduces the classical dilogarithm and its companions:

$$H(0, 1; z) = \int_0^z \frac{dt}{t} (-\ln(1-t)) = \text{Li}_2(z), \quad H(0, -1; z) = \int_0^z \frac{dt}{t} \ln(1+t) = -\text{Li}_2(-z), \quad (1.15)$$

$$H(1, 0; z) = \int_0^z \frac{\ln t}{1-t} dt = -\text{Li}_2(z) - \ln z \ln(1-z), \quad H(1, 1; z) = \frac{1}{2} \ln^2(1-z), \quad (1.16)$$

the last by (1.13) with $H(1; t) = -\ln(1-t)$ and $\int_0^z (-\ln(1-t))/(1-t) dt = \frac{1}{2} \ln^2(1-z)$. The pair $H(0, 1; z) = \text{Li}_2(z)$ and $H(0, -1; z) = -\text{Li}_2(-z)$ shows the general pattern: the words $H(\underbrace{0, \dots, 0}_{s-1}, 1; z) = \text{Li}_s(z)$ and $H(\underbrace{0, \dots, 0}_{s-1}, -1; z) = -\text{Li}_s(-z)$ recover the entire ladder of classical

polylogarithms, so the HPLs are a strict superset of the functions of Lewin's treatise. In Wolfram 15 these read, e.g., `HarmonicPolyLog[{0,1},x] = Li2(x)` and `HarmonicPolyLog[{0,-1},x] = -Li2(-x)`.

1.3.3 The condensed m -notation

Writing out long strings of 0's is wasteful, and Remiddi and Vermaseren introduced a condensed index that absorbs them. To each maximal block $\underbrace{0, \dots, 0}_{|m|-1} \sigma$ consisting of $|m| - 1$ zeros followed by a sign letter $\sigma \in \{1, -1\}$ one assigns a single integer

$$m = \sigma \cdot (\text{number of zeros} + 1),$$

so the sign of m records σ and the magnitude $|m|$ records how many kernels (one $f_{\pm 1}$ preceded by $|m| - 1$ copies of f_0) the block contributes. The resulting tuple is written as a subscript,

$$H_{m_1, \dots, m_k}(z) := H(\vec{a}; z),$$

and the *depth* of the HPL is k , the number of nonzero letters, while the weight is $w = \sum_i |m_i|$. Thus (1.15) becomes $H_2(z) = \text{Li}_2(z)$ and $H_{-2}(z) = -\text{Li}_2(-z)$, the polylogarithm ladder is $H_s(z) = \text{Li}_s(z)$, $H_{-s}(z) = -\text{Li}_s(-z)$, and a mixed word such as $H(0, 1, -1; z)$ is $H_{2, -1}(z)$. The m -notation is the input most reduction tables are organised by; the explicit word notation $H(\vec{a}; z)$ is the one that makes the integral representation (1.13) transparent. (Some authors reverse the global sign or the letter ordering; we follow the convention compatible with (1.14)–(1.15) and with `HarmonicPolyLog`, where indices read outermost-integration-first.)

1.3.4 The shuffle algebra

Theorem 1.8 (Shuffle product). *For a fixed argument z , the product of two harmonic polylogarithms is a $\mathbb{Q}$ -linear combination of harmonic polylogarithms of the same argument:*

$$H(\vec{a}; z) H(\vec{b}; z) = \sum_{\vec{c} \in \vec{a} \amalg \vec{b}} H(\vec{c}; z),$$

the sum running over all words $\vec{c}$ in the shuffle $\vec{a} \amalg \vec{b}$, i.e. all interleavings of $\vec{a}$ and $\vec{b}$ that preserve the internal order of each.

This is the iterated-integral analogue of integration by parts: a product of two iterated integrals over $[0, z]$ is the integral over the union of the two simplices, which decomposes into the $\binom{|\vec{a}|+|\vec{b}|}{|\vec{a}|}$ ordered shuffles. Concretely, at weight $1 + 1$,

$$H(0; z) H(1; z) = H(0, 1; z) + H(1, 0; z), \tag{1.17}$$

which is the statement $\ln z \cdot (-\ln(1 - z)) = \text{Li}_2(z) + (-\text{Li}_2(z) - \ln z \ln(1 - z)) = -\ln z \ln(1 - z)$, i.e. the integration-by-parts consistency that re-expresses the product of the two weight-one HPLs through the two weight-two ones. (The dilogarithm *reflection* formula is a different relation—the $z \mapsto 1 - z$ image of $H(0, 1; z)$, derived in (1.19) below.) The shuffle algebra is graded by weight, and crucially it lets one strip leading zeros: since $H(0; z) = \ln z$, repeated use of (1.8) expresses any $H(0, \vec{a}; z)$ through $\ln z$ and HPLs whose words begin with a nonzero letter, which is the regularisation alluded to after the all-zero regularization.

1.3.5 Values at $z = 1$: Euler sums and colored MZVs

Setting $z = 1$ turns admissible HPLs (those whose word does not begin with 1, so that the boundary integral converges) into constants. Because the kernels live on the fourth-root-of-unity locus $\{0, \pm 1\}$,

these constants are exactly the level-2, or *alternating*, multiple zeta values—the Euler sums. Words over $\{0, 1\}$ give the ordinary MZVs,

$$H(\underbrace{0, \dots, 0}_{s_1-1}, 1, \underbrace{0, \dots, 0}_{s_2-1}, 1, \dots; 1) = \zeta(s_1, s_2, \dots), \quad (1.18)$$

so for instance $H(0, 1; 1) = \text{Li}_2(1) = \zeta(2)$ and $H(0, 0, 1; 1) = \zeta(3)$. Admitting the letter -1 produces the alternating Euler sums: with $\bar{s}$ marking a sign,

$$H(0, -1; 1) = -\text{Li}_2(-1) = \frac{\pi^2}{12} = \zeta(\bar{2}), \quad H(0, 0, -1; 1) = -\text{Li}_3(-1) = \frac{3}{4}\zeta(3),$$

in the alternating-MZV notation $\zeta(\bar{s}_1, \dots) = \sum_{n_1 > \dots} \prod (\text{sgn})^{n_i} / n_i^{s_i}$. Thus the $\mathbb{Q}$ -algebra of HPL values at 1 is the algebra of alternating MZVs; its dimensions and the structure of its relations were mapped by Borwein–Bradley–Broadhurst–Lisonek and, on the motivic side, Deligne and Goncharov. This is the practical reason HPLs are the right basis: the boundary data of a Feynman integral land automatically in a well-understood number-theoretic ring.

1.3.6 Argument transformations and the alphabet

Changing variables transports the singular set of the differential forms. Negation and inversion, and the map $(1 - z)/(1 + z)$, preserve the extended three-letter singular set $\{0, 1, -1, \infty\}$, with appropriate endpoint and branch constants. Reflection $1 - z$ preserves the two-letter subalphabet $\{0, 1\}$ but sends the letter -1 to 2. Thus $H(-1; 1 - z) = \log(2 - z)$ cannot in general be expressed using only the original three-letter alphabet at z . Squaring likewise introduces letters $\pm i$ when applied to a word containing -1 .

The useful algorithm is to differentiate, partially fraction the transported kernels, integrate in the resulting alphabet, and fix the endpoint constant. For the two-letter dilogarithm this gives

$$\text{Li}_2(z) + \text{Li}_2(1 - z) = \frac{\pi^2}{6} - \log z \log(1 - z), \quad 0 < z < 1. \quad (1.19)$$

Analytic continuation requires a common choice of branches. Duplication is the special closed identity

$$H(0, 1; z^2) = 2(H(0, 1; z) - H(0, -1; z)).$$

These examples illustrate transport of alphabets, rather than universal closure under every fractional-linear or quadratic substitution.

1.3.7 Relation to Nielsen polylogarithms and to Li

The harmonic polylogarithms contain the Nielsen generalised polylogarithms $S_{n,p}(z)$ of Nielsen (1909), studied analytically by Kölbig (1986). Indeed $S_{n,p}(z)$ is, exactly the single HPL whose word is n zeros followed by p ones,

$$S_{n,p}(z) = H(\underbrace{0, \dots, 0}_n, \underbrace{1, \dots, 1}_p; z), \quad (1.20)$$

of weight $n + p$ and depth p ; the classical case is $S_{n,1}(z) = \text{Li}_{n+1}(z)$, consistent with (1.15) for $n = 1, p = 1$. Thus the chain of inclusions reads: classical Li_s (depth-1 HPLs over $\{0, 1\}$) $\subset$ Nielsen $S_{n,p}$ (two-letter words $0^n 1^p$) $\subset$ harmonic polylogarithms (arbitrary words over $\{0, 1, -1\}$) $\subset$ multiple polylogarithms (the $z = 1$ specialisations being MZVs and Euler sums). Each enlargement adds exactly the words the previous class could not name.

1.3.8 Finite truncations and Wolfram 15

Behind every iterated integral stands a nested *sum*, and behind every value at $z = 1$ stands the partial sum of that series. These partial sums are the multiple (finite) harmonic numbers

$$H_N^{(s_1, \dots, s_k)} = \sum_{N \geq n_1 > n_2 > \dots > n_k \geq 1} \prod_{i=1}^k \frac{1}{n_i^{s_i}},$$

generalising the ordinary $H_N = H_N^{(1)} = \sum_{n \leq N} 1/n$; their $N \rightarrow \infty$ limits are the admissible MZVs, and they are the natural objects in which the series expansions of ε -dependent Feynman integrals are stored. Wolfram 15 provides all of these as first-class functions: `HarmonicPolyLog` for $H(\vec{a}; z)$ (with the index/sign conventions used above; recall `HarmonicPolyLog[{1}, x]` = $-\ln(1 - x)$), `MultiplePolyLog` and `MultipleZeta` for the limiting transcendentals, and `MultipleHarmonicNumber` for the finite sums $H_N^{(\vec{s})}$. With `DSolve` and `AsymptoticDSolveValue` now returning harmonic polylogarithms for Fuchsian systems with rational coefficients, the entire Remiddi–Vermaseren workflow—solve the differential equation, expand at the boundary into Euler sums, continue by the transformations the transformations just discussed—is available without leaving the kernel.

Remark 1.9. The choice of alphabet $\{0, 1, -1\}$ is what fixes the singular set at $\{0, 1, -1, \infty\}$ and hence ties the boundary values to *alternating* MZVs. Allowing the additional kernels $f_{1/2}(t) = 1/(t - \frac{1}{2})$ or roots of unity of higher order enlarges the class to the two-dimensional harmonic polylogarithms and to Goncharov’s multiple polylogarithms with general singular points (the `GeneralizedPolyLog` of the previous section), at the price of boundary constants drawn from level- N colored MZVs rather than the level-2 Euler sums. For one-scale problems the three-letter alphabet is almost always enough, which is the empirical reason HPLs, not their more general cousins, became the working basis of the field.

1.4 Nielsen’s generalized polylogarithms

Long before multiple zeta values acquired their modern algebraic name, Nielsen (1909) introduced a two-parameter family of higher polylogarithms that already pervades Lewin’s *Polylogarithms and Associated Functions* (1981, Chapter 7). They are the simplest objects that go beyond the ordinary $\text{Li}_s(z)$ while remaining genuinely “one-variable”: a single argument z , two integer indices $n, p \geq 1$, and a closed integral representation. The reader who is comfortable with the dilogarithm reflection and Landen’s transformation will find here their direct higher-weight descendants, and – once we translate into the language of the previous sections – their place as one distinguished corner of the multiple-polylogarithm world.

1.4.1 Definition and elementary reductions

Definition 1.10. *For integers $n, p \geq 1$ the Nielsen generalized polylogarithm is*

$$S_{n,p}(z) = \frac{(-1)^{n+p-1}}{(n-1)! p!} \int_0^1 \frac{(\ln t)^{n-1} \ln^p(1-zt)}{t} dt, \quad (1.21)$$

analytically continued from $|z| < 1$ to $z \in \mathbb{C} \setminus [1, \infty)$ along the cut of $\ln(1-zt)$. Its weight is $n + p$ and its depth is p .

The normalizing factorials are chosen exactly so that two boundary specializations collapse onto familiar functions. Setting $p = 1$ and integrating $\ln(1 - zt)/t$ termwise against $(\ln t)^{n-1}$ recovers the classical Euler integral for the polylogarithm:

$$S_{n,1}(z) = \text{Li}_{n+1}(z), \quad n \geq 1. \quad (1.22)$$

Thus the entire first column of the Nielsen table is nothing but the ordinary polylogarithm shifted by one weight; in particular the lowest member is the dilogarithm,

$$S_{1,1}(z) = \text{Li}_2(z). \quad (1.23)$$

At the opposite corner $n = 1$ the inner factor $\ln t$ disappears and (1.21) reduces to a pure power of $\ln(1 - zt)$, giving the elementary $S_{0,p}(z) = \frac{(-1)^p}{p!} \ln^p(1 - z)$ in the degenerate $n = 0$ row (we keep $n \geq 1$ in the sequel, but this boundary is the seed for the recursions below).

1.4.2 Series and the multiple-polylogarithm identification

Expanding $\ln^p(1 - zt)$ as a power of the series $-\sum_{m \geq 1} (zt)^m/m$ and integrating against $(\ln t)^{n-1}/t$ turns (1.21) into a single power series in z whose coefficients are nested harmonic sums:

$$S_{n,p}(z) = \sum_{k \geq 1} \frac{z^k}{k^{n+1}} c_k^{(p)}, \quad c_k^{(p)} = \sum_{k > m_1 > \dots > m_{p-1} \geq 1} \frac{1}{m_1 \cdots m_{p-1}}, \quad (1.24)$$

so $c_k^{(1)} \equiv 1$ (reproducing (1.22)) and in general $c_k^{(p)}$ is a depth- $(p-1)$ harmonic sum, an instance of `MultipleHarmonicNumber`. Comparing (1.24) with the strictly nested sum of the article preamble identifies $S_{n,p}$ as a multiple polylogarithm with a single repeated argument:

$$\boxed{S_{n,p}(z) = \text{Li}_{n+1, \underbrace{1, \dots, 1}_{p-1}}(z, \underbrace{1, \dots, 1}_{p-1})} \quad (1.25)$$

of weight $n + p$ and depth p . Its iterated-integral word over the alphabet $\{0, 1\}$ (with $0 \leftrightarrow dt/t$, $1 \leftrightarrow dt/(1 - t)$) is

$$0^n 1^p = \underbrace{0 \cdots 0}_n \underbrace{1 \cdots 1}_p,$$

a single block of n zeros followed by a single block of p ones. This is the defining structural fact: among all multiple polylogarithms, the Nielsen family is exactly the “one initial zero-block, one trailing one-block” subclass. Everything peculiar about $S_{n,p}$ – the two-index labelling, the clean ladders, the duality below – is a shadow of that two-block word.

Remark 1.11 (Sign and normalization). With the positive kernel $dt/(1 - t)$ the word $0^n 1^p$ equals $S_{n,p}(z)$ with no extra sign. Replacing each 1-kernel by $dt/(t - 1)$ gives the factor $(-1)^p$. No factor $(-1)^{p-1}$ belongs to either convention.

1.4.3 Integral and differential relations

Differentiating (1.24) (or (1.21)) in z lowers the leading index by one and threads the function back into the ladder:

$$\frac{d}{dz} S_{n,p}(z) = \frac{1}{z} S_{n-1,p}(z) \quad (n \geq 1), \quad \frac{d}{dz} S_{1,p}(z) = \frac{1}{z} S_{0,p}(z) = \frac{(-1)^p}{p!} \frac{\ln^p(1 - z)}{z}, \quad (1.26)$$

the second relation being the $n = 1$ boundary, where the dt/t block is exhausted and the degenerate row $S_{0,p}(z) = \frac{(-1)^p}{p!} \ln^p(1-z)$ is the pure power of $\ln(1-z)$. The two clean ladder moves are the integral forms of these derivatives,

$$\text{zero-block ladder: } S_{n,p}(z) = \int_0^z S_{n-1,p}(u) \frac{du}{u} \quad (n \geq 1), \quad (1.27)$$

$$\text{one-block ladder: } S_{0,p}(z) = \int_0^z S_{0,p-1}(u) \frac{du}{1-u} \quad (p \geq 1), \quad (1.28)$$

with seed $S_{0,0}(z) = 1$ (the empty word): (1.27) peels the outermost 0 from the word $0^n 1^p$, and (1.28) peels the outermost 1 from the pure one-block 1^p ; in iterated-integral terms both are simply “integrate against the next letter of $0^n 1^p$.” The practical content is that the whole Nielsen table is generated from the empty word by first applying the 1-kernel integral $\int_0^z (\cdot) du/(1-u)$ a total of p times to build $S_{0,p}$ (so $S_{0,1}(z) = -\ln(1-z)$, $S_{0,2}(z) = \frac{1}{2} \ln^2(1-z)$, and so on), then the 0-kernel integral $\int_0^z (\cdot) du/u$ a total of n times to prepend the zeros; this is exactly the Fuchsian recursion that `HarmonicPolyLog` encodes, and it is why `DSolve` now returns Nielsen polylogs for the relevant rational-coefficient ODEs.

1.4.4 Special values

The point $z = 1$ and Hoffman’s duality

Setting $z = 1$ in (1.25) turns the multiple polylogarithm into a multiple zeta value with the same two-block word. The first column reproduces Euler,

$$S_{n,1}(1) = \zeta(n+1), \quad S_{1,2}(1) = \zeta(3), \quad (1.29)$$

and in general

$$S_{n,p}(1) = \zeta(n+1, \{1\}^{p-1}), \quad (1.30)$$

the MZV whose argument string is $n+1$ followed by $p-1$ ones (admissible because the leading entry $n+1 \geq 2$). Now the iterated-integral word $0^n 1^p$ of $S_{n,p}$ has a manifest symmetry: reversing it and exchanging $0 \leftrightarrow 1$ sends $0^n 1^p \mapsto 0^p 1^n$, the word of $S_{p,n}$. This reversal-complement is exactly Hoffman’s MZV *duality* (the Kontsevich integral symmetry $\omega \mapsto \bar{\omega}^{\text{rev}}$), so

$$\boxed{\zeta(n+1, \{1\}^{p-1}) = \zeta(p+1, \{1\}^{n-1}), \quad \text{equivalently} \quad S_{n,p}(1) = S_{p,n}(1).} \quad (1.31)$$

This is a striking statement: the value of a Nielsen polylogarithm at 1 is invariant under swapping its two indices, even though $S_{n,p}(z)$ and $S_{p,n}(z)$ are wholly different functions of z . For $p = 1$ it gives $\zeta(n+1) = \zeta(2, \{1\}^{n-1})$; the first nontrivial case is $S_{1,2}(1) = S_{2,1}(1)$, i.e. $\zeta(2, 1) = \zeta(3)$, Euler’s reflection. We confirmed (1.30) and (1.31) symbolically against `MultipleZeta` (e.g. $S_{1,2}(1) - \zeta(2, 1)$ and $S_{n,p}(1) - S_{p,n}(1)$ vanish to > 70 digits for all $n, p \leq 4$).

The points $z = -1$ and $z = \frac{1}{2}$ (Kölbig 1986)

At $z = -1$ the column reduction (1.22) gives $S_{n,1}(-1) = \text{Li}_{n+1}(-1) = -(1-2^{-n})\zeta(n+1)$, and the first off-column value is the elegant

$$S_{1,2}(-1) = \frac{1}{8} \zeta(3). \quad (1.32)$$

Higher entries $S_{n,p}(-1)$ reduce (Kölbig 1986) to the alternating-MZV basis at their weight; for example $S_{2,2}(-1)$ and $S_{1,3}(-1)$ already require $\text{Li}_4(\frac{1}{2})$ alongside π^4 , $\ln^4 2$, $\pi^2 \ln^2 2$ and $\zeta(3) \ln 2$, the

unmistakable level-2 weight-4 signature (we recovered both reductions by integer-relation search). At the binary point $z = \frac{1}{2}$ the cleanest closed form is

$$\boxed{S_{1,2}(\tfrac{1}{2}) = \tfrac{1}{8} \zeta(3) - \tfrac{1}{6} \ln^3 2.} \quad (1.33)$$

We obtained (1.33) by PSLQ over $\{\zeta(3), \pi^2 \ln 2, \ln^3 2\}$ – the relation $24 S_{1,2}(\frac{1}{2}) - 3 \zeta(3) + 4 \ln^3 2 = 0$ – and verified it to 40 digits. (Compare Lewin’s dilogarithm value $\text{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2} \ln^2 2$: the same flavour, one weight up.)

1.4.5 Functional equations

The reader who knows Lewin’s dilogarithm chapter expects three staples – reflection $z \leftrightarrow 1 - z$, inversion $z \leftrightarrow 1/z$, and the Landen ladder – to survive at higher weight. They do, as specializations of the harmonic-polylogarithm argument transformations of the previous section, but they grow extra lower-weight “tail” terms.

Reflection $z \mapsto 1 - z$

At weight 2 the only reflection is the dilogarithm law itself, here in Nielsen dress ($n = p = 1$):

$$S_{1,1}(z) + S_{1,1}(1 - z) = \frac{\pi^2}{6} - \ln z \ln(1 - z), \quad (1.34)$$

which is exactly Euler’s reflection for Li_2 . At weight 3 the genuinely new instance is the $S_{1,2}$ reflection; carrying out the integer-relation search over the natural weight-3 basis yields the clean

$$\boxed{S_{1,2}(z) = \zeta(3) - \text{Li}_3(1 - z) + \ln(1 - z) \text{Li}_2(1 - z) + \tfrac{1}{2} \ln z \ln^2(1 - z).} \quad (1.35)$$

We verified (1.35) to 40 digits at several rational and one negative argument. Two consistency checks are worth noting. Putting $z = 1$ collapses the right side to $\zeta(3) - \text{Li}_3(0) + 0 + 0 = \zeta(3)$, reproducing (1.29); and adding (1.35) to its $z \mapsto 1 - z$ image recovers the symmetric Euler relation $S_{1,2}(z) + S_{1,2}(1 - z) = 2\zeta(3) - \text{Li}_3(z) - \text{Li}_3(1 - z) + \dots$ with the logarithmic tails arranging themselves into the $\zeta(3) + \text{Li}_3$ shuffle. Identity (1.35) is precisely the statement that the iterated integral of 011 from 0 to z and from 0 to $1 - z$ differ by an explicit lower-weight boundary correction – the weight-3 avatar of the dilogarithm reflection.

Inversion $z \mapsto 1/z$ **and Landen**

The inversion does *not* close on $S_{n,p}$ alone at weight ≥ 3 : $S_{1,2}(1/z)$ is not a $\mathbb{Q}$ -combination of $S_{1,2}(z)$, $\zeta(3)$, $\pi^2 \ln z$ and $\ln^3 z$ (an integer-relation search over that ad hoc basis returns only spurious large-coefficient vectors). What is true is the structural statement that $z \mapsto 1/z$, like $z \mapsto 1 - z$ and the Landen map $z \mapsto z/(z - 1)$, is one of the S_3 (in fact $\mathfrak{S}$ -group) argument transformations under which the *whole* weight- w harmonic-polylogarithm space is closed. Concretely, writing $S_{n,p}$ in the word $0^n 1^p$ and applying the HPL substitution rules of the previous section expresses $S_{n,p}(1/z)$ – and, separately, $S_{n,p}(z/(z - 1))$ – as an explicit $\mathbb{Q}$ -linear combination of all weight- $(n + p)$ harmonic polylogarithms in z together with products of $\ln z$, $\ln(1 - z)$ and lower-weight S ’s and ζ ’s. Thus the Nielsen polylogarithms inherit the dilogarithm’s full transformation calculus, but only after one leaves the two-index Nielsen subfamily for the ambient HPL algebra; the inversion is the cheapest illustration that $\{S_{n,p}\}$ is *not* closed under the group of fractional-linear argument changes, whereas the HPLs are.

1.4.6 Placement inside the multiple-polylogarithm world

Equation (1.25) identifies $S_{n,p}$ as the “one trailing block” corner of the depth- p multiple polylogarithms: the words $0^n 1^p$ form a single two-parameter slice of the full $\{0, 1\}^*$ alphabet, the slice closest to the ordinary polylogarithm (recovered at $p = 1$) and stable under the two operations – evaluation at 1 to MZVs, and the duality swap $n \leftrightarrow p$ – that organize the modern theory. This is exactly why the Nielsen polylogarithms are the natural bridge between Lewin’s classical apparatus and the MZV algebra: every Lewin-style identity (reflection, Landen, the special values at $\pm 1, \frac{1}{2}$) is the $p \leq 2$ or $n \leq 2$ shadow of a multiple-polylogarithm identity, and conversely the duality $S_{n,p}(1) = S_{p,n}(1)$ is the first place a classical reader meets a genuinely MZV-theoretic symmetry with no single-variable analogue. In the Wolfram Language the family is a first-class function: `PolyLog[n,p,z]` is exactly $S_{n,p}(z)$, with `PolyLog[n,1,z]` agreeing with `PolyLog[n+1,z]` per (1.22), and it sits one specialization step from `MultiplePolyLog` via (1.25) and one from `MultipleZeta` via (1.30).

1.5 The common algebra and the motivic boundary

The nested series and outer-letter-first integral describe one function. Choosing the alphabet distinguishes general multiple polylogarithms, cyclotomic multiple polylogarithms, HPLs and the Nielsen subfamily. General Goncharov letters can be arbitrary complex numbers; restricting them to roots of unity gives only the cyclotomic class.

One must distinguish the argument patterns of the two products. Shuffle at a common endpoint gives $\text{Li}_1(z)^2 = 2 \text{Li}_{1,1}(z, 1)$, whereas stuffle gives $\text{Li}_1(z)^2 = 2 \text{Li}_{1,1}(z, z) + \text{Li}_2(z^2)$. Both follow directly from their respective representations and therefore give a nontrivial identity when compared.

Iterated integrals also carry a coproduct organized by subsequences of marked points. Its reduced components and the associated symbol organize functional equations by weight. They lose some period constants: vanishing symbol or reduced coproduct alone is not a proof of a zero function. Endpoint constants and branches must be fixed separately. For the dilogarithm the differential $d\text{Li}_2(z) = -\log(1-z)d\log z$ makes this elementary; differentiating Euler reflection and evaluating an endpoint proves the identity exactly.

Motivic MZVs [39] provide an algebraic setting in which dimension and spanning theorems can be proved. The numerical period map transports identities and upper bounds. Its injectivity, and thus the corresponding lower bounds for numerical periods, is a separate conjectural issue. The same caution will apply to the Gaussian and Eisenstein depth experiments.

Chapter 2

Cyclotomic values and the polygamma bridge

Roots of unity turn additive Fourier coordinates into arithmetic character coordinates. The same finite transform carries Hurwitz zeta, positive-order polygamma and Clausen values. It explains which half of a polylogarithm is elementary at a given weight and which character values remain to be studied.

2.1 The two-way exact DFT

We use the Clausen convention

$$\mathrm{Cl}_{2m}(\theta) = \Im \mathrm{Li}_{2m}(e^{i\theta}), \quad \mathrm{Cl}_{2m+1}(\theta) = \Re \mathrm{Li}_{2m+1}(e^{i\theta}).$$

The sine series at odd weight is a different function; its rational-angle values reduce to powers of π through Bernoulli polynomials.

Law 2.1 (forward direction; classical). *For every $n \geq 2$, every $q \geq 1$, and $\gcd(p, q) = 1$,*

$$\mathrm{Li}_n(e^{2\pi ip/q}) = \frac{(-1)^n}{q^n (n-1)!} \sum_{k=1}^{q-1} e^{2\pi i k p/q} \psi^{(n-1)}\left(\frac{k}{q}\right) + \frac{\zeta(n)}{q^n}.$$

This is identity 10.08.03.0007.01 of the `functions.wolfram.com PolyLog` reference [2]; it is provable by rearranging the absolutely convergent Hurwitz-zeta series, and it is the workhorse of the project's cyclotomic reducer (`cycloExact`): a finite polygamma sum, exact and fast, resolving Li_n at *every* root of unity with no PSLQ, no squarefree-radical basis, and no nested-radical limitation.

Result 2.2 (inverse direction). Inverting the discrete Fourier transform over $\mathbb{Z}/q$ gives, with $\zeta_q = e^{2\pi i/q}$,

$$\psi^{(n-1)}\left(\frac{k}{q}\right) = (-1)^n (n-1)! q^{n-1} \sum_{p=0}^{q-1} \zeta_q^{-pk} \mathrm{Li}_n(\zeta_q^p).$$

[historical experiment: verified to 40 digits at weights 2, 3 across $q \leq 21$; re-verified by mpmath at 70 dps for $n-1 = 1, 2, 3$ over the full store grid]

The bridge is therefore an *exact finite Fourier transform, in both directions*. Three non-obvious consequences are recorded.

1. **DFT-conjugacy of the relation lattices.** The distribution relation $\sum_j \text{Li}_n(z \zeta_m^j) = m^{1-n} \text{Li}_n(z^m)$ Fourier-transforms into the polygamma multiplication theorem, and polylogarithm inversion/reflection corresponds to the symmetry $k \leftrightarrow q - k$ of the ψ -grid. This is *why* the two relation lattices — computed independently by the project’s two reducers — carry identical dimension counts $\varphi(q)/2$ per denominator grid.
2. **The parity split, and Catalan as the $q = 4$ odd component.** The imaginary parts are the odd Fourier sector under $k \mapsto q - k$. The usual even-weight Clausen values belong to this sector, while odd-weight Clausen values are real parts in the even sector. In particular

$$G = \text{Im Li}_2(i) = \frac{\psi^{(1)}(1/4) - \psi^{(1)}(3/4)}{16},$$

which identifies the well-known “arguments 1/4 and 3/4” pairing (Math.SE 893486) as the $q = 4$ odd component of the parity split.

3. **Certificate transport.** Cyclotomic-polylogarithm proof certificates can in principle be transported to and from ψ -side lattice certificates through the exact $\mathbb{Q}(\zeta_q)$ DFT matrix. The ψ -side elementary atoms ($\pi^2 \csc^2$ values) are far lighter than the sine-product algebraic objects that constrain the Γ -certificate sweeps on large grids, so the transport is a proposed escape route from that bottleneck (not yet implemented).

2.2 Clausen-type laws for polygamma values at rational arguments

MathWorld’s *Polygamma Function* page lists fifteen closed forms in its special-cases block, eqs. (22)–(36) ($\psi^{(1)}, \psi^{(2)}, \psi^{(3)}$ at p/q with $q \in \{2, 3, 4, 6\}$) in Catalan/Clausen/ β atoms, several of them traditionally presented as isolated curiosities. All of them — and their extension to *every* rational argument — are instances of three sine/cosine-weighted Clausen-DFT laws, obtained by splitting the inverse DFT of Result 2.2 into its elementary (reflection) half and its Clausen half.

Law 2.3 (trigamma; Lewin 1991). *For $0 < p < q$, $\gcd(p, q) = 1$,*

$$\psi^{(1)}\left(\frac{p}{q}\right) = \frac{\pi^2}{2 \sin^2(\pi p/q)} + 2q \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \sin\left(\frac{2\pi j p}{q}\right) \text{Cl}_2\left(\frac{2\pi j}{q}\right).$$

[historical experiment: verified at $q = 3, 4, 5, 8, 12$ to 10^{-46} or better, then over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

This law is classical — MathWorld’s *Trigamma Function* eq. (14) cites Lewin (1991) [1] — and was independently re-derived here from the DFT bridge before the attribution was spotted; the project’s records explicitly re-frame the finding as *placement*, not novelty. MathWorld’s scattered coefficients are its small- q shadows: the $3\sqrt{3}$ of $q = 3$ is $2q \sin(2\pi/3)$, the $8G$ of $q = 4$ is $2q \text{Cl}_2(\pi/2)$.

Law 2.4 (tetragamma).

$$\psi^{(2)}\left(\frac{p}{q}\right) = -\pi^3 \cot\left(\frac{\pi p}{q}\right) \csc^2\left(\frac{\pi p}{q}\right) - 2q^2 \left(\zeta(3) + 2 \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \cos\left(\frac{2\pi j p}{q}\right) \text{Cl}_3\left(\frac{2\pi j}{q}\right) + \varepsilon_q \right),$$

where $\varepsilon_q = (-1)^p \text{Cl}_3(\pi)$ for even q , $\varepsilon_q = 0$ for odd q , and $\text{Cl}_3(\pi) = -\frac{3}{4}\zeta(3)$. [historical experiment: verified over the full store grid to 10^{-50} in-kernel and at 70 dps in mpmath]

Law 2.5 (pentagramma).

$$\psi^{(3)}\left(\frac{p}{q}\right) = \pi^4 \left(2 \cot^2\left(\frac{\pi p}{q}\right) \csc^2\left(\frac{\pi p}{q}\right) + \csc^4\left(\frac{\pi p}{q}\right) \right) + 12 q^3 \sum_{j=1}^{\lfloor (q-1)/2 \rfloor} \sin\left(\frac{2\pi j p}{q}\right) \text{Cl}_4\left(\frac{2\pi j}{q}\right),$$

equivalently, in difference form, $\psi^{(3)}(p/q) - \psi^{(3)}(1-p/q) = 24 q^3 \sum_j \sin(2\pi j p/q) \text{Cl}_4(2\pi j/q)$. [historical experiment: same verification battery]

The $\psi^{(3)}$ law is the Lewin law one weight up, by the same derivation; no literature citation has been located, and the project records it as “presumably in Lewin’s orbit.” MathWorld’s $162\sqrt{3}$ and $768\beta(4)$ coefficients are $12q^3 \sin(2\pi j/q)$ -instances of the one-sided law ($24q^3$ in the difference form).

Result 2.6 (new instances beyond the MathWorld table). Among the instances at $q \in \{5, 8, 12\}$:

$$\psi^{(1)}\left(\frac{1}{5}\right) = \frac{\pi^2}{2 \sin^2(\pi/5)} + 10 \left[\sin \frac{2\pi}{5} \text{Cl}_2\left(\frac{2\pi}{5}\right) + \sin \frac{4\pi}{5} \text{Cl}_2\left(\frac{4\pi}{5}\right) \right],$$

$$\psi^{(1)}\left(\frac{1}{12}\right) = 40 G + \frac{4\pi^2}{(\sqrt{3}-1)^2} + 30\sqrt{3} \text{Cl}_2\left(\frac{2\pi}{3}\right),$$

the latter after duplication/triplication canonicalization collapses the raw five-atom DFT form onto the canonical atom set. Also, MathWorld’s eq. (27) simplifies: from $\text{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$,

$$\psi^{(2)}\left(\frac{1}{3}\right) = -\frac{4\pi^3}{3\sqrt{3}} - 26 \zeta(3).$$

Result 2.7 (the polygamma–Clausen store). The corpus now carries **identities/polygamma-clausen-values.wl**: 54 entries $\psi^{(m)}(p/q)$ for $m \in \{1, 2, 3\}$, $q \in \{3, 4, 5, 6, 8, 12\}$, all p coprime to q (18 arguments $\times$ 3 orders), in canonicalized Clausen atoms. The residual atom basis across the store is

$$\begin{aligned} & \{G, \beta(4), \zeta(3)\} \cup \text{Cl}_2\left\{\frac{2\pi}{3}, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{\pi}{4}\right\} \\ & \cup \text{Cl}_3\left\{\frac{2\pi}{5}, \frac{\pi}{4}, \frac{\pi}{6}\right\} \cup \text{Cl}_4\left\{\frac{2\pi}{3}, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{\pi}{4}\right\}. \end{aligned}$$

Every entry is verified in-kernel to $|\text{lhs} - \text{rhs}| < 10^{-50}$ at 60 digits (including a disk round-trip) and independently by an mpmath script at 70 dps (tolerance 10^{-55}) that re-derives the raw DFT bridge, the three laws, all 17 Clausen canonicalization relations, and the L -function bridges of Section 2.3 from mpmath primitives, with positive and negative PSLQ controls.

2.3 The boundary of the odd-weight Clausen collapse

A folklore rule says that odd-weight Clausen values at rational angles collapse to rational multiples of zeta values ($\text{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$ being the standard example). The store emission *sharpened the rule’s scope*:

Proposition 2.8 (Crystallographic-angle reductions). *At denominators $q \in \{1, 2, 3, 4, 6\}$, the odd Clausen value $\text{Cl}_3(2\pi j/q) = \Re \text{Li}_3(e^{2\pi i j/q})$ reduces to a rational multiple of $\zeta(3)$. At $q = 5, 8, 12$ the*

character decomposition instead gives

$$\begin{aligned}\mathrm{Cl}_3(2\pi/5) &= -\frac{6}{25}\zeta(3) + \frac{\sqrt{5}}{4}L(3, \chi_5), \\ \mathrm{Cl}_3(\pi/4) &= -\frac{3}{256}\zeta(3) + \frac{1}{\sqrt{2}}L(3, \chi_8), \\ \mathrm{Cl}_3(\pi/6) &= \frac{1}{24}\zeta(3) + \frac{\sqrt{3}}{2}L(3, \chi_{12}).\end{aligned}$$

Here χ_D is the primitive even quadratic character of conductor D .

These are exact coordinate identities from the residue-class Fourier sum. No theorem asserting that the three L -values are independent of $\zeta(3)$ or π^3 is supplied. The source searches at height 10^{10} are numerical evidence in a stated basis, not an “if and only if” classification of all possible reductions.

2.4 The fundamental constants, by conductor

The residue classes of a cyclotomic series can be organized either by additive Fourier modes or by Dirichlet characters. Odd Clausen values use the odd character sector. Distribution relations first remove non-coprime residues; the remaining finite character transform has $\varphi(n)/2$ formal odd coordinates. Its coefficients are generally algebraic, so this coordinate count is not a proof of the dimension of the numerical $\mathbb{Q}$ -span of the periods.

For the levels 8, 12, 24, convenient generators are

$$V = \mathrm{Cl}_2(\pi/3), \quad G = \mathrm{Cl}_2(\pi/2), \quad W_8 = \mathrm{Cl}_2(\pi/4), \quad W_{24} = \mathrm{Cl}_2(\pi/12).$$

There are no primitive odd characters of conductor 12; its odd characters are induced from conductors 3 and 4. This explains the distribution reductions below. Conductors 3 and 4 each have one primitive odd character, while conductor 5 has two conjugate primitive odd characters. The source’s printed sequence of primitive-character counts started with two incorrect zeros; it is not used here.

Their canonical L -value (trigamma) forms are

$$G = \frac{\psi^{(1)}(\frac{1}{4}) - \pi^2}{8}, \quad \sqrt{3}V = \frac{3\psi^{(1)}(\frac{1}{3}) - 2\pi^2}{6} \quad \left(\text{i.e. } \psi^{(1)}(\frac{1}{3}) = \frac{2\pi^2}{3} + 2\sqrt{3}V\right). \quad (2.1)$$

2.5 Reduction tables and the settled level-24 question

Every $\mathrm{Cl}_2(2\pi a/n)$ at $n \in \{8, 12, 24\}$ reduces to the basis above (PSLQ, ≥ 110 digits, Champernowne canary; all confirmed in Wolfram):

$$\begin{aligned}\text{level 12 : } & \mathrm{Cl}_2(\frac{\pi}{6}) = \frac{2}{3}G + \frac{1}{4}V, \quad \mathrm{Cl}_2(\frac{5\pi}{6}) = \frac{2}{3}G - \frac{1}{4}V, \quad \mathrm{Cl}_2(\frac{2\pi}{3}) = \frac{2}{3}V; \\ \text{level 8 : } & \mathrm{Cl}_2(\frac{3\pi}{4}) = W_8 - \frac{1}{2}G; \\ \text{level 24 : } & \mathrm{Cl}_2(\frac{5\pi}{12}) = \frac{16W_8 + 24W_{24} - 4G - 3V}{24}, \quad \mathrm{Cl}_2(\frac{7\pi}{12}) = \frac{4W_8 + 6W_{24} - 3G}{6}, \\ & \mathrm{Cl}_2(\frac{11\pi}{12}) = \frac{24W_{24} - 8G - 3V}{24}.\end{aligned}$$

The displayed formulas prove that the level-24 values are generated by G, V, W_8, W_{24} . They give an upper bound of four on their rational span. The source did not prove that W_{24} lies outside the span of the other three, so the minimal number of numerical generators remains open. This distinction is central when a symbolic conductor count is compared with a PSLQ experiment.

2.6 A new closed form

Since $\text{Cl}_2(\pi/12) = W_{24}$ is itself fundamental, the cleanest carrier is the dilogarithm at the primitive 24th root $e^{i\pi/12}$:

$$\text{Li}_2(e^{i\pi/12}) = \frac{73\pi^2}{576} + i W_{24}, \quad (2.2)$$

the real part being $\frac{\pi^2}{6} - \frac{\pi}{2} \cdot \frac{\pi}{12} + \frac{1}{4}(\frac{\pi}{12})^2$. Pulling this back through $z = 1/(1 - e^{i\theta})$ gives Reshetnikov-style closed forms at the corresponding $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{6})$ points; the cleanest stays the root-of-unity form (2.2).

2.7 A weight parity swap

The same character parity that makes Cl_2 (weight 2, odd characters) carry deep constants reverses one weight up. For primitive odd χ , the positive odd integer values are algebraic multiples of π^s ; since $\text{Im Li}_3(e^{i\theta}) = \sum \sin(k\theta)/k^3$ pairs with odd characters at $s = 3$,

$$\text{Im Li}_3(e^{2\pi i a/n}) \in \mathbb{Q} \cdot \pi^3 \quad \text{for all } a, n, \quad (2.3)$$

e.g. $\text{Im Li}_3(e^{2\pi i/3}) = \frac{2\pi^3}{81}$, $\text{Im Li}_3(i) = \frac{\pi^3}{32}$, $\text{Im Li}_3(e^{i\pi/3}) = \frac{5\pi^3}{162}$, $\text{Im Li}_3(e^{2\pi i/5}) = \frac{4\pi^3}{125}$. Thus at weight 3 the *imaginary* part at roots of unity is elementary, and the deep constants ($\zeta(3)$ and the even-character $L(3, \chi)$) live in the *real* part—the exact reverse of weight 2. (On the line $\text{Re } z = 1$ the picture is different again: there $\text{Im Li}_3(1+i)$ is the central-binomial Gaussian tower constant, because $1+i$ is not a root of unity; see [polylog-vertical-line](#).)

Chapter 3

Algebraic arguments, integrals and ladders

Away from roots of unity, functional equations still move values onto structured sets of arguments. Vertical lines and half-angle substitutions give tractable examples; unit relations in number fields lead to longer ladders. The organizing principle is to understand the map of arguments before choosing the constants in an integer-relation search.

3.1 The problem and the key geometric fact

[MSE 1410276](#) records

$$\operatorname{Re} \operatorname{Li}_2(1 \pm i\sqrt{3}) = \frac{\pi^2}{24} - \frac{1}{2} \ln^2 2 - \frac{1}{4} \operatorname{Li}_2\left(\frac{1}{4}\right), \quad (3.1)$$

with $\operatorname{Im} \operatorname{Li}_2(1 + i\sqrt{3}) = \frac{\pi \ln 2}{2} + \frac{5\sqrt{3}}{72} [\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})]$ ([a/1410829](#), [a/1410705](#)). The point $1 + i\sqrt{3} = 2e^{i\pi/3}$, but the structurally decisive fact is that for any $z = 1 + iy$ the reflection point $1 - z = -iy$ is *purely imaginary*. This is exactly what makes the dilogarithm functional equations collapse.

3.2 The weight-2 vertical-line theorem

Theorem 3.1. *For $y > 0$,*

$$\operatorname{Re} \operatorname{Li}_2(1 + iy) = \frac{\pi^2}{6} - \frac{1}{2} \ln(1+y^2) \ln y - \frac{\pi}{2} \arctan y - \frac{1}{4} \operatorname{Li}_2(-y^2), \quad (3.2)$$

$$\operatorname{Im} \operatorname{Li}_2(1 + iy) = \frac{\pi}{4} \ln(1+y^2) - \arctan y \ln y + \operatorname{Ti}_2(y), \quad \operatorname{Ti}_2(y) = \operatorname{Im} \operatorname{Li}_2(iy). \quad (3.3)$$

Proof. The reflection law $\operatorname{Li}_2(z) + \operatorname{Li}_2(1-z) = \zeta(2) - \ln z \ln(1-z)$ at $z = 1 + iy$, $1-z = -iy$, gives $\operatorname{Li}_2(1+iy) = \zeta(2) - \ln(1+iy) \ln(-iy) - \operatorname{Li}_2(-iy)$. The duplication law $\operatorname{Li}_2(iy) + \operatorname{Li}_2(-iy) = \frac{1}{2} \operatorname{Li}_2(-y^2)$ together with $\overline{\operatorname{Li}_2(w)} = \operatorname{Li}_2(\bar{w})$ yields $\operatorname{Re} \operatorname{Li}_2(-iy) = \frac{1}{2} (\operatorname{Li}_2(iy) + \operatorname{Li}_2(-iy)) = \frac{1}{4} \operatorname{Li}_2(-y^2)$ and $\operatorname{Im} \operatorname{Li}_2(-iy) = -\operatorname{Ti}_2(y)$. Expanding $\ln(1+iy) = \frac{1}{2} \ln(1+y^2) + i \arctan y$ and $\ln(-iy) = \ln y - \frac{i\pi}{2}$ and taking real and imaginary parts gives (3.2)–(3.3). $\square$

On the line $\operatorname{Re} z = 1$ the dilogarithm thus splits into a *real* part carried by the even real dilogarithm $\operatorname{Li}_2(-y^2)$ and an *imaginary* part carried by the odd inverse-tangent integral $\operatorname{Ti}_2(y)$. Equation (3.1) is the case $y = \sqrt{3}$ (using $\operatorname{Li}_2(-3) = \operatorname{Li}_2(\frac{1}{4}) - \zeta(2) + 2 \ln 2 \ln \frac{2}{3}$), and the trigamma form of the imaginary part is just $\operatorname{Ti}_2(\sqrt{3}) = \frac{\pi}{6} \ln 3 + \frac{5\sqrt{3}}{72} [\psi_1(\frac{1}{3}) - \psi_1(\frac{2}{3})]$ substituted into (3.3).

3.3 The Clausen form at $y = \tan \theta$

Since $1 + i \tan \theta = e^{i\theta} / \cos \theta$, Theorem 3.1 combines with Ramanujan's $\text{Ti}_2(\tan \theta) = \theta \ln \tan \theta + \frac{1}{2} [\text{Cl}_2(2\theta) + \text{Cl}_2(\pi - 2\theta)]$ so that the $\theta \ln \tan \theta$ terms cancel, leaving a pure-Clausen imaginary part:

$$\text{Im Li}_2(1 + i \tan \theta) = -\frac{\pi}{2} \ln \cos \theta + \frac{1}{2} [\text{Cl}_2(2\theta) + \text{Cl}_2(\pi - 2\theta)]. \quad (3.4)$$

Corollary 3.2. $\text{Re Li}_2(1 + i) = \frac{\pi^2}{16}$ and $\text{Im Li}_2(1 + i) = \frac{\pi \ln 2}{4} + G$ (with G Catalan's constant), from $y = 1$ in Theorem 3.1 (using $\text{Li}_2(-1) = -\pi^2/12$, $\text{Ti}_2(1) = G$).

3.4 Weight 3: a half-lifted symmetry

At weight 3 the real/imaginary split survives only in part. The *real* part still closes at special y :

$$\text{Re Li}_3(1 + i) = \frac{35}{64} \zeta(3) + \frac{\pi^2 \ln 2}{32}, \quad \text{Re Li}_3(1 + i\sqrt{3}) = \frac{35\zeta(3) + 18 \text{Li}_3(\frac{1}{4}) + 6\pi^2 \ln 2 - 24 \ln^3 2}{144}, \quad (3.5)$$

the second being the exact weight-3 analogue of the $\text{Li}_2(\frac{1}{4})$ in (3.1). But the general real part does *not* close: differentiating, $\partial_y \text{Re Li}_3(1 + iy) = -(\text{Im Li}_2(1 + iy) - y \text{Re Li}_2(1 + iy))/(1 + y^2)$, and integrating the Ti_2 contribution produces $\int_0^y \arctan^2(t)/t dt$, a weight-three inverse-tangent-square integral. The displayed special cases at $y = 1, \sqrt{3}$ simplify, without giving an elementary-value classification for every rational angle. Meanwhile the *imaginary* part is no longer elementary at all: it is the central-binomial Gaussian tower constant

$$\text{Im Li}_3(1 + i) = \mathcal{F}_3 - \frac{\pi^3}{192}, \quad \mathcal{F}_3 = \sqrt{2} {}_4F_3(\{\frac{1}{2}\}^4; \{\frac{3}{2}\}^3; \frac{1}{2}), \quad (3.6)$$

the $n = 3$ entry of the $\text{Im Li}_n(1 + i)$ tower (companion report **gaussian-1plusi-tower**). So on $\text{Re } z = 1$ the imaginary part is precisely the deep object that the central-binomial ladder measures, while the real part remains tractable—the two halves of the polylogarithm part ways at weight 3.

3.5 The integral and its provenance

The object (Math.StackExchange [question 564816](#)) is

$$I = \int_0^{\pi/2} \arctan^2\left(\frac{6 \sin x}{3 + \cos 2x}\right) dx = 1.3309703968425666666125327 \dots \quad (3.7)$$

Cleo gave the dilogarithmic form

$$I = \frac{\pi}{2} \left(2 \text{Li}_2((1 - \sqrt{2})(2 - \sqrt{5})) - 2 \text{Li}_2((1 - \sqrt{2})(\sqrt{5} - 2)) \right. \\ \left. + \text{Li}_2(3 - \sqrt{8}) - \text{Li}_2(\sqrt{8} - 3) + \text{Li}_2(9 - \sqrt{80}) - \text{Li}_2(\sqrt{80} - 9) \right), \quad (3.8)$$

while Sangchul Lee gave the Legendre-chi form

$$I = \pi \left(\chi_2(3 - 2\sqrt{2}) + \chi_2(9 - 4\sqrt{5}) + 2\chi_2((\sqrt{2} - 1)(\sqrt{5} - 2)) \right), \quad (3.9)$$

where $\chi_2(x) = \frac{1}{2}(\text{Li}_2(x) - \text{Li}_2(-x)) = \sum_{k \geq 0} \frac{x^{2k+1}}{(2k+1)^2}$. These are identical: $\text{Li}_2(x) - \text{Li}_2(-x) = 2\chi_2(x)$ turns each Li_2 pair in (3.8) into a χ_2 , and $3 - \sqrt{8} = (\sqrt{2} - 1)^2$, $9 - \sqrt{80} = (\sqrt{5} - 2)^2$, $(1 - \sqrt{2})(2 - \sqrt{5}) = (\sqrt{2} - 1)(\sqrt{5} - 2)$, so both equal

$$I = \pi \left[\chi_2((\sqrt{2} - 1)^2) + \chi_2((\sqrt{5} - 2)^2) + 2\chi_2((\sqrt{2} - 1)(\sqrt{5} - 2)) \right]. \quad (3.10)$$

The arguments are a silver $\otimes$ golden pair: $\sqrt{2} - 1 = \tan \frac{\pi}{8} = \delta_S^{-1}$ (reciprocal silver ratio $\delta_S = 1 + \sqrt{2}$) and $\sqrt{5} - 2 = \phi^{-3}$ (golden $\phi = \frac{1+\sqrt{5}}{2}$, $\phi^3 = \sqrt{5} + 2$), so $I = \pi[\chi_2(\delta_S^{-2}) + \chi_2(\phi^{-6}) + 2\chi_2(\delta_S^{-1}\phi^{-3})]$.

3.6 The master formula (Reshetnikov–Lee) and a differentiation proof

The closed forms (3.8)–(3.9) are instances of a two-parameter master formula. Its seed is Reshetnikov’s single-arctangent ladder $\int_0^{\pi/2} \arctan(t \sin x) dx = 2\chi_2(u_t)$; inspired by it, Lee (a/562492, 2013) conjectured and proved

Theorem 3.3 (Lee 2013). *For real p, q ,*

$$J(p, q) := \int_0^{\pi/2} \arctan(p \sin x) \arctan(q \sin x) dx = \pi \chi_2(u_p u_q), \quad (3.11)$$

$$\text{where } u_p = \frac{\sqrt{1+p^2} - 1}{p} = \tan\left(\frac{1}{2} \arctan p\right).$$

Lee gives two proofs (a triple integral and a Fourier-orthogonality argument). We record a short third proof by differentiation, which is the most direct route and is machine-verifiable.

Proof. Both sides vanish on the axes, and $\partial_p J(p, 0) = \partial_p [\pi \chi_2(u_p u_0)] = 0$ (since $u_0 = 0$), symmetrically in q ; hence each side equals the double integral of its mixed partial over $[0, p] \times [0, q]$, and it suffices to match mixed partials. Differentiating under the integral and using the partial fraction $\frac{s^2}{(1+p^2 s^2)(1+q^2 s^2)} = \frac{1}{p^2 - q^2} \left(\frac{1}{1+q^2 s^2} - \frac{1}{1+p^2 s^2} \right)$ together with $\int_0^{\pi/2} \frac{dx}{1+c \sin^2 x} = \frac{\pi}{2\sqrt{1+c}}$,

$$\partial_p \partial_q J = \frac{\pi}{2(p^2 - q^2)} \left(\frac{1}{\sqrt{1+q^2}} - \frac{1}{\sqrt{1+p^2}} \right). \quad (3.12)$$

On the right, $\chi_2'(w) = \text{arctanh}(w)/w$ gives $\partial_p \partial_q [\pi \chi_2(u_p u_q)] = \pi u_p' u_q' / (1 - u_p^2 u_q^2)$. The substitution $p = \frac{2u_p}{1-u_p^2}$ yields $\sqrt{1+p^2} = \frac{1+u_p^2}{1-u_p^2}$ and $u_p' = \frac{(1-u_p^2)^2}{2(1+u_p^2)}$, whence both sides reduce to the same rational function $\frac{\pi(1-u_p^2)^2(1-u_q^2)^2}{4(1+u_p^2)(1+u_q^2)(1-u_p^2 u_q^2)}$. (Wolfram confirms this reconciliation symbolically: `FullSimplify` of the difference returns 0.) $\square$

3.7 Cleo’s integral as the perfect-square instance

The argument in (3.7) collapses to a sum of arctangents. With $\cos 2x = 1 - 2 \sin^2 x$,

$$\frac{6 \sin x}{3 + \cos 2x} = \frac{3 \sin x}{2 - \sin^2 x} = \frac{(1 + \frac{1}{2}) \sin x}{1 - (1 \cdot \frac{1}{2}) \sin^2 x},$$

the tangent-addition shape with roots $1, \frac{1}{2}$ of $t^2 - \frac{3}{2}t + \frac{1}{2}$, so for $0 \leq x \leq \frac{\pi}{2}$

$$\arctan\left(\frac{6 \sin x}{3 + \cos 2x}\right) = \arctan(\sin x) + \arctan\left(\frac{1}{2} \sin x\right), \quad (3.13)$$

so that $I = \int_0^{\pi/2} (\arctan(\sin x) + \arctan(\frac{1}{2} \sin x))^2 dx$. Applying Theorem 3.3 to the expanded square, with $u_1 = \sqrt{2} - 1$, $u_{1/2} = \sqrt{5} - 2$, reproduces (3.10). The coefficient pattern 1, 2, 1 is a perfect square: writing $a = \sqrt{2} - 1$, $b = \sqrt{5} - 2$,

$$\frac{I}{\pi} = \sum_{k \geq 0} \frac{((\sqrt{2} - 1)^{2k+1} + (\sqrt{5} - 2)^{2k+1})^2}{(2k+1)^2}. \quad (3.14)$$

The coefficients $a^{2k+1} + b^{2k+1}$ are the sin-power Fourier coefficients of $\arctan(\sin x) + \arctan(\frac{1}{2} \sin x)$; (3.14) is the Parseval form underlying Lee's Fourier proof.

3.8 Closed values and a bounded experimental survey

Three classical values supply useful exact specializations. With $\phi = (1 + \sqrt{5})/2$,

$$\begin{aligned} \chi_2(\phi^{-1}) &= \frac{\pi^2}{12} - \frac{3}{4} \log^2 \phi, \\ \chi_2(\phi^{-3}) &= \frac{\pi^2}{24} - \frac{3}{4} \log^2 \phi, \\ \chi_2(\sqrt{2} - 1) &= \frac{\pi^2}{16} - \frac{1}{4} \log^2(1 + \sqrt{2}). \end{aligned} \quad (3.15)$$

The source surveyed inverse metallic means m_n^{-k} with $1 \leq n, k \leq 6$, selected rational points, and $\tan(\pi/k)$ for $k = 5, 8, 10, 12$ and their squares. Searches at 130–140 digits found no further accepted reduction in their selected logarithmic and π^2 baskets. This does not classify all elementary values. Nor would independence of the individual terms alone prove that their sum cannot simplify.

3.9 New closed-form integrals from elementary χ_2

Conversely, (3.15) feeds the master formula to produce integrals that *do* close. The cleanest cross integral uses $u_2 = \phi^{-1}$, $u_{2/\sqrt{5}} = \phi^{-2}$, so $u_2 u_{2/\sqrt{5}} = \phi^{-3}$:

$$\boxed{\int_0^{\pi/2} \arctan(2 \sin x) \arctan\left(\frac{2}{\sqrt{5}} \sin x\right) dx = \frac{\pi^3}{24} - \frac{3\pi}{4} \ln^2 \phi} \quad (= 0.7463164407 \dots). \quad (3.16)$$

The diagonal $\int_0^{\pi/2} \arctan^2(\lambda \sin x) dx = \pi \chi_2(u_\lambda^2)$ closes at three clean λ , since $u_{\sqrt{\phi}}^2 = \phi^{-3}$, $u_{\sqrt{2+2\sqrt{2}}}^2 = \sqrt{2} - 1$, and $u_{2\phi^{3/2}}^2 = \phi^{-1}$:

$$\int_0^{\pi/2} \arctan^2(\sqrt{\phi} \sin x) dx = \frac{\pi^3}{24} - \frac{3\pi}{4} \ln^2 \phi, \quad \int_0^{\pi/2} \arctan^2(2\phi^{3/2} \sin x) dx = \frac{\pi^3}{12} - \frac{3\pi}{4} \ln^2 \phi. \quad (3.17)$$

The simplest non-elementary diagonal is the (1, 1) case $\int_0^{\pi/2} \arctan^2(\sin x) dx = \pi \chi_2(3 - 2\sqrt{2})$.

3.10 A sibling trichotomy

The master formula has two siblings under the companion half-angle substitution $v_p = \frac{1-\sqrt{1-p^2}}{p} = \tanh(\frac{1}{2} \operatorname{arctanh} p)$ (for $|p| < 1$). Replacing one or both arctangents by $\operatorname{arctanh}$ produces:

$$\int_0^{\pi/2} \begin{cases} \operatorname{arctan}(p \sin x) \operatorname{arctan}(q \sin x) \\ \operatorname{arctanh}(p \sin x) \operatorname{arctanh}(q \sin x) \\ \operatorname{arctan}(p \sin x) \operatorname{arctanh}(q \sin x) \end{cases} dx = \begin{cases} \pi \chi_2(u_p u_q), \\ \pi \chi_2(v_p v_q), \\ \pi \operatorname{Ti}_2(u_p v_q), \end{cases} \quad (3.18)$$

where $\operatorname{Ti}_2(z) = \Im \operatorname{Li}_2(iz) = \int_0^z \frac{\operatorname{arctan} t}{t} dt$ is the inverse-tangent integral: like pairs give the Legendre chi, the mixed pair gives the inverse-tangent integral. The linear shadows are $\int_0^{\pi/2} \operatorname{arctan}(\lambda \sin x) dx = 2\chi_2(u_\lambda)$ (Reshetnikov) and, with $\theta = \arcsin \lambda$,

$$\int_0^{\pi/2} \operatorname{arctanh}(\lambda \sin x) dx = \theta \ln \tan \frac{\theta}{2} + 2 \operatorname{Cl}_2(\theta) - \frac{1}{2} \operatorname{Cl}_2(2\theta), \quad (3.19)$$

the latter showing that the $\operatorname{arctanh}$ branch is *not* a clean $2\chi_2(v_\lambda)$ mirror—that coincidence is special to arctan . By $x \mapsto \frac{\pi}{2} - x$ the cos versions of all of these equal the sin versions. The weight-3 analogue $\int_0^{\pi/2} \operatorname{arctan}(p \sin x) \operatorname{arctan}(q \sin x) / \sin x dx$ does not factor by the same mechanism into a single χ_3 : its coefficient $\frac{4^N}{(2N+1)\binom{2N}{N}}$ couples the two summation indices only through $N = m + n$, so it does not factor through $u_p u_q$ (numerically no clean χ_3 relation appears up to height 10^8).

3.11 An alternate form, and minimality

Landen-mirror form. Applying the χ_2 Landen law

$$\chi_2(x) + \chi_2\left(\frac{1-x}{1+x}\right) = \frac{\pi^2}{8} - \frac{1}{2} \ln x \ln \frac{1-x}{1+x} \quad (0 < x < 1) \quad (3.20)$$

to the three arguments of (3.10) sends $(\sqrt{2}-1)^2 \mapsto \frac{1}{\sqrt{2}}$, $(\sqrt{2}-1)(\sqrt{5}-2) \mapsto \sqrt{5}-\sqrt{2}$, $(\sqrt{5}-2)^2 \mapsto \frac{2}{\sqrt{5}}$, giving a second three-term representation

$$\frac{I}{\pi} = \mathcal{E} - \left[\chi_2\left(\frac{1}{\sqrt{2}}\right) + 2 \chi_2(\sqrt{5}-\sqrt{2}) + \chi_2\left(\frac{2}{\sqrt{5}}\right) \right], \quad (3.21)$$

with the elementary constant $\mathcal{E} = \frac{\pi^2}{2} - \frac{1}{2} \ln 2 \ln(1+\sqrt{2}) + 3 \ln 2 \ln \phi - \frac{3}{2} \ln 5 \ln \phi + (\ln(1+\sqrt{2}) + 3 \ln \phi) \ln(\sqrt{5}-\sqrt{2})$. This representation trades the small original arguments for larger ones and explicit logarithms. It follows directly by applying the Landen law three times.

Search for shorter forms. The source tested 209 algebraic arguments in $\mathbb{Q}(\sqrt{2}, \sqrt{5}, \sqrt{10})$ at at least 120 digits, with residual threshold 10^{-105} . It found no accepted one-term or two-term representation in the tested baskets. The three-term formula is an exact evaluation, while its minimality and non-elementarity remain unproved.

3.12 Unit relations and the golden family

With $\phi = \frac{1+\sqrt{5}}{2}$, $\rho = \phi^{-1}$, the defining relation $1 - \rho^2 = \rho$ is the seed of the golden ladders. Reshetnikov's conjectures live among the values $\text{Li}_n(\phi^{-2k})$; all coefficients below were obtained or verified by PSLQ (mpmath, 130–420 digits, Champenowne canary) and, where the cancellation depth allows, cross-checked in Wolfram.

Remark 3.4. These golden combinations cancel $\sim 10^8$ -sized terms down to zero across 100+ digits. The historical Wolfram evaluation under-resolved some cancellations; independently controlled precision in mpmath supplied a comparator. Neither engine is an accuracy oracle. (Per-term polylogarithm values agree between the two engines.)

3.13 Reshetnikov's conjectures, verified

The tetralogarithm conjecture (MSE 1373123, MO 261408) holds: with $\alpha = \ln 2, \beta = \ln 3$,

$$\begin{aligned} & 720 \text{Li}_4\left(\frac{1}{2}\right) - 2160 \text{Li}_4\left(\frac{1}{3}\right) + 2160 \text{Li}_4\left(\frac{2}{3}\right) \\ & \quad + 270 \text{Li}_4\left(\frac{1}{4}\right) + 540 \text{Li}_4\left(\frac{3}{4}\right) + 135 \text{Li}_4\left(\frac{1}{9}\right) \\ & = 19\pi^4 + 30(\pi^2 - \beta^2)(10\alpha^2 - 12\alpha\beta + 3\beta^2) - 30\alpha^2(19\alpha^2 - 24\alpha\beta + 8\beta^2), \end{aligned} \quad (3.22)$$

The source recovered this relation in its weight-four basket; uniqueness of all numerical relations is not established. The three Li_5 golden conjectures also hold (verified to $\sim 10^{-257}$):

$$\begin{aligned} & 4455 \text{Li}_5(\phi^{-2}) - 1215 \text{Li}_5(\phi^{-4}) - 360 \text{Li}_5(\phi^{-6}) + 15 \text{Li}_5(\phi^{-12}) = 3015\zeta(5) - 702 \ln \phi^5 + 180\pi^2 \ln \phi^3 - 38\pi^4 \ln \phi, \\ & 45000 \text{Li}_5(\phi^{-2}) - 16875 \text{Li}_5(\phi^{-4}) - 144 \text{Li}_5(\phi^{-10}) + 9 \text{Li}_5(\phi^{-20}) = 28944\zeta(5) - 6000 \ln \phi^5 + 1600\pi^2 \ln \phi^3 - 356\pi^4 \ln \phi, \\ & 15660 \text{Li}_5(\phi^{-2}) - 19440 \text{Li}_5(\phi^{-4}) + 7680 \text{Li}_5(\phi^{-6}) \\ & \quad + 2430 \text{Li}_5(\phi^{-8}) - 15 \text{Li}_5(\phi^{-24}) = 5025\zeta(5) + 2088 \ln \phi^5 - 240\pi^2 \ln \phi^3 - 28\pi^4 \ln \phi. \end{aligned}$$

3.14 Completing Tito's family: the unwritten weight-6 and weight-7 ladders

Tito's "old answer" lists the golden family at weights 2–5 (all verified here), then states the weight-6 and weight-7 ladders only schematically, the integer coefficients "too tedious to write." We recover them by PSLQ. On the six-argument set $\{\phi^{-2}, \phi^{-4}, \phi^{-6}, \phi^{-8}, \phi^{-12}, \phi^{-24}\}$:

$$\begin{aligned} & 98820 \text{Li}_6(\phi^{-2}) - 104490 \text{Li}_6(\phi^{-4}) + 33120 \text{Li}_6(\phi^{-6}) + 7290 \text{Li}_6(\phi^{-8}) - 50 \text{Li}_6(\phi^{-12}) - 15 \text{Li}_6(\phi^{-24}) \\ & = -13032 \ln \phi^6 + 3240\pi^2 \ln \phi^4 - 424\pi^4 \ln \phi^2 + \mathbf{32155} \zeta(6), \end{aligned} \quad (3.23)$$

$$\begin{aligned} & 74707920 \text{Li}_7(\phi^{-2}) - 39497220 \text{Li}_7(\phi^{-4}) + 8346240 \text{Li}_7(\phi^{-6}) + 1377810 \text{Li}_7(\phi^{-8}) - 6300 \text{Li}_7(\phi^{-12}) - 945 \text{Li}_7(\phi^{-24}) \\ & = 2814912 \ln \phi^7 - 979776\pi^2 \ln \phi^5 + 213696\pi^4 \ln \phi^3 - 51448\pi^6 \ln \phi + \mathbf{44689995} \zeta(7). \end{aligned} \quad (3.24)$$

The $\zeta(6)$ and $\zeta(7)$ coefficients reproduce Tito's stated 32155 and 44689995 *exactly*—an independent confirmation that the reconstructions (3.23)–(3.24) are correct (residuals 10^{-117} , 10^{-114} by direct substitution; coefficients primitive).

3.15 Two weights beyond: explicit ladders at weight 8 and 9

Tito remarks that the index-24 ladder "can be extended even to $n = 9$ " but gives no formula. We find the explicit ladders. The searches on the six-argument set of (3.23)–(3.24) found reductions at

weights 2–7 and none at 8 or 9. The accepted higher-weight formulas use the *eight*-argument union of the index-12, 20, 24 sets, $\mathcal{A} = \{\phi^{-2}, \phi^{-4}, \phi^{-6}, \phi^{-8}, \phi^{-10}, \phi^{-12}, \phi^{-20}, \phi^{-24}\}$:

$$\begin{aligned} & -490795200 \operatorname{Li}_8(\phi^{-2}) + 636131475 \operatorname{Li}_8(\phi^{-4}) - 388684800 \operatorname{Li}_8(\phi^{-6}) + 82668600 \operatorname{Li}_8(\phi^{-8}) \\ & + 9507456 \operatorname{Li}_8(\phi^{-10}) + 5250000 \operatorname{Li}_8(\phi^{-12}) - 74277 \operatorname{Li}_8(\phi^{-20}) - 18900 \operatorname{Li}_8(\phi^{-24}) \\ & = -23641920 \ln \phi^8 + 7956480 \pi^2 \ln \phi^6 - 1434720 \pi^4 \ln \phi^4 + 184432 \pi^6 \ln \phi^2 - 148565886 \zeta(8), \end{aligned} \quad (3.25)$$

$$\begin{aligned} & 220857840000 \operatorname{Li}_9(\phi^{-2}) - 143129581875 \operatorname{Li}_9(\phi^{-4}) + 58302720000 \operatorname{Li}_9(\phi^{-6}) - 9300217500 \operatorname{Li}_9(\phi^{-8}) \\ & - 855671040 \operatorname{Li}_9(\phi^{-10}) - 393750000 \operatorname{Li}_9(\phi^{-12}) + 3342465 \operatorname{Li}_9(\phi^{-20}) + 708750 \operatorname{Li}_9(\phi^{-24}) \\ & = -2364192000 \ln \phi^9 + 1022976000 \pi^2 \ln \phi^7 - 258249600 \pi^4 \ln \phi^5 \\ & + 55329600 \pi^6 \ln \phi^3 - 14149132 \pi^8 \ln \phi + 125596001790 \zeta(9). \end{aligned} \quad (3.26)$$

Both are verified by direct substitution (residuals 0 at 150 digits for (3.25), 10^{-190} at 200 digits for (3.26)); coefficients are primitive. The weight-9 ladder appeared identically across three independent searches with different padding arguments (the extra arguments returned coefficient 0), a strong guard against over-fitting.

Remark 3.5. So the golden family does *not* cap at weight 7: the closure persists, but the argument set grows—four or five arguments per index at weight 5, six arguments at weights 6–7, eight arguments (the full union) at weights 8–9. Whether the growth can be sustained to arbitrarily high weight is exactly the question Abouzahra–Lewin address in the negative; our data are consistent with an eventual stop once the required argument set outgrows the available golden multiplicative relations.

3.16 Reducing the tetralogarithm relation to a functional identity

The tetralogarithm relation (3.22) follows from duplication together with the following cross-base functional identity. No independence claim about the six individual values is needed.

Two duplications $\operatorname{Li}_4(\frac{1}{4}) = 8(\operatorname{Li}_4(\frac{1}{2}) + \operatorname{Li}_4(-\frac{1}{2}))$ and $\operatorname{Li}_4(\frac{1}{9}) = 8(\operatorname{Li}_4(\frac{1}{3}) + \operatorname{Li}_4(-\frac{1}{3}))$ reduce the left side to the reflected arguments, and a single *cross-base Kummer relation* then cancels every tetralogarithm:

$$\begin{aligned} \operatorname{Li}_4(\tfrac{3}{4}) + 4 \operatorname{Li}_4(\tfrac{2}{3}) &= -\tfrac{16}{3} \operatorname{Li}_4(\tfrac{1}{2}) - 4 \operatorname{Li}_4(-\tfrac{1}{2}) + 2 \operatorname{Li}_4(\tfrac{1}{3}) - 2 \operatorname{Li}_4(-\tfrac{1}{3}) + \tfrac{19}{540} \pi^4 + \tfrac{5}{9} \pi^2 \ln^2 2 + \tfrac{1}{6} \pi^2 \ln^2 3 \\ &\quad - \tfrac{2}{3} \pi^2 \ln 2 \ln 3 - \tfrac{19}{18} \ln^4 2 - \tfrac{1}{6} \ln^4 3 + \tfrac{4}{3} \ln^3 2 \ln 3 - \ln^2 2 \ln^2 3 + \tfrac{2}{3} \ln 2 \ln^3 3. \end{aligned} \quad (3.27)$$

Substituting the two duplication identities into (3.27) cancels the tetralogarithms in (3.22). A complete proof from Kummer’s general functional equation requires its branch conditions; the displayed specialized identity was checked numerically in the source and is retained as the explicit intermediate relation.

3.17 Canonical index-20/24 ladders and a ζ -elimination chain

Reshetnikov’s Li_5 conjectures are the weight-5 entries of three first-degree golden ladders (Abouzahra–Lewin). In canonical (monic, a^{n-1} -normalized) form, with $\rho = \phi^{-1}$, $\ell = \ln \rho$, and $\operatorname{tw}(n) = D_1 \frac{\ell^n}{n!} + D_2 \zeta(2) \frac{\ell^{n-2}}{(n-2)!} + D_3 \zeta(4) \frac{\ell^{n-4}}{(n-4)!}$:

$$\begin{aligned} L_{20}(n) &= \frac{\operatorname{Li}_n(\rho^{20})}{20^{n-1}} - \frac{\operatorname{Li}_n(\rho^{10})}{10^{n-1}} - 3 \frac{\operatorname{Li}_n(\rho^4)}{4^{n-1}} + \frac{1}{2} \frac{\operatorname{Li}_n(\rho^2)}{2^{n-1}} + \operatorname{tw}(n), \\ L_{24}(n) &= \frac{\operatorname{Li}_n(\rho^{24})}{24^{n-1}} - \frac{4}{3} \frac{\operatorname{Li}_n(\rho^{12})}{12^{n-1}} - 2 \frac{\operatorname{Li}_n(\rho^8)}{8^{n-1}} \\ &\quad + \frac{7}{3} \frac{\operatorname{Li}_n(\rho^4)}{4^{n-1}} - \frac{205}{576} \frac{\operatorname{Li}_n(\rho^2)}{2^{n-1}} + \operatorname{tw}(n). \end{aligned}$$

The coefficient triples are respectively $(-1/2, 1/25, -89/4000)$ and $(179/576, -5/192, 629/41472)$, and

$$L_{20}(5) = \frac{201}{10000}\zeta(5), \quad L_{24}(5) = -\frac{1541}{110592}\zeta(5).$$

each vanishing for $n \leq 4$. The shared factor 67 in $201 = 3 \cdot 67$, $1541 = 23 \cdot 67$ is what lets the three indices jointly eliminate $\zeta(5)$. Combining them removes higher zetas: $M_{20} = L_{20} - \frac{1296}{625}L_{12}$ and $M_{24} = L_{24} + \frac{23}{16}L_{12}$ kill $\zeta(5)$ and reach weight 7, and $T = M_{24} + \frac{409375}{373248}M_{20}$ kills $\zeta(7)$ and reaches weight 9, with $T(9) = \frac{66452911}{41278242816000}\zeta(9) + \frac{3537283}{2063912140800}\zeta(8)\ell - \frac{11527}{2866544640}\zeta(6)\frac{\ell^3}{3!}$ (confirmed by direct substitution and an independent PSLQ at 460 digits). This is the Abouzahra–Lewin route to weight 9, distinct from the single eight-argument ladders (3.25)–(3.26).

3.18 A new ladder base: the minimal Pisot number

Beyond the golden ratio, the reciprocal of the minimal Pisot number $\omega = 0.7548776\dots$ (the real root of $x^3 + x^2 - 1$, the base Abouzahra–Lewin deferred to a never-published sequel) carries its own ladder. Its seeds are pure units— $1 - \omega = \omega^5$, $1 - \omega^2 = \omega^3$, $1 - \omega^3 = \omega^2$, $1 - \omega^5 = \omega$ —and the first-degree ladder climbs to weight 4 (with $\lambda = \ln \omega$):

$$\begin{aligned} \frac{1}{2} \text{Li}_2(\omega^2) - \text{Li}_2(\omega^5) - \lambda^2 &= 0, & -\text{Li}_3(\omega) + \frac{6}{4} \text{Li}_3(\omega^2) + \frac{3}{9} \text{Li}_3(\omega^3) - \text{Li}_3(\omega^5) - \frac{6\lambda^3}{3!} &= 0, \\ -71 \text{Li}_4(\omega) - \frac{120}{8} \text{Li}_4(\omega^2) + \frac{1053}{27} \text{Li}_4(\omega^3) - 17 \text{Li}_4(\omega^5) + \frac{5488}{343} \text{Li}_4(\omega^7) - \text{Li}_4(\omega^{14}) + 51\zeta(4) + \frac{2010\lambda^4}{4!} - \frac{84\zeta(2)\lambda^2}{2!} &= 0. \end{aligned}$$

The weight-four formula uses the extra legs ω^7, ω^{14} . The later Bloch-framework investigation identifies higher ladders for this field; the earlier assertion that the tower stops at weight four is discarded. A fixed argument set can stop yielding detected relations while larger sets continue to produce them.

3.19 Hunting more ladder bases, and a methodological caveat

Pure-unit seeds $1 - \rho^a = \rho^b$ offer a useful way to discover ladders. They do not exhaust general multiplicative relations, and a bounded search over polynomial coefficients does not cover all monic polynomials of a given degree. Root disguises are easy to miss: the root of $x^4 + x^2 - 1$ becomes a golden reciprocal after squaring. The following two cubic and quartic relations illustrate why negative searches require this care.

Two bases the earlier survey had dismissed as “weight-2 only” do, in fact, carry weight-3 ladders. With ρ the root of $x^3 + x - 1$ (the supergolden reciprocal) and of $x^4 + x - 1$ (the reciprocal of the second-smallest Pisot number $x^4 - x^3 - 1$),

$$\begin{aligned} -60 \text{Li}_3(\rho) + 54 \text{Li}_3(\rho^2) - 32 \text{Li}_3(\rho^3) - 3 \text{Li}_3(\rho^6) &= 24 \ln^3 \rho - 4\pi^2 \ln \rho - 36\zeta(3), \\ -12 \text{Li}_3(\rho) + 12 \text{Li}_3(\rho^3) - 12 \text{Li}_3(\rho^4) - 3 \text{Li}_3(\rho^6) &= 22 \ln^3 \rho - 2\pi^2 \ln \rho - 12\zeta(3), \end{aligned}$$

each verified in two engines (and by direct substitution to 400+ digits). The lesson is methodological: **a PSLQ “no relation” is not a proof of non-existence.** The first relation above was *missed* by a pure-tower PSLQ search at 300 digits, yet holds by direct substitution to 400—so the survey’s negative verdicts were false negatives. The reliable test is to take a candidate relation and substitute it at precision far above where it was found. Whether these two bases climb past weight 3 is left open here; the question is delicate precisely because PSLQ negatives cannot settle it.

3.20 The modified polylogarithm

Zagier’s one-valued modified polylogarithm (his eq. (33)) is

$$P_m(x) = \operatorname{Re}_m \left(\sum_{j=0}^{m-1} \frac{2^j B_j}{j!} (\log |x|)^j \operatorname{Li}_{m-j}(x) \right), \quad \operatorname{Re}_m = \begin{cases} \operatorname{Re}, & m \text{ odd}, \\ \operatorname{Im}, & m \text{ even}, \end{cases} \quad (3.28)$$

with B_j the Bernoulli numbers and $\operatorname{Li}_1(x) = -\log(1-x)$. It is continuous on the Riemann sphere away from $\{0, 1, \infty\}$, satisfies $P_m(\bar{x}) = (-1)^{m-1} P_m(x)$, and $P_m(1) = \zeta(m)$ for odd m , 0 for even m ; $P_2(i)$ is Catalan’s constant. We verified our implementation against Zagier’s worked examples $P_5(-\frac{1}{8}) - 126P_5(\frac{1}{2}) - 162P_5(-\frac{1}{2}) = \frac{403}{16}\zeta(5)$ (his eq. (825)) and the long thirteen-term relation (825)/(833), to 60 digits.

3.21 Ladders are Bloch-group elements

The Bloch-group viewpoint records the obstruction to simplifying a dilogarithm combination. At weight two the boundary sends $[x] \mapsto x \wedge (1-x)$; a vanishing boundary makes a combination a candidate for the Bloch group, modulo the five-term relation. Higher weights require further relations and compatibility conditions: a weight-one unit relation alone does not prove that its pseudointegrations are higher Bloch-group elements. Modified polylogarithms remove explicit logarithmic tails and make the following ladder combinations transparent:

$$\text{golden : } 15 P_3(\phi^{-2}) = 12 \zeta(3), \quad (3.29)$$

$$\text{supergolden : } -60 P_3(\rho) + 54 P_3(\rho^2) - 32 P_3(\rho^3) - 3 P_3(\rho^6) = -36 \zeta(3), \quad (3.30)$$

both exact. Compare (3.29) with the “raw” form $15 \operatorname{Li}_3(\phi^{-2}) = 10 \ln^3 \phi - 2\pi^2 \ln \phi + 12\zeta(3)$: the modified polylogarithm is simply the natural coordinate in which a ladder *is* its ζ -value. The higher golden ladders (weights 5 through 9 in the companion report) become, in the same way, the stated $\zeta(5), \zeta(6), \zeta(7), \zeta(9)$ multiples.

3.22 The bases as number fields

base ρ	minimal polynomial of $1/\rho$	field F	signature	disc
golden ϕ^{-1}	$x^2 - x - 1$	$\mathbb{Q}(\sqrt{5})$	(2, 0) totally real	5
minimal Pisot ω	$x^3 - x - 1$	$\mathbb{Q}(\theta)$	(1, 1)	-23
supergolden recip.	$x^3 - x^2 - 1$	$\mathbb{Q}(\sigma)$	(1, 1)	-31

The middle row is Zagier’s own running example (§4–5, §9C): the field $\mathbb{Q}(\theta)$, $\theta^3 - \theta - 1 = 0$, whose ladder he calls “particularly spectacular,” and whose defining unit relations $1 - \theta = -\theta^{-4}$, $1 + \theta = \theta^3, \dots$ are precisely the seeds we rediscovered. Our supergolden base gives a *different* complex cubic field, discriminant -31.

3.23 Regulator parity and the limits of ladder counting

For $m \geq 2$, Borel’s rank theorem gives rank $r_1 + r_2$ for $K_{2m-1}(F)$ when m is odd, and r_2 when m is even. The modified polylogarithm uses its real part at odd weight and imaginary part at even

weight. In particular P_{2m} vanishes on real arguments, even when the number field has nonzero regulator directions at its complex embeddings. This explains why a real-argument search can miss arithmetic content visible at a complex embedding.

Cyclotomic unit searches compute $\text{Norm } \Phi_k(\rho) = \text{Res}(p, \Phi_k)$ and select the unit cases. For a fixed finite seed set, imposing successive regulator constraints is a finite linear-algebra problem. A dimension count can give a lower bound for a kernel or exhaust that chosen construction. It does not prove there are no additional seeds, no dependencies among constraints, and no other functional identities at higher weight.

The later source identifies the minimal-Pisot ladders through weight six and points to Cohen–Lewin–Zagier’s published numerical sixteenth-order Lehmer ladder [36]. These supersede earlier negative searches. Its stronger assertion that the supergolden field has no weight-five real ladder is not justified by exhausting its selected seed set, and is omitted. Determining a maximal ladder weight requires a completeness theorem for the chosen class of relations.

3.24 Attributed dilogarithm and inverse-binomial examples

The following examples are attributed to Hakimoglu-Brown [37]. Their displayed numerical checks in the source are historical.

All to ≥ 38 digits (`tools/scratch/dilog-inverse-binomial/`); a representative selection:

- **Sun’s open conjecture** (eq. (24)), proved in the paper via the $\mathbb{Q}(\sqrt{3})$ evaluation (25):

$$\sum_{n \geq 0} \frac{\binom{2n}{n}}{(2n+1)8^n} \left(\sum_{0 \leq k < n} \frac{(-1)^k}{2k+1} - \frac{(-1)^n}{2n+1} \right) = -\frac{\sqrt{2}}{16} \pi^2.$$

- **Theorem 2** ($\mathbb{Q}(\sqrt{3})$): $8 \text{Li}_2(2 - \sqrt{3}) - \text{Li}_2(4\sqrt{3} - 7) = \frac{5\pi^2}{12} + \ln(2 - \sqrt{3}) \ln(2 + \sqrt{3})$, and $3 \text{Li}_2(7 - 4\sqrt{3}) = 6 \text{Li}_2(2 - \sqrt{3}) + 6 \text{Li}_2(\sqrt{3} - 2)$.
- **Lima’s identity** ($\mathbb{Q}(\sqrt{2})$, eq. (26)): $\text{Li}_2(\frac{1}{\sqrt{2}}) - \text{Li}_2(1 - \sqrt{2}) = \frac{\pi^2}{6} - \frac{\ln^2 2}{8} + \frac{1}{2} \ln 2 \ln(\sqrt{2} - 1)$.
- **Zagier’s four-term** (eq. (27)): $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{\pi^2}{12}$, with the (unnormalized) Rogers $L(x) = \text{Li}_2(x) + \frac{1}{2} \ln |x| \ln |1 - x|$; and its compressed three-term form.
- the rational-argument $\frac{\pi^2}{18} + 2 \ln 2 \ln 3 - 2 \ln^2 2 - \frac{2}{3} \ln^2 3 = \text{Li}_2(\frac{1}{4}) + \frac{1}{3} \text{Li}_2(\frac{1}{9})$, the inverse-tangent/ $\text{Cl}_2(\frac{\pi}{3})$ relation (30), and $\text{Li}_2(\frac{4}{3}) + \text{Li}_2(4) = \frac{1}{2}(\pi^2 - \ln^2 3)$ (real parts).
- the **engine**: ${}_4F_3(\{1, 1, 1, \frac{3}{2}\}; \{\frac{5}{2}, \frac{5}{2}, \frac{5}{2}\}; 1) = \frac{27}{2}(7\zeta(3) + (3 - 2G)\pi - 12)$ (D’Aurizio) and the cube-binomial ${}_4F_3$ of eq. (1).
- the paper’s **claimed-original elliptic** ${}_4F_3(1)$ **evaluations** (eqs. 32, 33, 35):

$$\begin{aligned} \text{Cl}_2(\frac{\pi}{3}) &= \frac{\sqrt{3}}{2} {}_4F_3(\{1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\}; \{\frac{3}{2}, \frac{5}{6}, \frac{7}{6}\}; 1), \\ {}_4F_3(\{1, \frac{1}{2}, \frac{2}{3}, \frac{1}{3}\}; \{\frac{3}{2}, \frac{5}{4}, \frac{3}{4}\}; 1) &= \frac{3\sqrt{3}}{4} (2 \ln(4 + 3\sqrt{2}) + \ln(\frac{7}{2} - 2\sqrt{3})), \\ \ln^2\left(\frac{5+\sqrt{13}}{2\sqrt{3}}\right) - \frac{\pi^2}{12} &= \frac{1}{180} {}_4F_3(\{1, 1, 1, \frac{3}{2}\}; \{\frac{7}{6}, \frac{11}{6}, 2\}; -\frac{1}{324}). \end{aligned}$$

All check out. One caution worth recording: equation (27) uses the *unnormalized* Rogers L (so the right side is $-\frac{\pi^2}{12}$, not a pure rational); with the $\frac{6}{\pi^2}$ -normalized L the identity reads $3L(-\frac{1}{6}) - L(\frac{1}{8}) + L(\frac{1}{9}) + L(\frac{1}{28}) = -\frac{1}{2}$.

Connection. The paper evaluates inverse-binomial series with $\binom{2n}{n}^2$ and $\binom{3n}{n}$ via beta integrals; this is the reciprocal-binomial sibling of the central-binomial ladder $\sum \binom{2n}{n} 4^{-n} (2n+1)^{-p}$ studied in `harvested-central-binomial-arcsin.wl`. The verified identities are stored (with attribution) in `src/PolyLog/identities/harvested-dilogarithm-ladders-2506.10206.wl`.

Chapter 4

Gaussian and Eisenstein depth

The classical transformations suggest a sharper question: how much nesting is necessary after known identities have been used? We first prove an infinite family of reductions. The other families illustrate how a successful experimental identity gives an upper bound on depth, while a failed search leaves open both a large-coefficient reduction and a missing argument family.

4.1 Depth conventions and Gaussian generators

Write $G = \beta(2)$, $\beta(s) = \sum_{n \geq 0} (-1)^n / (2n+1)^s$, and

$$g_{a,b} = \Im \operatorname{Li}_{a,b}(i, 1), \quad h_{a,b} = \Re \operatorname{Li}_{a,b}(i, 1), \quad \lambda_n = \Im \operatorname{Li}_n((1+i)/2).$$

Subscripts such as g_{41} mean $g_{4,1}$. Series depth counts nested indices. In the experimental tables, a “depth-one reduction” means a polynomial in the stated single-value basket, which is a different convention from the additive depth filtration on a motivic algebra. Products of two depth-two factors have additive depth at most four; their membership in an algebra generated by doubles does not make them depth at most two in that additive filtration.

4.2 An infinite even-weight reduction

Define

$$S_p = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^p}, \quad p \geq 1.$$

For $p = 1$ the sum is conditionally convergent; for $p > 1$ it is absolute.

Theorem 4.1 (Odd-denominator harmonic sums at even weight). *For every integer $m \geq 0$,*

$$S_{2m+1} = (2m+1)\beta(2m+2) - [t^{2m}] \frac{\pi}{4 \cos(\pi t/2)} \left(2 \log 2 + \sum_{j \geq 1} (2 - 2^{-2j}) \zeta(2j+1) t^{2j} \right). \quad (4.1)$$

In particular, each sum of even weight $2m+2$ belongs to the algebra generated by $\pi, \log 2$, odd zeta values and even beta values.

Proof. Put

$$I(t) = \int_0^1 \frac{x^t \log(1+x^2)}{1+x^2} dx, \quad J(t) = \int_0^1 \frac{x^t}{1+x^2} dx, \quad A(t) = \int_0^\infty \frac{x^t \log(1+x^2)}{1+x^2} dx.$$

These are analytic near $t = 0$. Splitting A at one and substituting $x = 1/u$ in the second part gives $A(t) = I(t) + I(-t) - 2J'(-t)$. The generating function $\sum H_n z^n = -\log(1-z)/(1-z)$, used first with an Abel factor and then by dominated convergence, gives $I^{(2m)}(0) = -(2m)!S_{2m+1}$. Also $J^{(2m+1)}(0) = -(2m+1)!\beta(2m+2)$, by the geometric series for $1/(1+x^2)$; its integrated absolute majorant is summable. Consequently

$$S_{2m+1} = (2m+1)\beta(2m+2) - \frac{[t^{2m}]A(t)}{2}.$$

Differentiating the beta integral in its second parameter yields

$$A(t) = \frac{\pi}{2\cos(\pi t/2)} \left[\psi(1) - \psi\left(\frac{1-t}{2}\right) \right], \quad |\Re t| < 1.$$

Its even part follows from $\psi(1) - \psi(1/2) = 2\log 2$ and $\psi^{(2j)}(1/2) = -(2j)!(2^{2j+1} - 1)\zeta(2j+1)$. Extracting the coefficient proves the formula. $\square$

The first cases are

$$S_1 = G - \frac{\pi}{2} \log 2, \tag{4.2}$$

$$S_3 = 3\beta(4) - \frac{\pi^3}{16} \log 2 - \frac{7\pi}{16} \zeta(3), \tag{4.3}$$

$$768S_5 = 3840\beta(6) - 5\pi^5 \log 2 - 42\pi^3 \zeta(3) - 372\pi \zeta(5), \tag{4.4}$$

$$92160S_7 = 645120\beta(8) - 61\pi^7 \log 2 - 525\pi^5 \zeta(3) - 5580\pi^3 \zeta(5) - 45720\pi \zeta(7). \tag{4.5}$$

The source's high-precision experiments suggested the last two identities; the generating integral proves them and every higher odd-index case at once. The proof gives a sufficient reduction. It does not prove that the even beta values are independent, nor that the odd-weight sums have minimal depth two. No such conclusion follows merely from the absence of a new odd beta value.

4.3 Candidate Gaussian reductions at weights five and six

The following equations collect the source's accepted numerical relations. They include the corrected dependence among the four weight-five generators, the five weight-six double reductions, and the triple reductions through weight five. Historical residuals range from about 10^{-193} to 10^{-690} . The source scripts for every such relation are not present in this checkout, so these remain numerically supported identities unless an analytic derivation is explicitly supplied. In particular the weight-five relation gives a spanning set of at most three directions, rather than an independence theorem.

$$S_4 = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n+1)^4} = \frac{4g_{41} - 3g_{32} - 9g_{23}}{7} + \frac{\pi^5}{224} - \frac{27}{224} G \zeta(3) - 2\beta(4) \log 2, \tag{4.6}$$

$$576g_{41} + 288g_{32} + 736g_{23} + 960g_{14} = 21G\zeta(3) - 480\beta(4)\log 2, \tag{4.7}$$

$$2048g_{51} = -64\pi^3 \zeta(3) - 527\pi \zeta(5) + 4096\beta(6), \tag{4.8}$$

$$1536g_{42} = 96\pi^3 \zeta(3) - 32\pi^2 \beta(4) + 1581\pi \zeta(5) - 8448\beta(6), \tag{4.9}$$

$$1024g_{33} = -3\pi^3 \zeta(3) + 64\pi^2 \beta(4) - 1581\pi \zeta(5) + 4608\beta(6), \tag{4.10}$$

$$23040g_{24} = -14\pi^4 G + 135\pi^3 \zeta(3) - 1440\pi^2 \beta(4) + 23715\pi \zeta(5) - 69120\beta(6), \tag{4.11}$$

$$92160g_{15} = -150\pi^5 \log 2 + 56\pi^4 G - 270\pi^3 \zeta(3) + 1920\pi^2 \beta(4) - 675\pi \zeta(5). \tag{4.12}$$

$$\Im \operatorname{Li}_{2,1,1}(i, 1, 1) = \lambda_4 + \frac{1}{2} \lambda_3 \log 2 - \frac{G(\pi^2 - 4 \log^2 2)}{32} - \frac{\pi(8 \log^3 2 + 105 \zeta(3))}{768}, \quad (4.13)$$

$$\Im \operatorname{Li}_{1,2,1}(i, 1, 1) = -3\lambda_4 - \lambda_3 \log 2 + \frac{G(\pi^2 - 4 \log^2 2)}{32} - \frac{3\pi^3 \log 2}{256} + \frac{\pi(2 \log^3 2 + 67 \zeta(3))}{128}, \quad (4.14)$$

$$\Im \operatorname{Li}_{1,1,2}(i, 1, 1) = 3\lambda_4 + \frac{1}{2} \lambda_3 \log 2 + \frac{5\pi^3 \log 2}{192} - \frac{163 \pi \zeta(3)}{256}. \quad (4.15)$$

4.4 The colored stuffle identity

Whenever the defining sums are absolutely convergent, sorting the two indices in a product proves

$$\operatorname{Li}_a(x) \operatorname{Li}_b(y) = \operatorname{Li}_{a,b}(x, y) + \operatorname{Li}_{b,a}(y, x) + \operatorname{Li}_{a+b}(xy). \quad (4.16)$$

At conditionally convergent boundary points, Abel limits give the same identity when the limiting products and constituent series exist.

4.5 Gaussian doubles: closed forms

Write $\operatorname{Li}_{a,b}$ at the three Gaussian base points beyond $(i, 1)$:

$$(i, i), \quad (-1, i), \quad (i, -1),$$

all generated to 700 digits by an accelerated evaluator (Section 10) and agreeing with Wolfram 15 to full precision. Under the level-4 complex conjugation $i \mapsto -i$ the imaginary parts are odd and the real parts even.

4.5.1 Family 1: the stuffle closed forms

Specializing (4.16) at the Gaussian points ($uv = -1$ at (i, i) ; $uv = -i$ at the mixed pair) gives a complete set of exact closed forms for the symmetric combinations:

$$\operatorname{Li}_{a,b}(i, i) + \operatorname{Li}_{b,a}(i, i) = \operatorname{Li}_a(i) \operatorname{Li}_b(i) - \operatorname{Li}_{a+b}(-1), \quad (4.17)$$

$$\operatorname{Li}_{a,b}(-1, i) + \operatorname{Li}_{b,a}(i, -1) = \operatorname{Li}_a(-1) \operatorname{Li}_b(i) - \operatorname{Li}_{a+b}(-i), \quad (4.18)$$

$$\operatorname{Li}_{a,b}(i, -1) + \operatorname{Li}_{b,a}(-1, i) = \operatorname{Li}_a(i) \operatorname{Li}_b(-1) - \operatorname{Li}_{a+b}(-i),$$

with the clean diagonal corollary

$$\operatorname{Li}_{a,a}(i, i) = \frac{1}{2} (\operatorname{Li}_a(i)^2 - \operatorname{Li}_{2a}(-1)). \quad (4.19)$$

Every single polylogarithm on the right is elementary in the level-4 vocabulary: $\operatorname{Im} \operatorname{Li}_n(i) = \beta(n)$ is the Dirichlet beta value, $\operatorname{Re} \operatorname{Li}_n(i)$ and $\operatorname{Li}_n(-1) = -\eta(n)$ are rational multiples of $\zeta(n)$ (or π^n at even n), and $\operatorname{Li}_n(-i) = \overline{\operatorname{Li}_n(i)}$. So Family 1 reduces every symmetric and diagonal Gaussian double to single L -values in closed form.

Remark 4.2. The three stuffle identities (4.17)–(4.18) were checked against the stored 700-digit values to $\sim 10^{-692}$ for every admissible a, b , and the closed forms were confirmed symbolically and numerically by Wolfram 15 at low weight, including the weight-6 diagonal $\operatorname{Li}_{3,3}(i, i) = \frac{1}{2} (\operatorname{Li}_3(i)^2 - \operatorname{Li}_6(-1))$, for which the kernel returns a form in $\beta(4)$, π^4 , $\zeta(3)$.

The consequence is structural: all depth-2 content of the Gaussian doubles lives in the *antisymmetric* parts $\operatorname{Li}_{a,b}(P) - \operatorname{Li}_{b,a}(P)$, which vanish on the diagonal. These are what the double shuffle of the next section reduces.

4.5.2 Family 2: antisymmetric reductions to the $(i, 1)$ algebra

Against the basis of single-polylog monomials together with the $(i, 1)$ generators $g_{a,b} = \text{Im Li}_{a,b}(i, 1)$, $h_{a,b} = \text{Re Li}_{a,b}(i, 1)$, the *low-weight* antisymmetric parts reduce, each reduction a new closed-form identity found by integer-relation search and confirmed in Wolfram 15. A representative instance, with $\mu_n = \text{Re Li}_n(\frac{1+i}{2})$ and $\lambda_n = \text{Im Li}_n(\frac{1+i}{2})$:

$$\text{Re}[\text{Li}_{1,2}(i, i) - \text{Li}_{2,1}(i, i)] = 3\zeta(3) - 4\log 2 \mu_2 - \pi \lambda_2 - 4h_{1,1} \log 2 + 3h_{1,2}. \quad (4.20)$$

Such reductions exist at weights 2–4 at the mixed points and at weight 3 at (i, i) ; at higher weight the antisymmetric parts do not reduce *against the $(i, 1)$ basis alone*. To go further—and to learn how many are genuinely new—one needs the shuffle.

4.6 Gaussian doubles: the double-shuffle dimension

4.6.1 A cautionary integer-relation count

A first attempt to count the genuinely-new depth-2 directions proceeds by integer-relation search: test each antisymmetric part for membership in the span of the $(i, 1)$ generators and single-polylog products, and tabulate those that fail. Carried out at 700 digits, this reports **30** independent new directions through weight 6. The number is wrong. It is an artifact of the spurious scale: the genuine reductions are of large height, and against a basis of a hundred-odd atoms the search cannot see them. (Three searches against bases of 46, 55, and the $(i, 1)$ -plus-single set returned “dimensions” of 3, 6, and 30 respectively—all different, all wrong, the count *decreasing* as the basis grew and the ceiling fell.)

4.6.2 The shuffle, and the correct count

The shuffle settles it. The key relation is that the shuffle of a (-1) -polylogarithm with an i -polylogarithm produces (i, i) and $(-1, i)$ doubles, for instance

$$\text{Li}_2(-1) \text{Li}_2(i) = \text{Li}_{2,2}(i, i) + 2 \text{Li}_{3,1}(i, i) + \overline{\text{Li}_{2,2}(-1, i)} + 2 \overline{\text{Li}_{3,1}(-1, i)} \quad (4.21)$$

(verified to $\sim 10^{-694}$), whereas the *stuffle* of the same product produces the $(-1, i)$, $(i, -1)$ doubles. Equating ties all three Gaussian points together. Collecting all 240 stuffle and shuffle relations from the products $\text{Li}_p(s) \text{Li}_q(t)$, $s, t \in \{i, -1, -i\}$, $2 \leq p + q \leq 6$, verifying each to $\sim 10^{-600}$, and computing the exact rational rank of the linear system on the doubles, modulo single polylogarithms and the $(i, 1)$ generators, gives:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
4	18	8	1
5	24	12	2
6	30	14	2
total		34	5

Observation 4.3. Modulo single polylogarithms and the $(i, 1)$ generators, the Gaussian double points (i, i) , $(-1, i)$, $(i, -1)$ contribute exactly **5** candidate depth-two generators through weight 6—one at weight 4, two at weight 5, two at weight 6. A representative basis is

$$\text{Re Li}_{3,1}(i, -1); \quad \text{Im Li}_{3,2}(i, -1), \text{Im Li}_{4,1}(i, -1); \quad \text{Re Li}_{4,2}(i, -1), \text{Re Li}_{5,1}(i, -1).$$

Every other Gaussian double of weight ≤ 6 is an explicit combination of these five, the $(i, 1)$ generators, and products of single L -values.

The rank is verified two ways: each relation reproduces its value to 10^{-600} , and the exact-rational rank is precision-independent. As a final gold-check at weight 4, fixing the single survivor $\text{Re Li}_{3,1}(i, -1)$ to its true value and solving the system predicts all other 17 weight-4 components to $\sim 10^{-692}$.

Remark 4.4 (the methodological moral). Observation 4.3 corrects the integer-relation count of 30 by a factor of six. The reductions are exact consequences of the small-height double-shuffle generators, but the *net* reductions are of large height—so high that an integer-relation search against any adequate basis cannot find them. This is the recurring trap of high-precision experimental mathematics, here biting in the direction of *over*-stating independence. The derived double shuffle, with its exact rank, is immune.

4.7 Triple sums and the scope of formal quotient dimensions

The word for $\text{Li}_{a,b,c}(i, 1, 1)$ is $0^{a-1}(-i)0^{b-1}(-i)0^{c-1}(-i)$ in the outer-letter-first convention. Shuffle relations express products of lower-weight functions as sums of such triples. At weight six the source reports rank seven for thirteen relations on ten triple coordinates, leaving three formal coordinates represented by $(3, 1, 2)$, $(3, 2, 1)$, $(4, 1, 1)$. Rank seven means that seven coordinates can be eliminated using the chosen relations and the three remaining coordinates. It does not mean that seven particular triples are individually products, or that the remaining three numerical values are independent.

The source also claimed binomial dimensions $\binom{n}{p}$ for the actual depth filtration from a total Hilbert series $1/(1-2t)$. A total Hilbert series alone cannot determine a depth grading. A grading by the length of a chosen motivic word basis must first be identified with the filtration under discussion; moreover a motivic lower bound does not give a numerical lower bound without injectivity of the period map. We therefore make no unconditional claim that genuine numerical depth three first occurs at weight six. A rigorous reduction or obstruction for these specific triples is an open problem. Panzer’s parity theorem [33] supplies exact lower-depth conjugation combinations, but does not assert independence of the complementary components.

4.8 The Eisenstein analogue

Everything above has a level-3 counterpart at the cube roots of unity. Write $\rho = e^{2\pi i/3}$, so $\rho^2 = \bar{\rho}$.

4.8.1 Family 1, with an arithmetic twist

The stuffle closed forms read

$$\text{Li}_{a,b}(\rho, \rho) + \text{Li}_{b,a}(\rho, \rho) = \text{Li}_a(\rho) \text{Li}_b(\rho) - \text{Li}_{a+b}(\rho^2), \quad (4.22)$$

$$\text{Li}_{a,b}(\rho^2, \rho) + \text{Li}_{b,a}(\rho, \rho^2) = \text{Li}_a(\rho^2) \text{Li}_b(\rho) - \zeta(a+b), \quad (4.23)$$

with the diagonal corollary $\text{Li}_{a,a}(\rho, \rho) = \frac{1}{2}(\text{Li}_a(\rho)^2 - \text{Li}_{2a}(\rho^2))$. The arithmetic twist is in (4.23): because $\rho^2 \cdot \rho = \rho^3 = 1$, the colored tail degenerates to the *ordinary* zeta value $\zeta(a+b)$, not a level-3 value. The single cube-root L -values at $n \geq 2$ split cleanly under the distribution relation $\text{Li}_n(\rho) + \text{Li}_n(\rho^2) = (3^{1-n} - 1)\zeta(n)$:

$$\text{Re Li}_n(\rho) = \frac{1}{2}(3^{1-n} - 1)\zeta(n), \quad \text{Im Li}_n(\rho) = \text{Sl}_n(2\pi/3), \quad (4.24)$$

At weight one, $\text{Li}_1(\rho) = -\frac{1}{2}\log 3 + i\pi/6$. For the displayed higher-weight formula, $\text{Sl}_n(\theta) = \sum_{k \geq 1} \sin(k\theta)/k^n$. The real part is a rational multiple of $\zeta(n)$; the sine series agrees with Cl_n at even weight and is elementary at rational angles at odd weight. Also $\text{Cl}_2(2\pi/3) = 2V/3$, where $V = \text{Cl}_2(\pi/3)$ is the Gieseking constant. All of (4.22)–(4.24) are confirmed in Wolfram 15.

4.8.2 The (ρ, ρ) frontier: one constant

Running the level-3 double shuffle as in Section 4.6—90 stuffle and shuffle relations from products $\text{Li}_p(s) \text{Li}_q(t)$, $s, t \in \{\rho, \rho^2\}$, each verified to $\sim 10^{-600}$, exact rational rank—collapses the (ρ, ρ) doubles still more sharply than the Gaussian:

weight	unknowns (Re+Im)	dim (stuffle only)	dim (double shuffle)
2	2	0	0
3	4	2	0
4	6	2	0
5	8	4	1
6	10	4	0
total		12	1

Observation 4.5. Modulo single polylogarithms and the $(\rho, 1)$ generators, the entire (ρ, ρ) family through weight 6 reduces to a *single* candidate depth-two generator— $\text{Im Li}_{3,2}(\rho, \rho)$, at weight 5—with weights 2, 3, 4 and 6 collapsing completely.

A gold-check at weight 5 fixes the survivor and predicts all seven other weight-5 components to $\sim 10^{-696}$. The pattern mirrors the Gaussian: there too the weight-5 survivors are imaginary $\text{Li}_{3,2}/\text{Li}_{4,1}$ directions.

Remark 4.6 (a genuine cube-root obstacle). The decisive shuffle $\text{Li}_p(\rho^2) \text{Li}_q(\rho) \rightarrow (\rho, \rho) + (\rho^2, \rho^2)$ lands only on points with $w = z_1 z_2 \neq 1$, so the computation never needs the mixed points (ρ^2, ρ) , (ρ, ρ^2) , where $w = \rho^3 = 1$. Those points presented a numerical obstacle in the earlier evaluator: both the Lerch-plus-Wynn evaluator and a direct cumulative-sum evaluator stall there at ~ 8 digits—the period-three, non-oscillating tail defeats Wynn’s ε -algorithm, unlike the even-period $i, -1$ cases, which converge to full precision. Folding the mixed cube-root points in (for the *full* level-3 frontier) requires a period-three-aware accelerated evaluator, or Wolfram; the following mixed-point method addresses that computational gap.

4.9 Why the standard evaluator degenerates at $w = 1$

With the convention the nested-series definition, the accelerated evaluator of the article writes the inner tail as a Lerch transcendent and sums the *outer* series with an acceleration factor $w^n = (z_1 z_2)^n$. At the mixed points $w = 1$, that factor is identically 1: the oscillation Wynn’s ε -algorithm relies on disappears, leaving a bare period-three power-law tail that the algorithm cannot resolve past a handful of digits. Summing over the *larger* index instead keeps an oscillating factor $z_1^{n_1}$, but for $z_1 = \rho^2$ this is a *period-three* oscillation, and a plain ε -algorithm (tuned for geometric tails) again stalls.

4.10 A robust $w = 1$ evaluator

Two ingredients fix it.

(a) **A smooth correction identity.** Writing $l = n_1$ (the larger index) and splitting the inner partial sum $P(l) = \sum_{n < l} z_2^n / n^b = \text{Li}_b(z_2) - R(l)$ with tail $R(l) = \sum_{n \geq l} z_2^n / n^b = z_2^l \Phi(z_2, b, l)$ (Lerch Φ), one gets the exact identity

$$\boxed{\text{Li}_{a,b}(z_1, z_2) = \text{Li}_a(z_1) \text{Li}_b(z_2) - \sum_{l \geq 1} \frac{w^l}{l^a} \Phi(z_2, b, l)} \quad (4.25)$$

At $w = 1$ the correction summand $\Phi(z_2, b, l)/l^a$ is *smooth*—every oscillating factor $z_1^l z_2^l = w^l$ collapses to 1—with the clean asymptotic $\Phi(z_2, b, l) \sim \sum_{j \geq 0} \binom{-b}{j} c_j(z_2) l^{-b-j}$, where $c_0 = \frac{1}{1-z_2}$ and $c_{j \geq 1} = \text{Li}_{-j}(z_2)$ (rational in z_2). The subtle point—the source of a factor-of-one bug that cost five digits in a first draft—is that $c_0 = \sum_{k \geq 0} z_2^k = \frac{1}{1-z_2}$ is *not* $\text{Li}_0(z_2) = \frac{z_2}{1-z_2}$; they differ by exactly 1. This gives an Euler–Maclaurin tail in Hurwitz zetas, $\sum_{l > M} \Phi/l^a = \sum_j \binom{-b}{j} c_j \zeta(a+b+j, M+1)$, a fast and accurate cross-check at low to medium precision.

(b) **Period-aware ρ -acceleration for production.** For high precision the production evaluator sums the outer cumulative series directly, samples the partial sums at the common period $N = \text{lcm}(\text{ord } z_1, \text{ord } z_2)$ (so every oscillating factor realigns to 1 at the sample points), and accelerates the resulting smooth, power-law-convergent sequence with **Wynn’s ρ -algorithm**—the algebraic-convergence counterpart of the ε -algorithm. The ρ -algorithm yields empirical convergence on the tested families; precision inflation and independent comparison are needed to assess its conditioning (`mpl_w1.py`).

4.11 Validation

The evaluator is checked by three independent, Wolfram-free routes plus one Wolfram cross-check.

1. **Exact symmetric closed form** (single polylogs only, an entirely separate code path):

$$\text{Li}_{a,b}(\rho^2, \rho) + \text{Li}_{b,a}(\rho, \rho^2) = \text{Li}_a(\rho^2) \text{Li}_b(\rho) - \zeta(a+b), \quad (4.26)$$

and the Gaussian analogue $\text{Li}_{a,b}(i, -i) + \text{Li}_{b,a}(-i, i) = \text{Li}_a(i) \text{Li}_b(-i) - \zeta(a+b)$. (The collision term is $\text{Li}_{a+b}(z_1 z_2) = \text{Li}_{a+b}(1) = \zeta(a+b)$.)

2. **Conjugation** $\text{Li}_{a,b}(\rho, \rho^2) = \overline{\text{Li}_{a,b}(\rho^2, \rho)}$.
3. **Two independent summation methods**—the ρ -accelerated cumulative sum and the Euler–Maclaurin smooth-correction series—agree to full working precision.
4. **Wolfram 15 MultiplePolyLog** at moderate precision.

All four agree; the symmetric closed form (4.26) holds to the full generated precision ($\sim 10^{-500}$) for every (a, b) .

4.12 Closed forms for the Eisenstein mixed-point values

With 500-digit values in hand, an integer-relation (PSLQ/LLL) search against the level-3 basis—monomials in the single L -values $\{\pi, \log 3, \zeta(3), \zeta(5), \text{Cl}_2(\frac{2\pi}{3}), \text{Cl}_4(\frac{2\pi}{3}), \text{Cl}_6(\frac{2\pi}{3})\}$ times the $(\rho, 1)$ generators $g_{a,b} = \text{Im Li}_{a,b}(\rho, 1)$, $h_{a,b} = \text{Re Li}_{a,b}(\rho, 1)$, and the lone (ρ, ρ) double-shuffle survivor $\text{Im Li}_{3,2}(\rho, \rho)$ —**reduces every mixed value through weight 4, and every weight-5 imaginary**

part, to explicit closed forms (each canary-clean, height ≤ 10 , re-verified to 10^{-425} against the *complete* weight-graded basis). The two simplest:

$$\operatorname{Im} \operatorname{Li}_{1,1}(\rho^2, \rho) = -\left(\operatorname{Cl}_2\left(\frac{2\pi}{3}\right) + \operatorname{Im} \operatorname{Li}_{1,1}(\rho, 1)\right), \quad \operatorname{Re} \operatorname{Li}_{1,1}(\rho^2, \rho) = 5 \operatorname{Re} \operatorname{Li}_{1,1}(\rho, 1) - \frac{1}{2} \log^2 3. \quad (4.27)$$

(The full list of ~ 16 reductions is in `harvest_eis_mixed.py`.) The weight-5 *real* parts and all of weight 6 do *not* reduce to this basis at height $\lesssim 10^4$.

4.13 The completed level-3 double-shuffle dimension

Two computations pin the dimension of the full level-3 depth-2 space—all four colored points $(\rho, \rho), (\rho^2, \rho^2), (\rho^2, \rho), (\rho, \rho^2)$ —modulo single polylogarithms and the $(\rho, 1)$ generators.

- The **exact-rank convergent double shuffle** (stuffle and shuffle of products $\operatorname{Li}_p(s) \operatorname{Li}_q(t)$, $s, t \in \{\rho, \rho^2\}$, together with the convergent zeta-products $\operatorname{Li}_p(s) \zeta(q)$ whose shuffle reaches the mixed points) gives an *upper bound* of **7** through weight 6 (the divergent Li_1 words, needing shuffle regularization, are omitted).
- **Combining** that exact rank with the PSLQ closed forms of Section 4.12 collapses it:

weight	reported residual count	survivors
2	0	—
3	0	—
4	0	—
5	2	$\operatorname{Im} \operatorname{Li}_{3,2}(\rho, \rho)$ (<i>article</i>), $\operatorname{Re} \operatorname{Li}_{4,1}(\rho^2, \rho)$ (new)
6	1	$\operatorname{Im} \operatorname{Li}_{5,1}(\rho^2, \rho)$ (new)

So the tested level-three quotient, after the numerical reductions, has reported residual count **3** through weight 6. Everything through weight 4 collapses entirely into the $(\rho, 1)$ -generator world; the mixed cube-root points contribute **exactly two new constants** beyond the article’s single (ρ, ρ) survivor—a *real* direction $\operatorname{Re} \operatorname{Li}_{4,1}(\rho^2, \rho)$ at weight 5 and an *imaginary* direction $\operatorname{Im} \operatorname{Li}_{5,1}(\rho^2, \rho)$ at weight 6.

Remark 4.7 (the article’s own lesson). Non-reduction by integer-relation search is strong evidence, not proof: a genuine reduction of large height is invisible to PSLQ, and the convergent double shuffle omits the regularized Li_1 relations. So the two new survivors are best read as *candidate* new level-3 depth-2 constants; regularized double shuffle can improve upper bounds; numerical independence would still need a separate theorem or a stated period conjecture. As an additional check, an *enriched*-basis probe (`probe_eis_candidates.py`)—adding Li_n at candidate level-3 CM half-points $\frac{1 \pm \rho}{2}, \frac{1}{1 - \rho}, \frac{\rho}{\rho - 1}, 1 - \rho$ (the level-3 analogue of the Gaussian $\frac{1+i}{2}$ that resolved the article’s Family-2)—fails to crack either candidate: $\operatorname{Re} \operatorname{Li}_{4,1}(\rho^2, \rho)$ resists a 105-atom basis (spurious scale $\sim 10^4$).

4.14 The Gaussian (i, −i) point and an exact parallel

The same evaluator handles the level-4 Gaussian point (i, −i) that the article’s level-4 run also skipped ($w = i \cdot (-i) = 1$). The picture is identical in shape.

- **Closed forms.** PSLQ against the level-4 basis (single L -values $\pi, \log 2, \beta(2), \beta(4), \beta(6), \zeta(3), \zeta(5)$, the half-point values $\operatorname{Re} \operatorname{Li}_n(\frac{1+i}{2}), \operatorname{Im} \operatorname{Li}_n(\frac{1+i}{2})$, and the $(i, 1)$ generators) reduces every $(i, -i)$ value through weight 3 and every weight-4 imaginary part. The weight-2 case is especially clean:

$$\operatorname{Li}_{1,1}(i, -i) = -\operatorname{Li}_2\left(\frac{1-i}{2}\right) \quad (\text{verified to } 10^{-50}). \quad (4.28)$$

- **Dimension.** The corrected exact-rank double shuffle (now including $(i, -i)$) gives an upper bound of 12; combined with the closed forms it collapses to the article’s five $(i, -1)$ constants plus **the same two new mixed directions**— $\operatorname{Re} \operatorname{Li}_{4,1}(i, -i)$ at weight 5 and $\operatorname{Im} \operatorname{Li}_{5,1}(i, -i)$ at weight 6. (One weight-3 residual, $\operatorname{Im} \operatorname{Li}_{2,1}(i, -1)$, is a value the article already reduces to the half-point λ/μ basis, outside the convergent double shuffle’s reach.)

Observation 4.8 (the parallel). In *both* towers the $w = 1$ mixed point leaves, beyond the earlier tested generators, two candidate mixed directions, and *in the same two positions*:

$\operatorname{Re} \operatorname{Li}_{4,1}(\text{mixed})$ at weight 5, $\operatorname{Im} \operatorname{Li}_{5,1}(\text{mixed})$ at weight 6.

That the level-3 and level-4 mixed points add new content at identical weights and index patterns—a *real* $\operatorname{Li}_{4,1}$ direction and an *imaginary* $\operatorname{Li}_{5,1}$ direction—is the report’s most suggestive finding, and a natural target for the regularized double shuffle to explain.

Chapter 5

Gamma grids and exact certificates

An experiment proposes a relation; a certificate explains why it holds. For gamma values at rational points, taking logarithms turns reflection and multiplication into rational rows. Their row combinations yield an exact, terminating reduction procedure relative to a specified law matrix. This computational completeness must be distinguished from the open arithmetic completeness conjecture.

5.1 The lattice model

Work on the denominator- N grid, in coordinates $x_k = \log \Gamma(k/N)$, $k = 1, \dots, N-1$. The two classical law families become exact $\mathbb{Q}$ -linear rows with symbolic right-hand sides over the atoms $\log \pi$, $\log n$, $\log \sin(\pi k/N)$:

- reflection, for $k < N/2$:

$$x_k + x_{N-k} = \log \pi - \log \sin\left(\frac{\pi k}{N}\right);$$

- Gauss multiplication, for each divisor $d \geq 2$ of N and $k = 1, \dots, N/d$:

$$\sum_{j=0}^{d-1} x_{k+jN/d} - [dk < N] x_{dk} = \frac{d-1}{2} (\log 2 + \log \pi) + \left(\frac{1}{2} - \frac{dk}{N}\right) \log d,$$

the instance at $z = k/N$ of the Gauss multiplication theorem [11],

$$\prod_{j=0}^{d-1} \Gamma\left(z + \frac{j}{d}\right) = (2\pi)^{(d-1)/2} d^{1/2-dz} \Gamma(dz),$$

with the convention $x_N := \log \Gamma(1) = 0$ in the boundary case $dk = N$.

The matrix is small ($O(N \sum_{d|N} 1/d)$ rows $\times$ $(N-1)$ columns). One exact row reduction of the augmented block $[M \mid I]$ over $\mathbb{Q}$ (columns ordered numerator-descending, so pivots land on large numerators) yields simultaneously:

1. a reduction $\Gamma(k/N) = \exp(\text{elementary}) \cdot \prod \Gamma(\text{basis})^{\mathbb{Q}}$ for *every* non-basis k at once, the elementary side computed exactly as (combination matrix) $\cdot$ (right-hand-side vector) — no `FullSimplify` anywhere;

2. the grid’s *basis* — its free formal gamma coordinates — and with it a termination guarantee: the lattice rank says the number of free coordinates in this presented system.

Proposition 5.1 (Completeness relative to the law matrix). *For a fixed denominator grid and a fixed finite family of reflection and multiplication rows, exact row reduction computes every linear consequence of those rows. The free columns give a basis of the formal quotient by that row space, and the augmented matrix records a certificate for each reduction.*

Proof. Elementary row operations preserve the row space and are invertible. The reduced echelon matrix therefore spans exactly the same consequences. Each pivot equation is a recorded rational combination of the original rows. The nonpivot columns are free in the presented quotient. $\square$

Unrestricted completeness of the standard relations for gamma monomials at rational arguments is Rohrlich’s conjecture. Koblitz–Ogus prove a sufficient algebraicity criterion, not its converse. Thus irreducibility of the numerical gamma basis is conditional, and a finite-grid computation also needs care if it is compared with relations using larger grids. See the discussion of standard relations in Rivoal and Roques, *Hadamard products of algebraic functions* (Section 3) [34].

Result 5.2 (benchmarks and coverage). Implementations: `tools/gamma-reducer.wl` (Wolfram) and `tools/gamma_reducer.py` (pure Python: exact $\mathbb{Q}$ arithmetic via `fractions.Fraction`, sparse atom-coefficient right-hand sides, `mpmath` verification). Both produce *identical* output (same rules, same bases).

workload (generation only)	Wolfram	Python
denominator-120 grid (103 rules, 16-element basis)	0.42 s	0.056 s
all 151 legacy-database grids, denominators ≤ 308 (15 588 rules)	75.3 s	7.44 s
<code>mpmath</code> verification of all 15 588 rules	—	8.32 s

Coverage of the 550 recovered `GammaRules.wdx` arguments: 550/550. The original prover needed multi-hour sessions for 550 rules with no terminus; either implementation beats it by orders of magnitude *with* a completeness guarantee. Sample output (the sine values collapse to radicals on their own):

$$\Gamma\left(\frac{5}{12}\right) = \frac{2^{2/3} \sqrt{\pi} \Gamma\left(\frac{1}{12}\right)}{(1 + \sqrt{3}) \Gamma\left(\frac{1}{6}\right)}.$$

Remark 5.3. The row-combination matrix produced by the augmented reduction is, at the same time, a *multiplier certificate* in the sense of Section 5.5: the reducer’s output is born proved, not merely verified.

5.2 Trig-root Γ identities: mechanism and census

The prompt for this investigation was the identity

$$\frac{\Gamma\left(\frac{2}{7}\right)\Gamma\left(\frac{11}{42}\right)}{\Gamma\left(\frac{1}{21}\right)} = \frac{8 \sin\left(\frac{\pi}{7}\right) \sqrt{\pi \sin\left(\frac{\pi}{21}\right) \sin\left(\frac{4\pi}{21}\right) \sin\left(\frac{5\pi}{21}\right)}}{2^{1/42} 7^{1/3} 3^{9/28}},$$

of the kind that has surprised mathematicians before the mechanism was recognized [historical experiment: verified `RootReduce`-exactly against the legacy table].

Result 5.4 (the mechanism, in lattice terms). When the lattice solve expresses a target $\log \Gamma(k/N)$, the elimination may pass through a combination in which the target appears with multiplicity $D > 1$ — i.e. the relation natively proves $\Gamma(k/N)^D = (\text{elementary}) \cdot (\Gamma\text{-monomial})$. Dividing by D puts coefficient $1/D$ on the elementary logsin atoms, and exponentiating yields a D -th root of a sine product; positivity of all factors fixes the branch. The root is a property of the *derivation path*, not of the value: the same $\Gamma(11/42)$ carries a square root in the harvested spelling above and cube roots in the lattice’s own reduction — both exactly correct.

The surprise dissolves further at grids whose sines are radical-expressible: for $N = 30, 60$ the renderer collapses $\sqrt{\sin \cdots}$ into golden-ratio radicals and nothing looks trig-rooted at all. The phenomenon is only *visible* when the sine arguments have large totient (21, 39, 78, ...), where nested square-root expressions need not exist; general radical expressibility is a different question.

Result 5.5 (census, denominator grids $N \leq 200$, primitive arguments). 856 lattice reductions carry fractional logsin coefficients. By root degree D (the lcm of the coefficient denominators):

D	2	3	4	6	8	16
rules	616	46	106	26	42	20

Demanding *integral* Γ -exponents (so only the trigonometric part is rooted, as in the prompt identity) leaves exactly 10 rules, all $D = 4$, at $\Gamma(71/78)$, $\Gamma(85/156)$, $\Gamma(65/114)$, $\Gamma(167/198)$ and conjugate-numerator siblings. The $D = 16$ cases live on the $N = 168$ grid with 26 sines under the radical.

Result 5.6 (showpieces). [historical experiment: verified to 30+ digits in both implementations] The fourth-root champion ($N = 78$; $\varphi(78) = 24$, so square-root-only expressions are unavailable):

$$\Gamma\left(\frac{71}{78}\right) = 2 \cdot 2^{55/312} \pi \sqrt{\cos \frac{\pi}{78}} \cos^{1/8} \frac{\pi}{39} \cos^{3/4} \frac{\pi}{26} \cos^{5/8} \frac{2\pi}{39} \cos^{11/8} \frac{4\pi}{39} \left(\cos \frac{\pi}{13} \sec \frac{7\pi}{78}\right)^{3/8} \Gamma\left(\frac{1}{39}\right) \\ / \left[13^{1/48} \cos^{1/4} \frac{5\pi}{78} \cos^{3/2} \frac{3\pi}{26} \cos^{7/8} \frac{5\pi}{39} \cos^{5/8} \frac{11\pi}{78} \Gamma\left(\frac{2}{39}\right) \Gamma\left(\frac{5}{78}\right) \right],$$

and the lattice’s cube-root reduction of the prompt’s own $\Gamma(11/42)$:

$$\Gamma\left(\frac{11}{42}\right) = \frac{7^{1/6} \sqrt{\pi} \Gamma\left(\frac{1}{42}\right) \Gamma\left(\frac{1}{21}\right) \left(\sec \frac{\pi}{42} \sec \frac{\pi}{21}\right)^{2/3} \sec^2 \frac{\pi}{14} \left(\cos \frac{2\pi}{21} \sec \frac{5\pi}{42}\right)^{1/3}}{8 \cdot 2^{5/21} 3^{3/7} \Gamma\left(\frac{2}{21}\right) \Gamma\left(\frac{3}{14}\right)}.$$

5.3 The trigamma analogue

The same construction applies one weight up. On the grid $y_k = \psi^{(1)}(k/N)$ the lattice rows are the reflection law $\psi^{(1)}(k/N) + \psi^{(1)}(1 - k/N) = \pi^2 \csc^2(\pi k/N)$ and the multiplication theorem $\psi^{(1)}(dk/N) = d^{-2} \sum_{j=0}^{d-1} \psi^{(1)}(k/N + j/d)$, with the pin $\psi^{(1)}(1) = \pi^2/6$ wherever an argument hits 1; the elementary atoms are π^2 and the $\pi^2 \csc^2(\pi k/N)$ values.

Result 5.7. `tools/psi_reducer.py` derives the complete trigamma reduction tables for grids $N = 3, \dots, 100$: 3428 rules, all mpmath-verified, in 7.7 s total, with basis dimensions equal to $\varphi(N)/2$ throughout — the same dimension count as the cyclotomic-polylogarithm lattice, which the exact DFT bridge between the two worlds explains (the DFT provides a change of algebraic coordinates between the value systems).

This subsumes, by a factor of ~ 50 , a legacy 70-entry hand/PSLQ-era workbook of $\psi^{(1)}(k/n)$ closed forms (denominators 2–16), which is preserved separately as a harvested store.

5.4 The harvest: legacy notebooks, verified and taxonomized

A kernel-free pipeline (notebook box parser, custom `TraditionalForm/InputForm` interpreter, `mpmath` 60-digit verification) mined the owner’s OneDrive notebook corpus into verified identity stores:

- `harvested-gammaprover-rules.wl` — the recovered 550-rule saturation-prover database, every rule verified in-kernel to $|N[\Gamma(\text{arg}) - \text{closed form}, 40]| < 10^{-30}$, zero failures;
- `harvested-gamma-values.wl` — 508 verified Γ reductions from four further notebook families (including a 451-rule table recovered from a single 700 KB output cell, and 637 sine-product reductions from a working-table pair identified during the sweep);
- `harvested-trigamma-values.wl` — all 70 identities of the trigamma workbook, `mpmath`-verified at 60 digits and re-verified in-kernel.

Result 5.8 (novelty taxonomy). Measured against the 670-rule `FunctionTransformations` Γ table [12] (denominators ≤ 120) that anchors the corpus: 314 distinct harvested arguments lie beyond the table — 280 ceiling-breakers from the prover database (denominators 123–308) plus 34 arguments that only the notebook harvest reaches: one further ceiling-breaker, $\Gamma(67/160)$, and 33 upper-half reflection twins of table arguments. (Eight more harvest ceiling-breakers duplicate prover-database arguments; an earlier recorded decomposition “280 + 8 + 34” double-counted them.)

5.5 Multiplier certificates

Before this program, every identity in the project’s stores carried one badge: *numerically verified* ($\gtrsim 25$ digits). The certificate program adds a qualitatively stronger badge: *derived* — a recorded, machine-checkable certificate exhibiting the value as an exact linear combination of verified functional-equation instances, replayed with no search and no numerics in the algebraic replay path of the historical checker. Its numerical branch guards remain outside the exact certificate guarantee.

5.5.1 The model

A certificate stores law *names* plus concrete substitution strings (parsed through a restricted substitution grammar; strings alone do not guarantee safe or valid instantiation) and two exact multiplier vectors over a canonical real/imaginary splitting of the instances. Validity means the claim $\text{lhs} - \text{rhs}$ equals the recorded linear combination *identically*: every inert Li/ψ atom coefficient cancels `RootReduce`-exactly, and the elementary residual is killed by a dedicated sound zero test (certified principal-branch log normalization, certified multiplicative log-basis reduction, formal coefficient comparison). Numerics appear in exactly two proposal/guard roles: proposing root-of-unity decompositions that exact algebra then certifies, and a numerical per-instance branch guard. That guard can reject bad instances, but is not an exact domain proof; branch validity is a separate obligation. Linear systems subsume orbit words, fixed-point arguments, and multi-value systems in one format.

A **derived** certificate proves its claim *relative to* a declared axiom layer: (a) the classical laws of the project’s functional-equation store [1] — Euler reflection, inversion, Landen, duplication/triplication/multiplication, the five-term family, the Kummer-class relations, Schwarz conjugation, the cyclotomic polygamma sum, and the polygamma reflection/multiplication laws, each numerically verified at recorded test points; (b) definitional pins ($\text{Li}_n(1) = \zeta(n)$, $\psi^{(m)}(1) = (-1)^{m+1} m! \zeta(m+1)$,

and the two Catalan pins below); (c) Euler’s $\zeta(2k) \in \mathbb{Q}\pi^{2k}$ and trigonometric values at rational multiples of π ; (d) **RootReduce** as the exact algebraic-number oracle.

5.5.2 Findings the linear algebra produced

Finding 5.9 (Schwarz conjugation is a necessary axiom). The anharmonic orbit of i is the six-point family $\{i, -i, 1 \pm i, (1 \pm i)/2\}$, and inversion and duplication at i produce the same sum relation. The full system of reflection/inversion/Landen/duplication instances over the orbit is rank-deficient by *exactly one* dimension, and the missing direction is precisely $\text{Li}_2(\bar{z}) = \overline{\text{Li}_2(z)}$. No combination of the single-variable transformation laws separates $\text{Li}_2(i)$ from $\text{Li}_2(-i)$ — the transformation laws are Möbius/algebraic in the argument and blind to conjugation. Obvious in hindsight; the linear algebra *measured* it.

Finding 5.10 (Catalan is definitional; the rest is not). $\beta(2)$ admits no derivation from the stored laws: it enters every certificate that needs it through one of two recorded definitional pins, both termwise series identities,

$$\text{Im Li}_2(i) = G \quad \text{or} \quad \psi^{(1)}\left(\frac{1}{4}\right) - \psi^{(1)}\left(\frac{3}{4}\right) = 16G.$$

By contrast $\text{Li}_2(-1) = -\pi^2/12$ (duplication at $z = 1$), the four golden-ratio dilogarithm values, $\text{Li}_3(\frac{1}{2})$, the golden trilogarithm $\text{Li}_3(1/\varphi^2)$, and $\text{Li}_2(2)$ are all honestly derived; the flagship compaction is $\text{Li}_2(\frac{1}{2})$ as a *one-instance* certificate (Euler reflection at its own fixed point), and each golden-ratio dilogarithm carries exactly the classical four-equation linear system, the coincidence $1 - 1/\varphi = 1/\varphi^2 = (1/\varphi)^2$ doing all the work. Within this law vocabulary, the tested imaginary orbit uses Catalan as its single additional named constant. The whole imaginary line also carries $\text{Ti}_2(y)$; this finite orbit calculation is not a universal reduction theorem.

Finding 5.11 (prime denominators need the full multiplication row). For the cyclotomic values $\text{Li}_n(e^{2\pi ip/q})$ the polygamma vocabulary pins the ψ -grid via proper-divisor multiplication instances when q is composite; for *prime* q the only ψ -sum relation is the full row $d = q$ — the multiplication theorem at $z = 1/q$,

$$\psi^{(m)}(1) = q^{-m-1} \sum_{k=1}^q \psi^{(m)}\left(\frac{k}{q}\right),$$

whose $k = q$ term is $\psi^{(m)}(1)$ itself — and the DFT coefficients are the source of the algebraic irrational coefficients — e.g. the $\sqrt{5}\zeta(3)$ terms in the $q = 5$ trilogarithm closed forms. Omitting $d = q$ cost precisely the prime- q certificates and nothing else.

Finding 5.12 (a trusted layer must be applied, not assumed). In the historical checker, a trigonometric normalization step was omitted: the $q = 8$ certificates initially failed because $\csc^2(\pi/8)$ survived as an opaque atom in the formal zero test while being numerically zero all along. The fix is policy, not code volume: trig at rational multiples of π is part of the declared trusted base and is applied explicitly during law instantiation and residual checking. A trusted constant layer is only trusted if the checker actually *invokes* it; silent reliance on auto-evaluation makes the trust boundary version-dependent.

5.5.3 Coverage ledger

At the repository snapshot, the certificate ledger replayed by the independent checker under the regression guard is 212 **derived** + 4 **definitional**, **zero failures**:

store	derived / attempted	notes
curated values	45 + 4 definitional	every concrete curated value
legacy polylog table, order-2 real	82/83	one honest holdout (below)
legacy Γ table, denominators ≤ 40	85/100	complete-lattice law pool

The Γ certificates use the complete denominator- q relation lattice of Section 5.1 as the law pool (20/20 on the pilot slice, versus 3/20 for the earlier heuristic pool). The polylog order-2 sweep was driven by three cumulative generator upgrades — atom-closure five-term pairs with structured one-parameter pair families $(x, 1 - x)$, $(x, x/(x - 1))$, $(x, -x)$; square-root argument seeding for duplication ladders; and a height-weighted cluster retry that pools orbit vocabulary across related rules without ever using a related *claim* (no circularity) — the largest certificate being 80 instances.

Search observation 5.13 (the honest holdout). $\text{Li}_2(-7/10)$, the deepest reduction ladder in its slice (10 residual dilogarithms), remains rank-deficient even at deep search caps: the orbit pairs demand an internal 17-flavoured cancellation that no implemented pair family reaches. The natural next lever is second-order stray matching or ladder relations beyond the stored law set. On the Γ side, the frontier is 15 no-certs of $q = 40$ grade (sine-product exact algebra blow-ups) and the $q > 40$ tail, awaiting cyclotomic-unit arithmetic or certificate transport through the exact DFT bridge to the lighter ψ -side atoms.

Chapter 6

Polygamma row sums at complex multiplication points

Rational gamma grids account for periods that reappear in modular forms. At imaginary quadratic arguments the polygamma series becomes a row of a lattice Eisenstein sum. Symmetry isolates elementary components; modular periods and class-field coordinates account for the non-elementary ones. Individual values and full row sums require different evidence.

6.1 Lattice normalization and row sums

For even $w \geq 4$ and $\Im\tau > 0$, define the absolutely convergent

$$G_w(\tau) = \sum_{(m,n) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{1}{(m + n\tau)^w} = 2\zeta(w)E_w(\tau).$$

Pair positive and negative n , and use the defining polygamma series:

$$G_w(\tau) = 2\zeta(w) + \frac{2}{(w-1)!} \sum_{n \geq 1} \left[\psi^{(w-1)}(n\tau) + \psi^{(w-1)}(1 - n\tau) \right]. \quad (6.1)$$

The hexagonal point in this chapter is $\rho = e^{i\pi/3}$. The modular invariant $\mathfrak{g}_2 = E_4/\eta^8$ below is unrelated to the second Stieltjes constant. Algebraic fits and historical numerical residuals are labeled as such; class polynomials are formed from normalized modular functions, rather than untreated modular-form values.

6.2 The elementary layer: reflection composed with conjugation

Before any CM input, one family of components is fixed by symmetry alone. The Schwarz reflection $\psi^{(m)}(\bar{z}) = \overline{\psi^{(m)}(z)}$ turns the reflection law into a statement about a single real component, leaving the other component untouched.

Result 6.1 (vertical-line trigamma law). On the line $\operatorname{Re} z = \frac{1}{2}$ the reflection law alone evaluates the real part of the trigamma function at every height: for all real t ,

$$\operatorname{Re} \psi^{(1)}\left(\frac{1}{2} + it\right) = \frac{\pi^2}{2 \cosh^2(\pi t)}.$$

In particular $\operatorname{Re} \psi^{(1)}(\frac{1+i}{2}) = \pi^2/(2 \cosh^2(\pi/2))$ [historical experiment: residual 0 at 50 dps]. Off the symmetric line, combining reflection, conjugation and the recurrence gives e.g.

$$\operatorname{Re} \psi^{(1)}(i) = -\frac{1}{2} - \frac{\pi^2}{2 \sinh^2 \pi} \quad [\text{historical experiment: residual 0 at 50 dps}].$$

The proof of the first identity is one line: on $\operatorname{Re} z = \frac{1}{2}$ one has $1 - z = \bar{z}$, so the reflection law $\psi^{(1)}(z) + \psi^{(1)}(1 - z) = \pi^2/\sin^2(\pi z)$ reads $2 \operatorname{Re} \psi^{(1)}(z) = \pi^2/\sin^2(\pi z)$ there.

The imaginary parts on this line stay opaque — they are odd-row lattice data in the sense of Section 6.9.

6.2.1 The FoxTrot identity demystified

MathWorld’s *Digamma Function* page [17] calls the following identity “surprising”; it arises from the FoxTrot series $\sum (-1)^{n-1} n^2/(n^3 + 1)$, whose partial fractions hit the roots of $n^3 - n + 1$:

$$-\psi\left(\frac{\omega}{2}\right) - \psi\left(-\frac{\omega^2}{2}\right) + \psi\left(\frac{1+\omega}{2}\right) + \psi\left(\frac{1-\omega^2}{2}\right) = 2\pi \operatorname{sech}\left(\frac{\sqrt{3}\pi}{2}\right), \quad \omega = e^{i\pi/3}. \quad (6.2)$$

Law 6.2 (line law for digamma differences). *For every shift $c \in (0, 1)$, on the line $\operatorname{Re} z = (1 - c)/2$,*

$$\operatorname{Re}[\psi(z + c) - \psi(z)] = \pi \operatorname{Re} \cot(\pi \bar{z}).$$

[historical experiment: verified for $c = \frac{1}{2}, \frac{1}{3}, \frac{2}{3}$ at generic heights t]

The mechanism is the same one-liner: on that line the shift satisfies $z + c = 1 - \bar{z}$, so reflection composed with conjugation eliminates the digamma values, leaving an elementary $\sinh/\cos$ closed form. The FoxTrot identity (6.2) is the instance $c = \frac{1}{2}$, $z = \frac{1}{4} + i\frac{\sqrt{3}}{4}$: its four arguments pair into $2 \operatorname{Re}[\psi(z + \frac{1}{2}) - \psi(z)]$, and on $\operatorname{Re} z = \frac{1}{4}$ one has $\pi \operatorname{Re} \cot(\pi \bar{z}) = \pi \operatorname{sech}(2\pi t)$. The identity is thus the point $t = \sqrt{3}/4$ on a line of identities, itself one member of a two-parameter family — no complex multiplication required.

Differentiating in t ladders the law up the polygamma tower [historical experiment: weights 2, 3 verified at 10^{-50}]:

$$\begin{aligned} \operatorname{Im}[\psi^{(1)}(\tfrac{3}{4} + it) - \psi^{(1)}(\tfrac{1}{4} + it)] &= 2\pi^2 \operatorname{sech}(2\pi t) \tanh(2\pi t), \\ \operatorname{Re}[\psi^{(2)}(\tfrac{3}{4} + it) - \psi^{(2)}(\tfrac{1}{4} + it)] &= 8\pi^3 \operatorname{sech}^3(2\pi t) - 4\pi^3 \operatorname{sech}(2\pi t). \end{aligned}$$

At each weight the alternating component (Re at odd weight, Im at even weight) stays opaque, exactly as in Section 6.9.

6.3 Maximal symmetry: vanishing identities and Γ -periods

At the two CM points with extra symmetry — $\tau = i$ (square lattice, $j = 1728$) and $\tau = \rho = e^{i\pi/3}$ (hexagonal lattice, $j = 0$) — classical theory pins every G_{2k} exactly. The square lattice admits no weight-6 form ($i \cdot (\mathbb{Z} + \mathbb{Z}i) = \mathbb{Z} + \mathbb{Z}i$ forces $G_6(i) = i^{-6}G_6(i) = -G_6(i)$), and the hexagonal lattice no weight-4 form; $G_4(i)$ and $G_6(\rho)$ are the lemniscatic and equianharmonic period powers — monomials in $\Gamma(1/4)$, resp. $\Gamma(1/3)$, by Chowla–Selberg [13].

6.3.1 Vanishing identities

Result 6.3 (the CM gems).

$$\sum_{n \geq 1} [\psi^{(5)}(ni) + \psi^{(5)}(1 - ni)] = -120 \zeta(6) \quad [\text{historical experiment: residual } 1.7 \times 10^{-49}],$$

$$\sum_{n \geq 1} [\psi^{(3)}(n\rho) + \psi^{(3)}(1 - n\rho)] = -6 \zeta(4), \quad \rho = e^{i\pi/3} \quad [\text{historical experiment: residual } 6.8 \times 10^{-50}].$$

Each individual bracket is a transcendental-looking complex number (elementary in $\coth/\operatorname{csch}$ of $\pi n\sqrt{3}/2$ and the like after the bilateral cotangent expansion, but nobody would guess the totals); the sums collapse to rational multiples of $\zeta(6)$, $\zeta(4)$ *because lattice rotation multiplies the corresponding value by a nontrivial root of unity*. Via (6.1) these are exactly the statements $G_6(i) = 0$ and $G_4(\rho) = 0$.

6.3.2 Chowla–Selberg made polygamma-visible

Result 6.4 (period identities).

$$2\zeta(4) + \frac{1}{3} \sum_{n \geq 1} [\psi^{(3)}(ni) + \psi^{(3)}(1 - ni)] = \frac{\Gamma(1/4)^8}{960 \pi^2} \quad [\text{historical experiment: residual } 2.1 \times 10^{-50}],$$

$$2\zeta(6) + \frac{1}{60} \sum_{n \geq 1} [\psi^{(5)}(n\rho) + \psi^{(5)}(1 - n\rho)] = \frac{\Gamma(1/3)^{18}}{8960 \pi^6} \quad [\text{historical experiment: residual } 2.0 \times 10^{-60}].$$

Polygamma values at purely imaginary (resp. hexagonal) points, summed over the integer multiples of the point, know the lemniscate constant and its equianharmonic sibling. The atoms $\Gamma(1/4)^2/\pi$ and $\Gamma(1/3)$ are precisely the CM-period currency of the repository’s hypergeometric reducer (`cmPeriodReduce` in `src/Hypergeometric`), so these identities tie the polygamma world directly to that corpus.

6.3.3 The complete low-weight table

The symmetry argument and the classical Hurwitz recursion [15]

$$(2n+1)(2n-1)(n-3) G_{2n} = 3 \sum_{m=2}^{n-2} (2m-1)(2n-2m-1) G_{2m} G_{2n-2m}$$

extend both subsections through weight 12 with no new constants [historical experiment: all entries verified, residuals 10^{-60} – 10^{-263}]:

weight $2k$	square lattice $\tau = i$	hexagonal lattice $\tau = \rho$
4	$\Gamma(1/4)^8/(960\pi^2)$	0
6	0	$\Gamma(1/3)^{18}/(8960\pi^6)$
8	$\frac{3}{7}G_4(i)^2 = 3\Gamma(1/4)^{16}/(7 \cdot 960^2\pi^4)$	0
10	0	0
12	$\frac{1}{143}(18G_4(i)^3 + 25G_6(i)^2)$ type	$\frac{25}{143}G_6(\rho)^2 \propto \Gamma(1/3)^{36}$

Every bold zero is a vanishing identity $\sum_{n \geq 1} [\psi^{(w-1)}(n\tau) + \psi^{(w-1)}(1 - n\tau)] = -(w-1)! \zeta(w)$, and every non-zero entry is a $\psi^{(w-1)}$ -sum evaluation. The pattern is exhaustive: $G_{2k}(i) = 0$ iff $2k \not\equiv 0 \pmod{4}$, $G_{2k}(\rho) = 0$ iff $2k \not\equiv 0 \pmod{6}$, and all non-zero values are $\mathbb{Q}$ -monomials in the two period powers — so the *entire* Eisenstein tower over both lattices is polygamma-explicit.

6.4 Beyond maximal symmetry: $2i$ and $i\sqrt{2}$

The same row-sum at the next two CM points, with closed forms found by PSLQ [historical experiment: 120-dps canaries]:

Result 6.5.

$$\begin{aligned} G_4(2i) &= \frac{11}{15360} \frac{\Gamma(1/4)^8}{\pi^2} \quad [\text{historical experiment: canary 0 at 120 dps}], \\ G_4(i\sqrt{2}) &= \frac{\Gamma(1/8)^4 \Gamma(3/8)^4}{4608 \pi^2} \quad [\text{historical experiment: canary } 3.9 \times 10^{-121}]. \end{aligned}$$

The weight-6 row at the same points [historical experiment: 160-dps canaries]:

$$G_6(2i) = \frac{\Gamma(1/4)^{12}}{2^{14} \cdot 5 \cdot \pi^3}, \quad G_6(i\sqrt{2}) = \frac{\Gamma(1/8)^6 \Gamma(3/8)^6}{2^{12} \cdot 3^3 \cdot 5 \cdot \pi^3}.$$

The first value confirms the isogeny picture: $2i$ sits over the same field $\mathbb{Q}(i)$, so only the rational factor changes ($G_4(i)/G_4(2i) = 16/11$). The second imports a genuinely *new period*: discriminant -8 , field $\mathbb{Q}(\sqrt{-2})$ — the polygamma sum $\sum_n [\psi^{(3)}(ni\sqrt{2}) + \psi^{(3)}(1-ni\sqrt{2})]$ evaluates in $\Gamma(1/8)\Gamma(3/8)$. The fundamental unit $1 + \sqrt{2}$ was offered in the PSLQ basis and declined (exponent 0). Together with Section 6.3 this makes three distinct Chowla–Selberg periods polygamma-visible.

6.5 Class number 2: discriminants -20 and -15

At $\mathbb{Q}(\sqrt{-5})$ (class number 2) a monomial-only logarithmic basket can miss the value: the algebraic factor in G_4/Ω^4 lives in the Hilbert class field and is not a monomial. The structure that works is the Hecke/Damerell pair [14] over the two reduced forms $x^2 + 5y^2$ ($\tau_1 = i\sqrt{5}$) and $2x^2 + 2xy + 3y^2$ ($\tau_2 = (-1 + i\sqrt{5})/2$).

Result 6.6 (disc -20).

$$\begin{aligned} \frac{G_4(\tau_1)}{G_4(\tau_2)} &= \frac{29 + 12\sqrt{5}}{44} = \frac{(3 + 2\sqrt{5})^2}{44} \quad [\text{historical experiment: canary 0 at 160 dps}], \\ G_4(\tau_1) G_4(\tau_2) &= \frac{11 \varphi^4 \Gamma(3/20)^8 \Gamma(7/20)^8}{2^{14} \cdot 3^4 \cdot 5^4 \cdot \pi^4} \quad [\text{historical experiment: canary } 5.3 \times 10^{-160}], \end{aligned}$$

where φ is the golden ratio. The ratio is algebraic of degree 2 with norm $N(29 + 12\sqrt{5}) = 121 = 11^2$; the class product is a period monomial carrying the prime 11 and a golden unit.

Result 6.7 (disc -15). With $\tau_1 = (1 + i\sqrt{15})/2$ for $x^2 + xy + 4y^2$ and the conjugate pair $\tau_{2,3} = (\pm 1 + i\sqrt{15})/4$ representing $2x^2 + xy + 2y^2$, write $\hat{G}_4 = \frac{1}{2}[G_4(\tau_2) + G_4(\tau_3)] = \text{Re } G_4(\tau_2)$ for the class-pair average. Then [historical experiment: 160-dps canaries]:

$$\frac{G_4(\tau_1)}{\hat{G}_4} = \frac{2(4 + \sqrt{5})^2}{77}, \quad G_4(\tau_1) \hat{G}_4 = \frac{77 \left(\Gamma(\frac{1}{15})\Gamma(\frac{2}{15})\Gamma(\frac{4}{15})\Gamma(\frac{8}{15})\right)^4}{2^{13} \cdot 3^6 \cdot 5^5 \cdot \pi^4}.$$

Unlike disc -20 , the non-principal value here is genuinely complex (the form $(2, 1, 2)$ has $|\tau_2| = 1$, where conjugation acts nontrivially on G_4); the class-pair average is the Galois-natural object.

The Γ -block in Result 6.7 is exactly the set of residues $\{1, 2, 4, 8\}$ with $\chi_{-15} = +1$; the golden unit is declined by PSLQ (exponent 0); and the prime content $77 = 7 \cdot 11$ of the product matches the ratio's denominator ($N(4 + \sqrt{5}) = 11$). The heuristic that emerges, used at every later discriminant: *run the ratio first; its norm names the primes the class-product basis needs.*

6.6 Class number 3: discriminant -23

$\mathbb{Q}(\sqrt{-23})$ has $h = 3$, with Hilbert class field generated over $\mathbb{Q}(\sqrt{-23})$ by a root of $x^3 - x - 1$. The three reduced forms give $\tau_1 = (1 + i\sqrt{23})/2$ and $\tau_{2,3} = (\pm 1 + i\sqrt{23})/4$; $G_4(\tau_1)$ is real and $G_4(\tau_{2,3})$ form a conjugate pair.

Result 6.8 (disc -23 class product). [historical experiment: canary 1.4×10^{-160}]

$$G_4(\tau_1) G_4(\tau_2) G_4(\tau_3) = \frac{11 \cdot 17 P^4}{2^{24} \cdot 3^6 \cdot 23^5 \cdot \pi^{16}}, \quad P = \prod_{\substack{k \bmod 23 \\ \left(\frac{k}{23}\right) = +1}} \Gamma\left(\frac{k}{23}\right) \quad (11 \text{ factors}).$$

Result 6.9 (the $\mathfrak{g}_2$ class polynomial from row-sums). With $\mathfrak{g}_2 = E_4/\eta^8$ phase-corrected by the canonical cube root (the $q^{1/3}$ ambiguity of η^8 at these points), the three values $\mathfrak{g}_2(\tau_i)$ computed from the polygamma row-sums are exactly the roots of the classical class polynomial

$$x^3 + 155x^2 + 650x + 23375 = 0 \quad [\text{historical experiment: coefficients integer to } 10^{-80}],$$

whose constant term $23375 = 5^3 \cdot 11 \cdot 17$ forecast the primes 11, 17 in the class product — the norm heuristic of Section 6.5, one degree up.

Search observation 6.10 (weight-4 sums are not Galois-clean). The dimensionless symmetric functions e_1^3/e_3 and e_2^3/e_3^2 of the raw G_4 values were not recognized as rational in the source search (height 10^{30} , 300 digits). Unlike the value *product* (a norm) and the weight-0 $\mathfrak{g}_2$ invariants, sums of weight-4 CM values across ideal classes are not Galois-stable — the period correction under $\text{Gal}(H/K)$ is nontrivial at weight 4. Class polynomials must be formed from $\mathfrak{g}_2 = E_4/\eta^8$ (or j), not from the G_4 values themselves.

6.7 Class number 4: discriminant -39 and genus structure

The four reduced forms of discriminant -39 , namely $(1, 1, 10)$, $(2, \pm 1, 5)$, $(3, 3, 4)$, give $\tau \in \{(1 + i\sqrt{39})/2, (\pm 1 + i\sqrt{39})/4, (3 + i\sqrt{39})/6\}$.

Result 6.11 (disc -39 class product). [historical experiment: 160-dps canary]

$$\prod_{i=1}^4 G_4(\tau_i) = \frac{17 \cdot 23 \cdot 29 P^4}{2^{28} \cdot 3^9 \cdot 5^4 \cdot 13^8 \cdot \pi^{16}}, \quad P = \prod_{\substack{k \bmod 39 \\ \chi_{-39}(k) = +1}} \Gamma\left(\frac{k}{39}\right) \quad (12 \text{ factors}).$$

A wrinkle relative to Section 6.6: $\mathfrak{g}_2$ is *not* a class invariant here ($3 \mid 39$, the classical obstruction [16]; observed directly — no cube-root phase choice makes the $\mathfrak{g}_2$ -sums rational). But j always is, and the row-sum values deliver the Hilbert class polynomial outright.

Result 6.12 (Hilbert class polynomial of disc -39).

$$H_{-39}(x) = x^4 + 331531596 x^3 - 429878960946 x^2 + 109873509788637459 x + 20919104368024767633$$

[historical experiment: $e_1, \dots, e_3$ PSLQ-exact integers; e_4 integral to 10^{-137}].

Worth dwelling on: a quartic with 20-digit integer coefficients recovered from sums of polygamma values at quadratic-irrational complex points. The polygamma row-sum is a perfectly serviceable special-function front door to the classical CM tower.

Result 6.13 (genus structure inside the quartet). The class group is $\mathbb{Z}/4$, with the ambiguous form $(3, 3, 4)$ as the order-2 class. The raw cross-class ratio v_1/v_4 is a quartic candidate; no degree-two polynomial was found up to height 10^{12} , but the genus-wise ratio — principal genus $\{1, A^2\}$ over $\{A, A^3\}$ — is exactly quadratic, in the real part of the genus field $\mathbb{Q}(\sqrt{-3}, \sqrt{13})$ as genus theory demands [historical experiment: 160-dps canary; $\sqrt{3}, \sqrt{39}$ declined by PSLQ]:

$$\frac{G_4(\tau_1) G_4(\tau_4)}{G_4(\tau_2) G_4(\tau_3)} = \frac{103299 + 4440\sqrt{13}}{181424}, \quad N = \left(\frac{9}{16}\right)^2, \quad 181424 = 16 \cdot 17 \cdot 23 \cdot 29.$$

The denominator reuses the class-product primes 17, 23, 29.

6.8 Class number 5: discriminant -47

The five reduced forms $(1, 1, 12)$, $(2, \pm 1, 6)$, $(3, \pm 1, 4)$ give the quintet $\tau \in \{(1 + i\sqrt{47})/2, (\pm 1 + i\sqrt{47})/4, (\pm 1 + i\sqrt{47})/6\}$.

Result 6.14 (disc -47 class product). [historical experiment: 200-dps canary]

$$\prod_{i=1}^5 G_4(\tau_i) = \frac{11^2 \cdot 23 \cdot 29 P^4}{2^{52} \cdot 3^6 \cdot 47^9 \cdot \pi^{36}}, \quad P = \prod_{\substack{k \bmod 47 \\ \binom{k}{47} = +1}} \Gamma\left(\frac{k}{47}\right) \quad (23 \text{ factors}).$$

Result 6.15 (Hilbert class polynomial of disc -47). The row-sum j -values deliver the quintic Hilbert class polynomial [historical experiment: all symmetric functions integral to 10^{-129} or better]:

$$H_{-47}(x) = x^5 + 2257834125 x^4 - 9987963828125 x^3 + 5115161850595703125 x^2 - 14982472850828613281250 x + 16042929600623870849609375.$$

The CM tower is now polygamma-explicit through class number 5 with one uniform recipe: *row-sum* $\rightarrow$ *class-product log-PSLQ (norm-forecast primes)* $\rightarrow$ *j-symmetric-function integrality*.

6.9 The opaque components

The components *not* fixed by reflection-conjugation or by the even-row mechanism — $\text{Im } \psi^{(1)}(i)$, $\text{Re } \psi(i)$ beyond γ , $\text{Im } \psi^{(1)}(\frac{1+i}{2})$ — are *odd-row* lattice data: for example

$$\text{Im } \psi^{(1)}(i) = -2 \sum_{k \geq 1} \frac{k}{(k^2 + 1)^2}$$

is an odd-power analogue of the coth family, which the bilateral (even) cotangent expansion cannot reach. These are Eichler-integral/quasi-modular territory — the E_2 -flavoured layer; compare $\sum_{k \geq 1} k/(e^{2\pi k} - 1) = \frac{1}{24} - \frac{1}{8\pi}$, which is $E_2(i) = 3/\pi$ in Lambert-series clothing.

Search observation 6.16. A PSLQ basket for these constants against products drawn from

$$\{1, \pi, \coth \pi, \pi^2/\sinh^2 \pi, \pi \coth \pi, \dots\}$$

found nothing at small coefficient heights — consistent with genuinely new constants. Recorded as an honest negative; the Lambert-series face ($\sum \sigma_5(n)q^n$ at $q = e^{-2\pi}$ and $q = -e^{-\pi\sqrt{3}}$) is the natural next probe surface.

Chapter 7

Zeta jets, Stieltjes antiderivatives and negative polygamma

The gamma function is the first spectral derivative of Hurwitz zeta at zero. Further derivatives give a natural language for integrating Stieltjes functions. Integration in the Hurwitz parameter moves the spectral argument downward, creating the negative-order polygamma ladder. This chapter develops that common mechanism before tabulating its moments and rational values.

7.1 Below the digamma: $\psi^{(-2)}$ at rationals

One integration below the digamma sits $\psi^{(-2)}(z) = \int_0^z \log \Gamma(t) dt$. Integrating Kummer’s Fourier series of $\log \Gamma$ [3] predicts that at rational points this grid lands in $\{\text{Clausen values}\}/\pi \oplus \zeta'(-1) \oplus \text{logs}$: the weight-2 Clausen world enters through the negative-order polygamma ladder, with the Glaisher-layer constant $\zeta'(-1)$ as the genuinely new atom. PSLQ confirms [historical experiment: 100-digit canaries; residuals $\leq 10^{-60}$]:

Result 7.1 (log-gamma integrals in the Clausen lattice).

$$\begin{aligned} \int_0^{1/4} \log \Gamma(t) dt &= \frac{3}{32} + \frac{\log 2}{8} + \frac{\log \pi}{8} + \frac{G}{4\pi} - \frac{9}{8} \zeta'(-1), \\ \int_0^{1/3} \log \Gamma(t) dt &= \frac{1}{9} + \frac{\log 2}{6} - \frac{\log 3}{72} + \frac{\log \pi}{6} + \frac{\text{Cl}_2(2\pi/3)}{4\pi} - \frac{4}{3} \zeta'(-1), \\ \int_0^{1/6} \log \Gamma(t) dt &= \frac{5}{72} + \frac{7 \log 2}{72} + \frac{\log 3}{144} + \frac{\log \pi}{12} + \frac{\text{Cl}_2(\pi/3)}{4\pi} - \frac{5}{6} \zeta'(-1). \end{aligned}$$

The structural punchline: the quarter-point integral carries Catalan ($\text{Cl}_2(\pi/2) = G$) and the third-point integral carries $\text{Cl}_2(2\pi/3) = \frac{2}{3} \text{Cl}_2(\pi/3)$, two-thirds of Gieseking’s constant — the “level-3 Catalan” that had surfaced independently in the repository’s Eisenstein-ladder work — now arriving from a completely different direction.

Search observation 7.2 (recorded on purpose). (i) The half-point probe $\int_0^{1/2} \log \Gamma$ initially found no relation — because $\log \Gamma(1/2) = \frac{1}{2} \log \pi$ made the PSLQ basis $\mathbb{Q}$ -dependent, and PSLQ returned that degeneracy instead of the target. Basis atoms must be $\mathbb{Q}$ -independent. (ii) A $\psi^{(-3)}(1/4)$ probe via Cauchy’s repeated-integration kernel produced only a large-coefficient candidate (heights $\sim 10^6$) with a weak canary (10^{-46} at 100 dps) — *rejected* as a PSLQ false positive. The weight-3 level needs a properly prepared basis (presumably $\zeta'(-2)$ -equivalents plus Barnes- G -layer logarithms).

7.2 Binet’s second formula and Malmsten’s integrals: the Lambert-kernel face

Binet’s second formula

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(t/z)}{e^{2\pi t} - 1} dt$$

is the integrated Abel–Plana formula with the *Lambert kernel* $1/(e^{2\pi t} - 1)$ — the same kernel whose moment sums are the Eisenstein q -series of the companion CM article. Malmsten/Vardi log-log integrals $\int_0^1 \ln \ln(1/x) R(x) dx$ are the *Mellin avatars* of Kummer’s Fourier series: expanding the rational kernel R in a geometric series produces $L'(\chi, 1)$ -type sums, and Kummer’s formula converts odd-character $L'(1)$ into $\log \Gamma$ at rationals.

Result 7.3. Binet’s formula replays exactly at $z = 1, \frac{1}{2}, \frac{1}{4}$ [historical experiment: residuals $\leq 10^{-52}$], turning every Γ -store atom into a closed-form arctan–Lambert integral; e.g.

$$\int_0^\infty \frac{\arctan(4t)}{e^{2\pi t} - 1} dt = \frac{1}{2} \left[\log \Gamma\left(\frac{1}{4}\right) - \frac{1}{2} \log 2 + \frac{1}{4} - \frac{1}{2} \log(2\pi) \right].$$

Result 7.4 (Vardi’s integral [5], re-derived blind). PSLQ over $\{\pi\gamma, \pi \log 2, \pi \log \pi, \pi \log \Gamma(1/4)\}$, with the γ atom *declined* (coefficient 0) [historical experiment: canary exact at 80 dps]:

$$\int_0^1 \frac{\ln \ln(1/x)}{1+x^2} dx = \frac{\pi}{4} \ln\left(\frac{4\pi^3}{\Gamma(1/4)^4}\right).$$

The $q = 3$ Malmsten variant [historical experiment: canary 3.4×10^{-81} ; γ declined again]:

$$\int_0^1 \frac{\ln \ln(1/x)}{1+x+x^2} dx = \frac{\pi}{6\sqrt{3}} \ln\left(\frac{2^8 \pi^8}{3^3 \Gamma(1/3)^{12}}\right).$$

Both integrals carry exactly the harvested- Γ -store atoms: the log-log family is a systematic *integral-representation generator* for the identity stores — every odd Dirichlet character (each denominator q) yields one such evaluation through Kummer’s formula, with the Γ -lattice reducer supplying the closed form mechanically.

7.3 The log-gamma moment ladder

Result 7.5 (Hurwitz ladder; classical (Whittaker–Watson [8])).

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{2} \sum_{n \geq 2} \frac{n-1}{n(n+1)} \zeta(n, z+1).$$

Since $\zeta(n, z+1)$ is $\psi^{(n-1)}(z+1)$ up to factorials, this is a $\mathbb{Q}$ -linear ladder between $\log \Gamma$ and the *entire polygamma tower at the same point* — a candidate certificate law binding the Γ and trigamma stores into one linear system.

Result 7.6 (second moment; classical (Espinosa–Moll [7])). With $L = \log(2\pi)$,

$$\int_0^1 [\log \Gamma(x)]^2 dx = \frac{\gamma^2}{12} + \frac{\pi^2}{48} + \frac{\gamma L}{6} + \frac{L^2}{3} - (\gamma + L) \frac{\zeta'(2)}{\pi^2} + \frac{\zeta''(2)}{2\pi^2}.$$

The quadratic-moment layer adds $\zeta''(2)$ to the atom inventory.

Search observation 7.7 (the cubic moment stays open).

$$\int_0^1 [\log \Gamma(x)]^3 dx = 5.7403888072294742800195716881024614629610130074549 \dots$$

(50 digits, tanh-sinh quadrature cross-calibrated by the second moment). PSLQ over the natural weight-mixed basket $(\gamma^3, \dots, L^3, \pi^2\gamma, \pi^2L, \zeta(3), (\gamma+L)\zeta'(2)/\pi^2, (\gamma+L)\zeta''(2)/\pi^2, \zeta'''(2)/\pi^2, \zeta'(2)^2/\pi^4, \zeta'(3), \gamma\zeta'(2)/\pi^2)$ returns only uniform-height junk at 60 dps — consistent with Espinosa–Moll and Bailey et al. The Kummer-cube resonance $k_1 \pm k_2 \pm k_3 = 0$ suggests a depth-two candidate (Tornheim–Witten-derivative flavour); finding the right *named* atom is the open problem, not search depth.

7.4 The Stieltjes layer

Differentiating $L(s, \chi) = q^{-s} \sum_k \chi(k) \zeta(s, k/q)$ at $s = 1$ gives the exact ladder

$$L'(\chi, 1) = -\ln q L(1, \chi) - \frac{1}{q} \sum_k \chi(k) \gamma_1\left(\frac{k}{q}\right),$$

so every Malmsten log-log integral is a statement about *first generalized Stieltjes constants* $\gamma_1(k/q)$ — Blagouchine’s program, equivalent to Deninger’s $\zeta''(0, a)$ layer via the functional equation.

Result 7.8 (even-character Malmsten integral). [historical experiment: verified to 10^{-51} ; even character mod 5; φ the golden ratio]

$$\int_0^1 \ln \ln \frac{1}{x} \cdot \frac{(1-x)(1-x^2)}{1-x^5} dx = -(\gamma + \ln 5) \frac{2 \ln \varphi}{\sqrt{5}} - \frac{\gamma_1(\frac{1}{5}) - \gamma_1(\frac{2}{5}) - \gamma_1(\frac{3}{5}) + \gamma_1(\frac{4}{5})}{5}.$$

Result 7.9 (the parity dichotomy). The dichotomy is empirically sharp. The *odd* γ_1 -combination reduces exactly as differentiated-Kummer predicts [historical experiment: verified blind]:

$$\gamma_1(\frac{1}{3}) - \gamma_1(\frac{2}{3}) = -\frac{\pi}{\sqrt{3}} \left(\gamma + 4 \ln 2 - \frac{1}{2} \ln 3 + 4 \ln \pi - 6 \ln \Gamma\left(\frac{1}{3}\right) \right),$$

which is *Malmsten’s 1846 reflection identity* [4] (long misattributed to Almkvist–Meurman). The even combination did not reduce in the tested standard basket [historical experiment: PSLQ junk at coefficient height 10^8 — honest negative]; pushing Blagouchine’s rational-arguments theorem [6] through it collapses everything except a ζ'' -block [historical experiment: derived and verified to 10^{-50}]:

$$\gamma_1(\frac{1}{5}) - \gamma_1(\frac{2}{5}) - \gamma_1(\frac{3}{5}) + \gamma_1(\frac{4}{5}) = \sqrt{5} \left[\zeta''(0, \frac{1}{5}) - \zeta''(0, \frac{2}{5}) - \zeta''(0, \frac{3}{5}) + \zeta''(0, \frac{4}{5}) \right] - 2\sqrt{5} \ln \varphi (\gamma + \ln 10\pi).$$

The canonical atom for the even layer is thus the even-character $\zeta''(0, l/q)$ combination (Deninger’s layer [9]); Malmsten integrals, even γ_1 -combinations, and even $L'(\chi, 1)$ are all elementary-equivalent to it. Two practical records: Blagouchine’s particular value $\gamma_1(1/4)$ was re-verified exactly against `mpmath`’s generalized Stieltjes constants, and the reduction coefficients live in $\mathbb{Q}(\sqrt{q})$ — rational-coefficient PSLQ baskets *miss* the relation, so a $\sqrt{q}$ -weighted basket or direct derivation is required.

7.5 The common Hurwitz-jet calculus

The generalized Stieltjes constants $\gamma_n(x)$ are defined by the Laurent expansion of the Hurwitz zeta function at its pole,

$$\zeta(s, x) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(x) (s-1)^n, \quad (7.1)$$

so that $\gamma_0(x) = -\psi(x)$ and $\gamma_n(1) = \gamma_n$ are the classical Stieltjes constants, $\gamma_0 = \gamma$. Although universally called *constants*, the $\gamma_n(x)$ are real-analytic functions of the argument on $(0, \infty)$, and it is this function-theoretic point of view we adopt. Following Blagouchine [22, 6] we write

$$\Gamma_n(p) := \int_1^p \gamma_n(x) dx, \quad \text{so that } \Gamma_n(1) = 0,$$

for the antiderivative normalized at 1. (The notation is Blagouchine's; the clash with the Euler Γ -function is resolved by the subscript, and indeed $\Gamma_0(p) = -\ln \Gamma(p)$, as Theorem 7.10 makes plain.) Throughout, $\zeta^{(k)}(s_0, a) := \partial_s^k \zeta(s, a)|_{s=s_0}$ and $\zeta^{(k)}(s_0) := \zeta^{(k)}(s_0, 1)$ denote derivative jets with respect to the *order*; $\Lambda_q := \gamma + \ln(2\pi q)$; $H_n = \sum_{i \leq n} 1/i$ and $H_n^{(2)} = \sum_{i \leq n} 1/i^2$ are harmonic numbers; A is the Glaisher–Kinkelin constant, $\ln A = \frac{1}{12} - \zeta'(-1)$; and

$$S_j\left(\frac{p}{q}\right) := \sum_{k=1}^{q-1} \sin \frac{2\pi kp}{q} \zeta^{(j)}\left(0, \frac{k}{q}\right), \quad S_0\left(\frac{p}{q}\right) = \frac{1}{2} \cot \frac{\pi p}{q}, \quad S_1\left(\frac{p}{q}\right) = \sum_{k=1}^{q-1} \sin \frac{2\pi kp}{q} \ln \Gamma\left(\frac{k}{q}\right)$$

(the closed forms for S_0, S_1 use $\zeta(0, a) = \frac{1}{2} - a$, $\zeta'(0, a) = \ln \Gamma(a) - \frac{1}{2} \ln(2\pi)$, and $\sum_k \sin(2\pi kp/q) = 0$).

The starting observation is elementary but organizing: differentiating (7.1) in x against $\partial_x \zeta(s, x) = -s \zeta(s+1, x)$ identifies every derivative and antiderivative of every γ_n with a derivative jet of the Hurwitz zeta function at an *integer* value of s . Antiderivatives live at $s = 0$; the functions themselves at $s = 1$; their x -derivatives at $s = 2$ [25] — a single ladder.

The preceding gamma-integral examples are the first rung of this calculus. We now prove the integration identity, build its spectral jets, and use reflection and distribution to compute rational arguments. All source residuals in this chapter are historical; locally rerun checks are recorded separately in the verification directory.

7.6 Underived Stieltjes grids and character coordinates

Expanding the Hurwitz distribution identity at $s = 1$ gives

$$\begin{aligned} \sum_{j=0}^{d-1} \gamma_1((x+j)/d) &= d\gamma_1(x) + d \log d \psi(x) - \frac{d}{2} \log^2 d, \\ \sum_{j=0}^{d-1} \gamma_2((x+j)/d) &= d \left[\gamma_2(x) - 2 \log d \gamma_1(x) - \log^2 d \psi(x) + \frac{1}{3} \log^3 d \right]. \end{aligned}$$

For a nonprincipal character,

$$L'(1, \chi) = -\log q L(1, \chi) - \frac{1}{q} \sum_{p=1}^q \chi(p) \gamma_1(p/q).$$

For a real quadratic character χ_D , $\zeta_{\mathbb{Q}(\sqrt{D})}(s) = \zeta(s)L(s, \chi_D)$ shows $L'(1, \chi_D)/L(1, \chi_D) = \gamma_{\mathbb{Q}(\sqrt{D})} - \gamma$, where the first term is the field's Euler–Kronecker constant. The source's exact finite matrices for $3 \leq q \leq 30$ left $\varphi(q)/2 - 1$ coordinates after distribution and first-constant reflection. Their negative searches constrain the selected baskets, without proving independence of Euler–Kronecker constants. The reflection formulas below retain the conductor factor $\gamma + \log(2\pi q)$: the earlier printed $q = 5$ specialization without $\log 5$ was corrected in the source reports.

7.7 The ladder

Theorem 7.10 (antiderivative ladder). *For $n \geq 0$ and $p > 0$,*

$$\frac{d}{dp} \zeta^{(n+1)}(0, p) = (-1)^{n+1}(n+1) \gamma_n(p), \quad (7.2)$$

and consequently

$$\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1} \left[\zeta^{(n+1)}(0, p) - \zeta^{(n+1)}(0) \right]. \quad (7.3)$$

Proof. Differentiating $\partial_x \zeta(s, x) = -s \zeta(s+1, x)$ is legitimate termwise for $\operatorname{Re} s > 1$ and extends by analytic continuation. Expand both sides around $s = 0$: on the right, $-s \zeta(s+1, x) = -s \left[\frac{1}{s} + \sum_k \frac{(-1)^k}{k!} \gamma_k(x) s^k \right] = -1 - \sum_k \frac{(-1)^k}{k!} \gamma_k(x) s^{k+1}$, whose s^{n+1} -coefficient is $(-1)^{n+1} \gamma_n(x)/n!$. On the left the s^{n+1} -coefficient is $\partial_x \zeta^{(n+1)}(0, x)/(n+1)!$. Equating gives (7.2); integrating from 1 to p gives (7.3). $\square$

The differential form (7.2) is Lemma 2 of Chakraborty–Kanemitsu–Kuzumaki [23] (for $n = 1$ it is Deninger's R' -law [9]); the integrated form (7.3) is eq. (2.16) of Condon [26]. For $n = 0$, $\zeta'(0, p) = \ln \Gamma(p) - \frac{1}{2} \ln(2\pi)$ recovers $\Gamma_0(p) = -\ln \Gamma(p)$.

Theorem 7.11 (interpolation of power sums of logarithms). *For $k \geq 1$ and integer $N \geq 0$,*

$$\Gamma_{k-1}(N+1) = -\frac{1}{k} \sum_{j=1}^N \ln^k j.$$

Proof. From $\zeta(s, N+1) = \zeta(s) - \sum_{j=1}^N j^{-s}$, $\zeta^{(k)}(0, N+1) = \zeta^{(k)}(0) - \sum_j (-\ln j)^k = \zeta^{(k)}(0) - (-1)^k \sum_j \ln^k j$. Insert into (7.3) with $n = k-1$. $\square$

Thus Γ_{k-1} is an analytic interpolant of the power sums $\sum \ln^k j$ (up to the factor $-1/k$), exactly as $\ln \Gamma$ interpolates $\ln N!$: the heuristic “differentiate $1^r + \cdots + N^r = \zeta(-r) - \zeta(-r, N+1)$ k times at $r = 0$ ” becomes precise through Theorem 7.10.

Corollary 7.12 (unit translates). *For $k \geq 0$ and integer $n \geq 1$,*

$$\int_n^{n+1} \gamma_k(x) dx = -\frac{\ln^{k+1} n}{k+1}; \quad \text{in particular} \quad \int_1^2 \gamma_k(x) dx = 0.$$

(The case $n = 1$ is eqs. (2.18)–(2.19) of [26].)

7.8 Zeta jets and the conversion dictionary

Making (7.3) explicit requires the constants $\zeta^{(k)}(0)$ and, for the moment evaluations of Section 7.10, the jets at negative integers. All of them follow mechanically from the functional equation $\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s)$ by expanding $\Gamma(1-s) = \exp(\gamma s + \sum_{k \geq 2} \zeta(k) s^k/k)$ and the Laurent series of $\zeta(1-s)$ at the pole.

Proposition 7.13 (jets at 0). *With $L := \ln(2\pi)$,*

$$\begin{aligned}\zeta'(0) &= -\frac{L}{2}, \\ \zeta''(0) &= \frac{\gamma^2}{2} + \gamma_1 - \frac{\pi^2}{24} - \frac{L^2}{2}, \\ \zeta'''(0) &= \gamma^3 + 3\gamma\gamma_1 + \frac{3}{2}\gamma_2 - \zeta(3) + \frac{3}{2}\gamma^2 L + 3\gamma_1 L - \frac{\pi^2}{8} L - \frac{L^3}{2}, \\ \zeta^{(4)}(0) &= \frac{3}{2}\gamma^4 + 4\gamma^3 L + 6\gamma^2\gamma_1 + 3\gamma^2 L^2 + \frac{\pi^2}{4}\gamma^2 + 12\gamma\gamma_1 L + 6\gamma\gamma_2 + 6\gamma_1 L^2 + \frac{\pi^2}{2}\gamma_1 \\ &\quad + 6\gamma_2 L + 2\gamma_3 - \frac{L^4}{2} - \frac{\pi^2}{4} L^2 - 4\zeta(3)L - \frac{19\pi^4}{480}.\end{aligned}$$

Proposition 7.14 (fifth jet at 0). *With $L := \ln(2\pi)$,*

$$\begin{aligned}\zeta^{(5)}(0) &= 2\gamma^5 + \frac{15}{2}\gamma^4 L + \gamma^3 \left(10L^2 + \frac{5\pi^2}{6}\right) + \gamma^2 \left(5L^3 + \frac{5\pi^2}{4} L + 10\zeta(3)\right) \\ &\quad + \gamma\gamma_1 \left(30L^2 + \frac{5\pi^2}{2}\right) + \gamma_1 \left(10\gamma^3 + 30\gamma^2 L + 10L^3 + \frac{5\pi^2}{2} L + 20\zeta(3)\right) \\ &\quad + \gamma_2 \left(15\gamma^2 + 30\gamma L + 15L^2 + \frac{5\pi^2}{4}\right) + \gamma_3 (10\gamma + 10L) + \frac{5}{2}\gamma_4 \\ &\quad - \zeta(3) \left(10L^2 + \frac{5\pi^2}{6}\right) - 12\zeta(5) - \frac{L^5}{2} - \frac{5\pi^2}{12} L^3 - \frac{19\pi^4}{96} L.\end{aligned}$$

The values $\zeta''(0)$ and $\zeta'''(0)$ are classical (Ramanujan [21]; Apostol [19]; Choudhury [24]). Apostol's Theorem 3 [19] in fact gives a closed form of $\zeta^{(n)}(0)$ at every order, in terms of the Laurent coefficients of $\Gamma(s)\zeta(s)$ at $s = 1$, and prints the example $n = 4$ in that vocabulary; we have not located the fully expanded forms above — nor any explicit form of $\zeta^{(5)}(0)$ — in print. The pattern continues to every order by the same expansion, which packages into the following closed recurrence.

Remark 7.15 (explicit recurrence for the jets at 0). In $\zeta(s) = \chi(s)\zeta(1-s)$ with $\chi(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s)$, the simple zero of χ at $s = 0$ cancels the pole of $\zeta(1-s)$; factor it out as $\chi(s) = \frac{s}{2} e^{g(s)}$ with $g(s) := \ln(2\chi(s)/s) = \ln[(2\pi)^s \frac{\sin(\pi s/2)}{\pi s/2} \Gamma(1-s)]$, analytic on $|s| < 1$ and real on $(-1, 1)$ (the bracket is positive there), $g(0) = 0$. From $\ln \Gamma(1-s) = \gamma s + \sum_{k \geq 2} \zeta(k) s^k/k$ and $\ln[\sin(\pi s/2)/(\pi s/2)] = -\sum_{m \geq 1} \zeta(2m)(s/2)^{2m}/m$,

$$g(s) = (\gamma + L)s + \sum_{k \geq 2} c_k s^k, \quad c_k = \frac{\zeta(k)}{k} \cdot \begin{cases} 1, & k \text{ odd}, \\ 1 - 2^{1-k}, & k \text{ even}, \end{cases}$$

so $c_2 = \frac{\pi^2}{24}$, $c_3 = \frac{\zeta(3)}{3}$, $c_4 = \frac{7\pi^4}{2880}$, $c_5 = \frac{\zeta(5)}{5}$. Since $s\zeta(1-s) = -1 + \sum_{m \geq 1} \gamma_{m-1} s^m/(m-1)!$ is entire, the Leibniz rule gives

$$\zeta^{(n)}(0) = \frac{1}{2} \sum_{k=0}^n \binom{n}{k} B_k(1! c_1, 2! c_2, \dots, k! c_k) h_{n-k}, \quad h_0 = -1, \quad h_m = m \gamma_{m-1} \quad (m \geq 1),$$

where $c_1 = \gamma + L$ and B_k is the complete Bell polynomial, $B_k(g'(0), \dots, g^{(k)}(0)) = (e^g)^{(k)}(0)$: $B_0 = 1$, $B_1 = c_1$, $B_2 = c_1^2 + 2c_2$, $B_3 = c_1^3 + 6c_1 c_2 + 6c_3$. The cases $n = 0, 1$ recover $\zeta(0) = -\frac{1}{2}$ and $\zeta'(0) = -\frac{L}{2}$; $n = 2, \dots, 5$ reproduce the two propositions verbatim.

Proposition 7.16 (jets at -1 and -2). *Expanding the functional equation around $s = -1$ (where $\zeta(2 - \varepsilon)$ is regular) and $s = -2$ (a trivial zero) gives, with $L = \ln(2\pi)$:*

$$\zeta(-1) = -\frac{1}{12}, \quad \zeta'(-1) = \frac{\zeta'(2)}{2\pi^2} + \frac{1 - \gamma - L}{12}, \quad (7.4)$$

$$\zeta''(-1) = -\frac{\zeta''(2)}{2\pi^2} + \frac{\zeta'(2)}{\pi^2}(\gamma + L - 1) - \frac{(\gamma + L)^2}{12} + \frac{\gamma + L}{6} + \frac{\pi^2}{144}, \quad (7.5)$$

$$\zeta(-2) = 0, \quad \zeta'(-2) = -\frac{\zeta(3)}{4\pi^2}, \quad \zeta''(-2) = \frac{\zeta(3)}{4\pi^2}(3 - 2\gamma - 2L) + \frac{\zeta'(3)}{2\pi^2}. \quad (7.6)$$

Remark 7.17 (Glaisher spelling). Since $\ln A = \frac{1}{12} - \zeta'(-1)$, eq. (7.4) is equivalent to

$$\zeta'(2) = \zeta(2)(\gamma + \ln 2 + \ln \pi - 12 \ln A),$$

the form returned by Mathematica's `FunctionExpand[Zeta'[2]]`. Either spelling may be used in all the moment formulas below; we keep $\zeta'(2), \zeta''(2), \zeta'(3)$ as the canonical second-layer constants.

7.9 Reflection formulas

Hurwitz's formula (see, e.g., [20, Ch. 12]), for $0 < p/q < 1$,

$$\zeta\left(1 - s, \frac{p}{q}\right) = \frac{2\Gamma(s)}{(2\pi q)^s} \sum_{k=1}^q \left[\cos\left(\frac{\pi s}{2} - \frac{2\pi k p}{q}\right) \right] \zeta\left(s, \frac{k}{q}\right),$$

expanded around $s = 0$, expresses every $\gamma_n(p/q)$ through the jets $\zeta^{(j)}(0, k/q)$, $j \leq n + 1$. Taking the part odd under $p \mapsto q - p$ isolates the sine sums S_j and yields reflection formulas. Write $E(s) := \exp(-\Lambda_q s + \sum_{k \geq 2} (-1)^k \zeta(k) s^k / k) = \sum_i e_i s^i$ for the prefactor expansion $(2\Gamma(s)(2\pi q)^{-s})^2 = \frac{2}{s} E(s)$.

Theorem 7.18 (reflection, general mechanism). *For $n \geq 1$ and $0 < p < q$,*

$$\gamma_n\left(\frac{p}{q}\right) - \gamma_n\left(1 - \frac{p}{q}\right) = 4n! \sum_{i+2l+1+j=n+1} e_i \frac{(-1)^l (\pi/2)^{2l+1}}{(2l+1)!} \frac{S_j(p/q)}{j!}.$$

Explicitly, with $\Lambda = \Lambda_q = \gamma + \ln 2\pi q$ (the letter distinguishes it from the Glaisher constant A):

$$\gamma_2\left(\frac{p}{q}\right) - \gamma_2\left(1 - \frac{p}{q}\right) = 2\pi S_2 - 4\pi \Lambda S_1 + \left(2\Lambda^2 + \frac{\pi^2}{6}\right) \pi S_0, \quad (7.7)$$

$$\gamma_3\left(\frac{p}{q}\right) - \gamma_3\left(1 - \frac{p}{q}\right) = 2\pi S_3 - 6\pi \Lambda S_2 + \left(6\Lambda^2 + \frac{\pi^2}{2}\right) \pi S_1 - \left(2\Lambda^3 + \frac{\pi^2}{2} \Lambda + 4\zeta(3)\right) \pi S_0, \quad (7.8)$$

$$\begin{aligned} \gamma_4\left(\frac{p}{q}\right) - \gamma_4\left(1 - \frac{p}{q}\right) &= 2\pi S_4 - 8\pi \Lambda S_3 + (12\Lambda^2 + \pi^2) \pi S_2 - (8\Lambda^3 + 2\pi^2 \Lambda + 16\zeta(3)) \pi S_1 \\ &\quad + \left(2\Lambda^4 + \pi^2 \Lambda^2 + 16\Lambda \zeta(3) + \frac{19\pi^4}{120}\right) \pi S_0. \end{aligned} \quad (7.9)$$

Proof. Split the cosine, $\cos(\pi s/2 - \theta) = \cos(\pi s/2) \cos \theta + \sin(\pi s/2) \sin \theta$. Under $p \mapsto q - p$ the cosine sums are even and the sine sums odd, so the odd part of $\zeta(1 - s, p/q)$ equals $\frac{2}{s} E(s) \sin(\pi s/2) \sum_j S_j s^j / j!$. By (7.1), $\zeta(1 - s, p/q) = -\frac{1}{s} + \sum_n \gamma_n(p/q) s^n / n!$, whose odd part has s^n -coefficient $\frac{1}{2n!} [\gamma_n(p/q) - \gamma_n(1 - p/q)]$. Matching coefficients of s^n (i.e. s^{n+1} of $E \cdot \sin \cdot \sum S_j s^j / j!$) gives the general formula; instantiating $e_0 = 1$, $e_1 = -\Lambda$, $e_2 = \frac{\Lambda^2}{2} + \frac{\pi^2}{12}$, $e_3 = -\frac{\Lambda^3}{6} - \frac{\pi^2}{12} \Lambda - \frac{\zeta(3)}{3}$, $e_4 = \frac{\Lambda^4}{24} + \frac{\pi^2}{24} \Lambda^2 + \frac{\zeta(3)}{3} \Lambda + \frac{\pi^4}{288} + \frac{\pi^4}{360}$ yields (7.7)–(7.9). $\square$

Formula (7.7) is eq. (89) of Blagouchine [6], where a footnote records that it also appears in unpublished work of D. Connon; (7.8) and (7.9) appear to be new. The same expansion, one order down, recovers Malmsten's reflection formula for γ_1 [6]; an elementary alternative proof of that $n = 1$ case, via Kummer's Fourier expansion of $\ln \Gamma$ rather than Hurwitz's functional equation, is given by Kouba [30]. For $q = 3$ and $q = 4$ the sums collapse ($S_j(\frac{1}{3}) = \frac{\sqrt{3}}{2}[\zeta^{(j)}(0, \frac{1}{3}) - \zeta^{(j)}(0, \frac{2}{3})]$, $S_j(\frac{1}{4}) = \zeta^{(j)}(0, \frac{1}{4}) - \zeta^{(j)}(0, \frac{3}{4})$), so (7.7)–(7.9) can be *inverted* to express the odd jet differences through the Stieltjes differences $\gamma_n(p) - \gamma_n(1 - p)$ — the key to Section 7.11.

7.10 Moments

Theorem 7.19 (Hurwitz transform and variants). *For $m \geq 1$ and $\operatorname{Re} s < 1$,*

$$\int_0^1 x^m \zeta(s, x) dx = - \sum_{j=1}^m \frac{m!}{(m-j+1)!} \frac{\zeta(s-j)}{(s-1)(s-2)\cdots(s-j)}, \quad (7.10)$$

and $\int_0^1 \zeta(s, x) dx = 0$. The same recursion with boundary terms $\zeta(\sigma, \frac{1}{2}) = (2^\sigma - 1)\zeta(\sigma)$ and $\zeta(\sigma, 2) = \zeta(\sigma) - 1$ gives the analogous closed forms for $\int_{1/2}^1$ and $\int_1^2$.

Proof. Since $\zeta(s, x) = -\partial_x \zeta(s-1, x)/(s-1)$, integration by parts gives $I_m(s) = -\zeta(s-1)/(s-1) + \frac{m}{s-1} I_{m-1}(s-1)$ with $I_0 \equiv 0$ (the boundary term at $x = 0$ vanishes for $\operatorname{Re} s < m+1$ since $\zeta(\sigma, x) = x^{-\sigma} + O(1)$); unrolling yields (7.10), which then extends meromorphically in s . $\square$

For the unit interval, (7.10) is Theorem 3.7 of Espinosa–Moll [28]. Expanding (7.10) at $s = 1 + t$ against $\int_0^1 x^m \zeta(1+t, x) dx = \frac{1}{(m+1)t} + M_0 - M_1 t + \frac{M_2}{2} t^2 - \cdots$, where $M_k = \int_0^1 x^m \gamma_k(x) dx$, reads off every Stieltjes moment. The expansion $1/[(t-1)\cdots(t-j+1)] = \frac{(-1)^{j-1}}{(j-1)!} (1 + H_{j-1}t + \frac{H_{j-1}^2 + H_{j-1}^{(2)}}{2} t^2 + \cdots)$ gives the coefficients in closed form:

Theorem 7.20 (general first moment). *For every integer $m \geq 1$,*

$$\int_0^1 x^m \gamma_1(x) dx = \sum_{j=1}^m (-1)^{j-1} \binom{m}{j-1} \left[\frac{\zeta''(1-j)}{2} + H_{j-1} \zeta'(1-j) + \frac{H_{j-1}^2 + H_{j-1}^{(2)}}{2} \zeta(1-j) \right].$$

The γ_n -moment carries the analogous formula with the order- $(n+1)$ jet leading and complete homogeneous symmetric functions of $1, \frac{1}{2}, \dots, \frac{1}{j-1}$ as coefficients.

Combining Theorems 7.19 and 7.20 with the dictionary of Proposition 7.16 produces the following table. Items (7.11)–(7.14) are restated from the corpus that motivated this work; the remaining

evaluations appear to be new.

$$\int_{1/2}^1 \gamma_1(x) dx = \frac{\gamma_1}{2} + \frac{\gamma^2}{4} - \frac{\pi^2}{48} + \frac{\ln^2 2}{2} - \frac{\ln^2 \pi}{4}, \quad (7.11)$$

$$\int_1^2 x \gamma_1(x) dx = 1 + \frac{\gamma_1}{2} + \frac{\gamma^2}{4} - \frac{\pi^2}{48} - \frac{\ln^2 2\pi}{4}, \quad (7.12)$$

$$\int_0^1 x^2 \gamma_1(x) dx = \frac{\gamma_1}{2} + \frac{\gamma^2}{3} - \frac{\pi^2}{36} - \frac{\ln^2 2\pi}{6} + \frac{\gamma}{6} \ln 2\pi - \frac{\zeta'(2)}{\pi^2} (\gamma + \ln 2\pi) + \frac{\zeta''(2)}{2\pi^2}, \quad (7.13)$$

$$\begin{aligned} \int_{1/2}^1 x \gamma_1(x) dx &= \frac{\gamma_1}{2} + \frac{5\gamma^2}{16} - \frac{5\pi^2}{192} + \frac{\ln^2 2}{8} - \frac{\ln 2 \ln \pi}{6} - \frac{3 \ln^2 \pi}{16} + \gamma \left(\frac{\ln 2}{12} + \frac{\ln \pi}{8} \right) \\ &\quad - \frac{\zeta'(2)}{4\pi^2} (3\gamma + 2 \ln 2 + 3 \ln \pi) + \frac{3 \zeta''(2)}{8\pi^2}, \end{aligned} \quad (7.14)$$

$$\int_0^1 x \gamma_1(x) dx = \frac{\zeta''(0)}{2} = \frac{\gamma^2}{4} + \frac{\gamma_1}{2} - \frac{\pi^2}{48} - \frac{\ln^2 2\pi}{4}, \quad (7.15)$$

$$\begin{aligned} \int_0^1 x^3 \gamma_1(x) dx &= \frac{3\gamma^2}{8} + \frac{\gamma_1}{2} - \frac{\pi^2}{32} + \frac{\gamma \ln 2\pi}{4} - \frac{\ln^2 2\pi}{8} - \frac{3(\zeta(3) + 2\zeta'(2))}{4\pi^2} (\gamma + \ln 2\pi) \\ &\quad + \frac{3(\zeta'(3) + \zeta''(2))}{4\pi^2}, \end{aligned} \quad (7.16)$$

$$\int_1^2 x^2 \gamma_1(x) dx = \frac{9}{4} + \frac{3\gamma_1}{2} + \frac{5\gamma^2}{6} - \frac{5\pi^2}{72} + \frac{\gamma \ln 2\pi}{6} - \frac{2 \ln^2 2\pi}{3} - \frac{\zeta'(2)}{\pi^2} (\gamma + \ln 2\pi) + \frac{\zeta''(2)}{2\pi^2}, \quad (7.17)$$

$$\int_0^1 x \gamma_2(x) dx = -\frac{\zeta'''(0)}{3} = \frac{\zeta(3)}{3} - \frac{\gamma^3}{3} - \gamma\gamma_1 - \frac{\gamma_2}{2} - \frac{\gamma^2}{2} \ln 2\pi - \gamma_1 \ln 2\pi + \frac{\pi^2}{24} \ln 2\pi + \frac{\ln^3 2\pi}{6}, \quad (7.18)$$

$$\int_1^2 x \gamma_2(x) dx = \int_0^1 x \gamma_2(x) dx - 2, \quad \int_{1/2}^1 \gamma_2(x) dx = -\Gamma_2\left(\frac{1}{2}\right). \quad (7.19)$$

At $m = 3$ the constants $\zeta(3)$ and $\zeta'(3)$ enter through (7.6), and each further unit of m opens one more column of jets at the next negative integer; the second moment of γ_1 over $[1/2, 1]$ and the higher γ_2 -moments are equally mechanical but bulkier, and we omit them.

7.11 Explicit values of Γ_n

7.11.1 The first antiderivative

Theorem 7.21 (Γ_1 in closed form). *For all $p > 0$,*

$$\Gamma_1(p) = c + \frac{1}{2} \zeta''(0, p), \quad c := -\frac{1}{2} \zeta''(0) = \frac{\pi^2}{48} + \frac{\ln^2 2\pi}{4} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2}.$$

Moreover $\int_0^1 \Gamma_1(x) dx = c$, equivalently (7.15) (since $\int_0^1 \zeta''(0, x) dx = 0$ [9, 25]).

Theorem 7.22 (half-integral and even-sum values).

$$\begin{aligned}\Gamma_1\left(\frac{1}{2}\right) &= \frac{\pi^2}{48} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2} - \frac{\ln^2 2}{4} - \frac{\ln 2 \ln 2\pi}{2} + \frac{\ln^2 2\pi}{4}, \\ \Gamma_1\left(\frac{1}{3}\right) + \Gamma_1\left(\frac{2}{3}\right) &= 2c - \frac{\ln^2 3}{4} - \frac{\ln 3 \ln 2\pi}{2}, \quad \Gamma_1\left(\frac{1}{4}\right) + \Gamma_1\left(\frac{3}{4}\right) = 2c - \frac{3\ln^2 2}{4} - \frac{\ln 2 \ln 2\pi}{2}, \\ \Gamma_1\left(\frac{1}{6}\right) + \Gamma_1\left(\frac{5}{6}\right) &= 2c - \frac{\ln 2 \ln 3}{2}.\end{aligned}$$

Proof. $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$ gives $\zeta''(0, \frac{1}{2}) = -\frac{1}{2}\ln^2 2 - \ln 2 \ln(2\pi)$; the multiplication identities $\zeta(s, \frac{1}{3}) + \zeta(s, \frac{2}{3}) = (3^s - 1)\zeta(s)$, $\zeta(s, \frac{1}{4}) + \zeta(s, \frac{3}{4}) = (4^s - 2^s)\zeta(s)$, $\zeta(s, \frac{1}{6}) + \zeta(s, \frac{5}{6}) = (6^s - 3^s - 2^s + 1)\zeta(s)$, twice differentiated at $s = 0$, give the even sums (for $q = 6$ the $\zeta''(0)$ -coefficient $\ln^2 6 - \ln^2 3 - \ln^2 2 = 2\ln 2 \ln 3$ and the $\zeta'(0)$ -coefficient $\ln 6 - \ln 3 - \ln 2 = 0$ produce the strikingly short third identity). Apply Theorem 7.21. $\square$

Theorem 7.23 (third and quarter values).

$$\begin{aligned}\Gamma_1\left(\frac{1}{3}\right) &= \frac{\pi^2}{72} - \frac{\ln^2 2\pi}{3} + \frac{\ln^2 3}{24} - \frac{2}{3} \ln 3 \ln 2\pi - \frac{\gamma}{12} (4\gamma + 8\ln 2\pi - \ln 3) + (\gamma + \ln 6\pi) \ln \Gamma\left(\frac{1}{3}\right) \\ &\quad - \frac{\gamma_1}{2} + \frac{\sqrt{3}}{12\pi} \left(\gamma_2\left(\frac{1}{3}\right) - \gamma_2\left(\frac{2}{3}\right) \right), \\ \Gamma_1\left(\frac{1}{4}\right) &= \frac{\pi^2}{96} - \frac{3}{8} \ln^2 \pi - \frac{9}{4} \ln 2 \ln 2\pi - \frac{\gamma}{8} (3\gamma + 8\ln 2 + 6\ln \pi) + (\gamma + \ln 8\pi) \ln \Gamma\left(\frac{1}{4}\right) \\ &\quad - \frac{\gamma_1}{2} + \frac{1}{8\pi} \left(\gamma_2\left(\frac{1}{4}\right) - \gamma_2\left(\frac{3}{4}\right) \right),\end{aligned}$$

and $\Gamma_1(\frac{2}{3})$, $\Gamma_1(\frac{3}{4})$ follow from Theorem 7.22. Furthermore, with the abbreviation $D_j := \zeta^{(j)}(0, \frac{1}{3}) - \zeta^{(j)}(0, \frac{2}{3})$,

$$\Gamma_1\left(\frac{1}{6}\right) - \Gamma_1\left(\frac{5}{6}\right) = \frac{\ln 2 \ln 18}{6} + \ln 2 \left(2\ln \Gamma\left(\frac{1}{3}\right) + \frac{\ln 3}{6} - \ln 2 - \ln \pi \right) + D_2,$$

which together with the even sum of Theorem 7.22 gives $\Gamma_1(\frac{1}{6})$ and $\Gamma_1(\frac{5}{6})$ explicitly — in the same obstruction atom $\gamma_2(\frac{1}{3}) - \gamma_2(\frac{2}{3})$, since inverting (7.7) at $q = 3$ expresses $D_2 = \frac{2}{\sqrt{3}}S_2(\frac{1}{3})$ through it.

Proof. By Theorem 7.21 everything reduces to the single jets $\zeta''(0, p)$ at $p = \frac{1}{3}, \frac{1}{4}, \frac{1}{6}$. Their even parts are in Theorem 7.22; their odd parts are the inversions of (7.7) at $q = 3, 4$:

$$S_2 = \frac{1}{2\pi} \left[\gamma_2\left(\frac{p}{q}\right) - \gamma_2\left(1 - \frac{p}{q}\right) + 4\pi\Lambda S_1 - \left(2\Lambda^2 + \frac{\pi^2}{6}\right)\pi S_0 \right],$$

with $S_1(\frac{1}{3}) = \frac{\sqrt{3}}{2}(2\ln \Gamma(\frac{1}{3}) - \ln \frac{2\pi}{\sqrt{3}})$, $S_0(\frac{1}{3}) = \frac{1}{2\sqrt{3}}$, $S_1(\frac{1}{4}) = 2\ln \Gamma(\frac{1}{4}) - \ln(\pi\sqrt{2})$, $S_0(\frac{1}{4}) = \frac{1}{2}$. For $q = 6$, the Dirichlet character χ_{-3} gives $\zeta(s, \frac{1}{6}) - \zeta(s, \frac{5}{6}) = (6^s + 3^s)L(s, \chi_{-3})$ while $\zeta(s, \frac{1}{3}) - \zeta(s, \frac{2}{3}) = 3^s L(s, \chi_{-3})$; eliminating the jets of $L(s, \chi_{-3})$ at $s = 0$ ($L(0, \chi_{-3}) = \frac{1}{3}$, $L'(0, \chi_{-3}) = 2\ln \Gamma(\frac{1}{3}) + \frac{\ln 3}{6} - \ln 2 - \ln \pi$) yields the displayed transport identity. $\square$

7.11.2 The second and third antiderivatives

Theorem 7.24 (Γ_2 values).

$$\Gamma_2\left(\frac{1}{2}\right) = \frac{\gamma_2}{2} + \frac{\gamma^3}{3} - \frac{\zeta(3)}{3} + \ln^3 2 - \frac{\ln^3 \pi}{6} + \frac{\ln \pi}{24} (12\gamma^2 - \pi^2 + 24\ln^2 2) + (\gamma + \ln \pi) \gamma_1,$$

and at the third and quarter points, with $Z_3(\frac{1}{q}) := \zeta'''(0, \frac{1}{q}) + \zeta'''(0, 1 - \frac{1}{q})$ and $D_3(\frac{1}{q}) := \zeta'''(0, \frac{1}{q}) - \zeta'''(0, 1 - \frac{1}{q})$,

$$\Gamma_2(\frac{1}{q}) = -\frac{1}{3} \left[\frac{Z_3 + D_3}{2} - \zeta'''(0) \right], \quad q \in \{3, 4\},$$

where the even parts are elementary,

$$Z_3(\frac{1}{3}) = -\frac{\ln^3 3}{2} - \frac{3 \ln^2 3 \ln 2\pi}{2} + 3 \ln 3 \zeta''(0), \quad Z_3(\frac{1}{4}) = \partial_s^3 [(4^s - 2^s) \zeta(s)]_{s=0},$$

and the odd parts are the inversions of (7.8): $D_3 = \frac{2}{\sqrt{3}} S_3$ at $q = 3$, $D_3 = S_3$ at $q = 4$, with

$$S_3 = \frac{1}{2\pi} \left[\gamma_3(\frac{1}{q}) - \gamma_3(1 - \frac{1}{q}) + 6\pi\Lambda S_2 - \left(6\Lambda^2 + \frac{\pi^2}{2}\right)\pi S_1 + \left(2\Lambda^3 + \frac{\pi^2}{2}\Lambda + 4\zeta(3)\right)\pi S_0 \right]$$

and S_2 as in the proof of Theorem 7.23. Thus $\Gamma_2(\frac{1}{3})$ and $\Gamma_2(\frac{1}{4})$ are explicit in the constants $\{\gamma, \gamma_1, \gamma_2, \zeta(3), \ln 2, \ln 3, \ln \pi, \ln \Gamma(\frac{1}{q}), \gamma_2(\frac{1}{q}) - \gamma_2(1 - \frac{1}{q}), \gamma_3(\frac{1}{q}) - \gamma_3(1 - \frac{1}{q})\}$, the two Stieltjes differences being the only non-classical ingredients.

Proof. $\Gamma_2(p) = -\frac{1}{3}[\zeta'''(0, p) - \zeta'''(0)]$ by Theorem 7.10. At $p = \frac{1}{2}$ use $\zeta'''(0, \frac{1}{2}) = \partial_s^3 [(2^s - 1)\zeta(s)]_{s=0} = \ln^3 2 \zeta(0) + 3 \ln^2 2 \zeta'(0) + 3 \ln 2 \zeta''(0)$ and Proposition 7.13; at $p = \frac{1}{3}, \frac{1}{4}$ split into even and odd parts as displayed. $\square$

Corollary 7.25 (Γ_2 even sums). *With $L = \ln 2\pi$,*

$$\begin{aligned} \Gamma_2(\frac{1}{3}) + \Gamma_2(\frac{2}{3}) &= \frac{2}{3} \zeta'''(0) - \ln 3 \zeta''(0) + \frac{L}{2} \ln^2 3 + \frac{\ln^3 3}{6}, \\ \Gamma_2(\frac{1}{4}) + \Gamma_2(\frac{3}{4}) &= \frac{2}{3} \zeta'''(0) - \ln 2 \zeta''(0) + \frac{3L}{2} \ln^2 2 + \frac{7 \ln^3 2}{6}, \\ \Gamma_2(\frac{1}{6}) + \Gamma_2(\frac{5}{6}) &= \frac{2}{3} \zeta'''(0) + L \ln 2 \ln 3 + \frac{\ln 2 \ln 3 \ln 6}{2}, \end{aligned}$$

so that $\Gamma_2(\frac{2}{3}), \Gamma_2(\frac{3}{4})$ follow from Theorem 7.24; as in Theorem 7.22, the $\zeta''(0)$ -term vanishes at $q = 6$ because $\ln 6 - \ln 3 - \ln 2 = 0$. (Generally, with the inclusion-exclusion combination $\zeta(s, \frac{1}{q}) + \zeta(s, 1 - \frac{1}{q}) = (\sum_i c_i m_i^s) \zeta(s)$, the even sum equals $\frac{2}{3} \zeta'''(0) + \frac{1}{6} \sum_i c_i \ln^3 m_i + \frac{L}{2} \sum_i c_i \ln^2 m_i - \zeta''(0) \sum_i c_i \ln m_i$.)

Theorem 7.26 (Γ_3 values).

$$\begin{aligned} \Gamma_3(\frac{1}{2}) &= -\frac{7 \ln^4 2}{4} + \frac{\ln^4 \pi}{8} - \frac{3\gamma^4}{8} - \frac{\gamma_3}{2} + \frac{19\pi^4}{1920} + \gamma_1 \left(3 \ln^2 2 - \frac{3\gamma^2}{2} - \frac{3 \ln^2 \pi}{2} - \frac{\pi^2}{8} \right) \\ &\quad - \frac{3\gamma_2}{2} (\gamma + \ln \pi) + \zeta(3) \ln \pi + \gamma^2 \left(\frac{3 \ln^2 2}{2} - \frac{3 \ln^2 \pi}{4} - \frac{\pi^2}{16} \right) - \gamma^3 \ln \pi - 3\gamma\gamma_1 \ln \pi \\ &\quad - 3 \ln \pi \ln^3 2 - \frac{3 \ln^2 2 \ln^2 \pi}{2} - \frac{\pi^2 \ln^2 2}{8} + \frac{\pi^2 \ln^2 \pi}{16}, \end{aligned}$$

and $\Gamma_3(\frac{1}{3}), \Gamma_3(\frac{1}{4})$ are given by $\Gamma_3(\frac{1}{q}) = \frac{1}{4}[\zeta^{(4)}(0, \frac{1}{q}) - \zeta^{(4)}(0)]$ with the even part $\partial_s^4 [(q\text{-orbit}) \zeta(s)]_{s=0}$ as in Theorem 7.24 and the odd part inverted from (7.9); the resulting expressions are explicit in the same vocabulary augmented by $\gamma_4(\frac{1}{q}) - \gamma_4(1 - \frac{1}{q})$.

7.11.3 The fourth antiderivative at $1/2$

Theorem 7.27 ($\Gamma_4(1/2)$).

$$\begin{aligned}\Gamma_4\left(\frac{1}{2}\right) &= \frac{\gamma_4}{2} + 3 \ln^5 2 - \frac{12 \zeta(5)}{5} - \frac{\ln^5 \pi}{10} + \frac{2\gamma^5}{5} + \frac{3\gamma^4 \ln \pi}{2} \\ &\quad + \gamma_1 \left(2\gamma^3 - 12 \ln^3 2 + 2 \ln^3 \pi + 4\zeta(3) + \frac{\pi^2 \ln \pi}{2} - 12 \ln \pi \ln^2 2 + 6\gamma^2 \ln \pi \right) \\ &\quad + \gamma\gamma_1 \left(\frac{\pi^2}{2} - 12 \ln^2 2 + 6 \ln^2 \pi \right) + \gamma_2 \left(3\gamma^2 - 6 \ln^2 2 + 3 \ln^2 \pi + \frac{\pi^2}{4} + 6\gamma \ln \pi \right) \\ &\quad + 2\gamma_3 (\gamma + \ln \pi) + \zeta(3) \left(4 \ln^2 2 - 2 \ln^2 \pi - \frac{\pi^2}{6} \right) \\ &\quad + \gamma^2 \left(\ln^3 \pi - 6 \ln^3 2 + 2\zeta(3) - 6 \ln \pi \ln^2 2 + \frac{\pi^2 \ln \pi}{4} \right) + \gamma^3 \left(2 \ln^2 \pi - 4 \ln^2 2 + \frac{\pi^2}{6} \right) \\ &\quad + \frac{\pi^2 \ln^3 2}{2} + 2 \ln^2 2 \ln^3 \pi + 6 \ln^3 2 \ln^2 \pi + 7 \ln \pi \ln^4 2 \\ &\quad - \frac{19\pi^4 \ln \pi}{480} - \frac{\pi^2 \ln^3 \pi}{12} + \frac{\pi^2 \ln \pi \ln^2 2}{2}.\end{aligned}$$

Proof. $\Gamma_4(1/2) = -\frac{1}{5}[\zeta^{(5)}(0, \frac{1}{2}) - \zeta^{(5)}(0)]$, with

$$\zeta^{(5)}\left(0, \frac{1}{2}\right) = \partial_s^5 [(2^s - 1)\zeta(s)]_{s=0} = \sum_{k=0}^4 \binom{5}{k} \ln^{5-k} 2 \zeta^{(k)}(0)$$

(the $k = 5$ term carries the factor $2^0 - 1 = 0$) and Proposition 7.14. $\square$

The pattern of Theorems 7.24–7.27 is general: each rung Γ_n adds the next Stieltjes constant γ_n and the functional-equation zeta terms, and one degree to the logarithmic monomials at $p = \frac{1}{2}$; at $p = \frac{1}{3}, \frac{1}{4}$ it adds the single new obstruction atom $\gamma_{n+1}(p) - \gamma_{n+1}(1-p)$, delivered by the order- $(n+1)$ reflection formula of Theorem 7.18. At denominators such as 5, 8, 12 the even character decomposition introduces additional coordinates (the multiplication identities no longer close over the principal character), and the values $\Gamma_n(p/q)$ additionally involve the even-character combinations of $\zeta^{(n+1)}(0, \cdot)$ — the obstruction layer studied, at $n = 1$, in [6].

7.12 The negative-order polygamma floor

The ladder of Section 7.7 integrates the coefficient functions of the Laurent expansion at $s = 1$ in their argument. The same Hurwitz-jet calculus has a second, equally natural direction, and this section — a second main act rather than an appendix — follows it downward. Integrating in the *argument* at the s -jet floor produces the polygamma functions of negative integer order (“negapolygamma”, a term Adamchik [18] credits to Gosper): for real $z > 0$,

$$\psi^{(-1)}(z) = \ln \Gamma(z), \quad \psi^{(-n)}(z) = \int_0^z \psi^{(-(n-1))}(t) dt = \frac{1}{(n-2)!} \int_0^z (z-t)^{n-2} \ln \Gamma(t) dt,$$

which is precisely Mathematica’s `PolyGamma[-n, z]` (all identities below are stated on $0 < z < 1$, where the rational arguments live, and extend to $z > 0$ by the same formulas wherever both sides are defined). Everything built for the Stieltjes side transfers wholesale: the ladder identity has a one-line analogue, the reflection mechanism of Section 7.9 reappears at the jet floor with bare

Clausen values in place of the sine sums S_j , the obstruction constants again organize along Dirichlet characters — and the bridge identities for those constants close a circle the Stieltjes side leaves open. The tools: a one-line ladder, reflection and multiplication laws, bridge identities, and explicit rational-argument values.

7.12.1 The ladder

Theorem 7.28. *For $n \geq 2$ and $z > 0$,*

$$\psi^{(-n)}(z) = \frac{\zeta'(1-n, z) - \zeta'(1-n)}{(n-1)!} + Q_n(z),$$

where Q_n is an explicit polynomial with coefficients in $\mathbb{Q}[\ln 2\pi, \zeta'(-1), \dots, \zeta'(2-n)]$:

$$\begin{aligned} Q_2(z) &= \frac{z}{2} \ln 2\pi + \frac{z(1-z)}{2}, \\ Q_3(z) &= \frac{z^2}{4} \ln 2\pi - \frac{z^3}{4} + \frac{3z^2}{8} - \frac{z}{24} - z\zeta'(-1), \\ Q_4(z) &= \frac{z^3}{12} \ln 2\pi - \frac{11z^4}{144} + \frac{11z^3}{72} - \frac{5z^2}{144} - \frac{z^2}{2}\zeta'(-1) - \frac{z}{2}\zeta'(-2), \end{aligned}$$

and in general Q_n has leading term $-H_{n-1}z^n/n!$ and $\ln 2\pi$ -coefficient $z^{n-1}/(2(n-1)!)$.

Proof. $\int_0^z \zeta(s, t) dt = -[\zeta(s-1, z) - \zeta(s-1)]/(s-1)$ for $\operatorname{Re} s < 1$; differentiating at $s = -k$,

$$\int_0^z \zeta'(-k, t) dt = \frac{\zeta'(-k-1, z) - \zeta'(-k-1)}{k+1} + \frac{\zeta(-k-1, z) - \zeta(-k-1)}{(k+1)^2},$$

with $\zeta(-m, z) = -B_{m+1}(z)/(m+1)$. Iterate from $\psi^{(-1)} = \zeta'(0, z) + \frac{1}{2} \ln 2\pi$: each step shifts the single ζ' -atom by one and deposits Bernoulli and constant terms into the polynomial. $\square$

An equivalent expansion is Adamchik's Proposition 2 [18]; the spelling above isolates the one z -dependent transcendental.

7.12.2 Reflection

Theorem 7.29 (reflection at the jet floor). *For every integer $k \geq 0$ and $0 < z < 1$,*

$$\zeta'(-k, z) + (-1)^k \zeta'(-k, 1-z) = (-1)^{\lfloor k/2 \rfloor} \frac{k!}{(2\pi)^k} \operatorname{Cl}_{k+1}(2\pi z),$$

where $\operatorname{Cl}_{2m}(\theta) = \operatorname{Im} \operatorname{Li}_{2m}(e^{i\theta})$, $\operatorname{Cl}_{2m+1}(\theta) = \operatorname{Re} \operatorname{Li}_{2m+1}(e^{i\theta})$.

Proof. In Hurwitz's formula, written as $\zeta(1-s, z) = \frac{2\Gamma(s)}{(2\pi)^s} [\cos(\frac{\pi s}{2})C_s(z) + \sin(\frac{\pi s}{2})S_s(z)]$ with C_s, S_s the cosine/sine Dirichlet series, the combination even (odd) under $z \mapsto 1-z$ keeps only the C (S) term. At the integer $s = k+1$, exactly one of $\cos \frac{\pi s}{2}$, $\sin \frac{\pi s}{2}$ vanishes, and it is the one attached to the parity-matching combination; differentiating in s there, the product rule retains only the term in which d/ds falls on the vanishing trigonometric factor, so a single *bare* Clausen value survives and no derivative of the Dirichlet series enters. Collecting constants gives the display. $\square$

For $k \geq 1$ this is the real part of Adamchik's Proposition 1 [18], stated there for every positive integer order with the Bernoulli term as the imaginary part (equivalently Miller–Adamchik [31], eq. (21)); the bare-Clausen real spelling, the $k = 0$ inclusion, and the mechanism note are the reformulation here. $k = 0$ is Kummer-flavoured ($\operatorname{Cl}_1(2\pi z) = -\ln(2 \sin \pi z)$). The parity-*opposite*

combination does not close: it produces the series $\sum \ln n \{\cos | \sin\}(2\pi n z)/n^{k+1}$ — the generalized-Glaisher atoms of §7.12.4.

Propagating Theorem 7.29 through the ladder yields reflection formulas at the $\psi^{(-n)}$ -level — a relation which Espinosa and Moll [27] record as not known to them (*"We have been unable to find a generalization of the other well-known functional relation for the polygamma function"*):

Theorem 7.30 (negapolygamma reflection).

$$\begin{aligned}
\psi^{(-2)}(z) - \psi^{(-2)}(1-z) &= \frac{\text{Cl}_2(2\pi z)}{2\pi} + \left(z - \frac{1}{2}\right) \ln 2\pi, \\
\psi^{(-3)}(z) + \psi^{(-3)}(1-z) &= -\frac{\text{Cl}_3(2\pi z)}{4\pi^2} + \left(\frac{z^2}{2} - \frac{z}{2} + \frac{1}{4}\right) \ln 2\pi - \zeta'(-1) - \zeta'(-2) + \frac{1}{12}, \\
\psi^{(-4)}(z) - \psi^{(-4)}(1-z) &= -\frac{\text{Cl}_4(2\pi z)}{8\pi^3} + \left(\frac{z^3}{6} - \frac{z^2}{4} + \frac{z}{4} - \frac{1}{12}\right) \ln 2\pi \\
&\quad + \left(\frac{1}{2} - z\right) (\zeta'(-1) + \zeta'(-2)) + \frac{z}{12} - \frac{1}{24}, \\
\psi^{(-5)}(z) + \psi^{(-5)}(1-z) &= \frac{\text{Cl}_5(2\pi z)}{16\pi^4} + \left(\frac{z^4}{24} - \frac{z^3}{12} + \frac{z^2}{8} - \frac{z}{12} + \frac{1}{48}\right) \ln 2\pi \\
&\quad + \left(\frac{z-z^2}{2}\right) (\zeta'(-1) + \zeta'(-2)) \\
&\quad - \frac{\zeta'(-1)}{6} - \frac{\zeta'(-2)}{4} - \frac{\zeta'(-3)}{6} - \frac{\zeta'(-4)}{12} + \frac{z^2-z}{24} + \frac{49}{4320}.
\end{aligned}$$

7.12.3 Multiplication

Differentiating the Hurwitz distribution relation at $s = -k$ gives, for all $k \geq 0$, $m \geq 2$,

$$\sum_{j=0}^{m-1} \zeta' \left(-k, z + \frac{j}{m} \right) = m^{-k} \left[\zeta'(-k, mz) - \ln m \frac{B_{k+1}(mz)}{k+1} \right] \quad (7.20)$$

($k = 0$ is Gauss's multiplication theorem for $\ln \Gamma$), and through the ladder,

$$\sum_{j=0}^{m-1} \psi^{(-n)} \left(z + \frac{j}{m} \right) = m^{1-n} \left[\psi^{(-n)}(mz) - \ln m \frac{B_n(mz)}{n!} \right] + \rho_{n,m}(z)$$

with elementary $\rho_{n,m}$; for $n = 2$,

$$\rho_{2,m}(z) = \frac{(m-1)(2z+1)}{4} \ln 2\pi + \frac{m^2-1}{12m} - \frac{m^2-1}{m} \zeta'(-1),$$

so only the *rational* part of the remainder is $(m^2-1)/(12m)$. The ψ -level law is, in the "balanced" convention of their negapolygamma [29], Theorem 3.2 of Espinosa–Moll [27]; the form above is its transcription to the Wolfram convention.

7.12.4 Bridges

At rational arguments the layers $\zeta'(-k, p/q)$ decompose along Dirichlet characters, and two laws organize the resulting constants.

Proposition 7.31 (trivial-zero bridges). *Let χ be primitive or principal, $k \geq 1$. If $\chi(-1) = (-1)^k$ (so that $L(-k, \chi) = 0$, since $B_{k+1}(\chi) \neq 0$ exactly when $\chi(-1) = (-1)^{k+1}$), then $L'(-k, \chi)$ is an elementary multiple of $L(k+1, \bar{\chi})$. For the principal character and χ_{-4} :*

$$\zeta'(-2k) = \frac{(-1)^k (2k)!}{2(2\pi)^{2k}} \zeta(2k+1), \quad \beta'(1-2k) = (-1)^{k+1} (2k-1)! \frac{2^{2k-1}}{\pi^{2k-1}} \beta(2k),$$

so in particular $\beta'(-1) = 2G/\pi$ (G Catalan's constant) and $\beta'(-3) = -48\beta(4)/\pi^3$.

Theorem 7.32 (harmonic-number bridge). *Let χ be a primitive character mod q with $\chi(-1) = (-1)^{k+1}$ (so that $L(-k, \chi) \neq 0$; for χ principal read ζ and require k odd). Then*

$$\frac{L'(k+1, \chi)}{L(k+1, \chi)} + \overline{\left(\frac{L'(-k, \chi)}{L(-k, \chi)} \right)} = \gamma + \ln \frac{2\pi}{q} - H_k.$$

For real χ the conjugation bar is invisible; for complex χ it is essential — equivalently, the bridge pairs χ at $s = k+1$ with $\bar{\chi}$ at $s = -k$, and the un-conjugated sum has the purely imaginary defect $2i \operatorname{Im}[L'(-k, \chi)/L(-k, \chi)]$.

Proof. Write the completed function $\Lambda(s) = (q/\pi)^{(s+\delta)/2} \Gamma(\frac{s+\delta}{2}) L(s, \chi)$ with $\delta = (1 - \chi(-1))/2$, so $\Lambda(s) = \varepsilon \bar{\Lambda}(1-s)$, whose logarithmic derivative at real s gives $\Lambda'/\Lambda(s) = -\bar{\Lambda}'/\bar{\Lambda}(1-s)$; at $s = k+1$,

$$\frac{L'(k+1, \chi)}{L(k+1, \chi)} + \overline{\left(\frac{L'(-k, \chi)}{L(-k, \chi)} \right)} = -\ln \frac{q}{\pi} - \frac{1}{2} \psi\left(\frac{k+1+\delta}{2}\right) - \frac{1}{2} \psi\left(\frac{\delta-k}{2}\right).$$

Since $\chi(-1) = (-1)^{k+1}$ means $\delta \equiv k+1 \pmod{2}$, the argument $(\delta-k)/2$ is a (possibly negative) half-integer, where $\cot \pi x = 0$; the reflection formula thus gives $\psi(\frac{\delta-k}{2}) = \psi(\frac{k+2-\delta}{2})$, and $\{\frac{k+1+\delta}{2}, \frac{k+2-\delta}{2}\}$ is a duplication pair $\{x, x+\frac{1}{2}\}$ with $2x = k+1$. Hence the two digammas sum to $2\psi(k+1) - 2\ln 2 = 2(H_k - \gamma) - 2\ln 2$, and the display follows. $\square$

Consequently, at each level $-k$ every character direction of the grid $\{\zeta'(-k, j/q)\}$ either reduces (trivial-zero bridge: to the $\zeta(k+1)$ /Catalan-family values) or carries exactly one new constant (the parity-opposite L'), the harmonic-number bridge converting freely between its $s = k+1$ and $s = -k$ spellings. Exact rational row-reduction of the full known-relation lattice (all reflection rows and all-divisor multiplication rows) makes this quantitative:

Proposition 7.33 (jet-grid residual dimension). *For every $3 \leq q \leq 30$ and $1 \leq k \leq 4$, the residual dimension of the grid $\{\zeta'(-k, j/q) : 1 \leq j \leq q-1\}$ modulo the lattice generated by the reflection and multiplication rows equals*

$$r_{q,k} = \#\{\chi \text{ primitive of conductor } f \mid q, f > 1, \chi(-1) = (-1)^{k+1}\} = \begin{cases} \varphi(q)/2 - 1, & k \text{ odd,} \\ \varphi(q)/2, & k \text{ even,} \end{cases}$$

with no exceptions.

The epistemic status deserves precision. For each fixed (q, k) in the stated range this *is* a theorem: the relation lattice is a finite, explicitly known matrix over $\mathbb{Q}$, and the rank computation is exact rational row reduction (no floating point; the script and its 112-cell output belong to the historical Smithereens snapshot and were not imported here). The character-count *law*, as a statement for all q , is at present a verified conjecture, with the expected mechanism visible in the computation: decomposing the grid along Dirichlet characters, the reflection rows annihilate

exactly the $(-1)^k$ -parity eigenspace (whose L -jets are the Clausen material already on the reflection right-hand side), the divisor multiplication rows pin the imprimitive and principal directions (the constant $\zeta'(-k)$ belongs to the right-hand-side inventory: a generalized Glaisher constant $\ln A_k$ for k odd, a multiple of $\zeta(k+1)$ for k even), and the surviving directions are precisely the parity-opposite primitive characters of Theorem 7.32. For k odd the count coincides with the residual law of the γ_1 -grids — the two lattices see the same even-character obstruction space.

7.12.5 Rational values

Theorem 7.34 (the half-argument tower). *For every $n \geq 2$,*

$$\psi^{(-n)}\left(\frac{1}{2}\right) = \frac{(2^{1-n} - 2) \zeta'(1-n) - 2^{1-n} \ln 2 \cdot B_n/n}{(n-1)!} + Q_n\left(\frac{1}{2}\right);$$

in lowest cases

$$\psi^{(-2)}\left(\frac{1}{2}\right) = \frac{3}{2} \ln A + \frac{5 \ln 2}{24} + \frac{\ln \pi}{4} \quad (\text{classical}), \quad \psi^{(-3)}\left(\frac{1}{2}\right) = \frac{7 \zeta(3)}{32\pi^2} + \frac{\ln A}{2} + \frac{\ln 2\pi}{16}.$$

Theorem 7.35 (third- and quarter-argument values at $n = 3, 4$).

$$\begin{aligned} \psi^{(-3)}\left(\frac{1}{4}\right) &= \frac{35 \zeta(3)}{256\pi^2} + \frac{\beta'(3)}{4\pi^3} + \frac{\ln A}{4} + \frac{\ln 2\pi}{128} - \frac{\gamma}{128}, \\ \psi^{(-3)}\left(\frac{1}{3}\right) &= \frac{13 \zeta(3)}{72\pi^2} + \frac{\sqrt{3} L'(3, \chi_{-3})}{8\pi^3} + \frac{\ln A}{3} + \frac{7 \ln 2\pi}{324} - \frac{\gamma}{162}, \end{aligned}$$

via the exact spellings $\beta'(-2) = \frac{1}{4}(3 - 2\gamma - 2 \ln \frac{\pi}{2}) + 16\beta'(3)/\pi^3$ and $L'(-2, \chi_{-3}) = \frac{2}{9}(\ln \frac{3}{2\pi} + \frac{3}{2} - \gamma) + \frac{9\sqrt{3}}{2\pi^3} L'(3, \chi_{-3})$; and at the next level the parity alternates, so that $\psi^{(-4)}(\frac{1}{4})$ is explicit with no new constant:

$$\psi^{(-4)}\left(\frac{1}{4}\right) = \frac{\zeta'(-3, \frac{1}{4}) - \zeta'(-3)}{6} + Q_4\left(\frac{1}{4}\right), \quad \zeta'(-3, \frac{1}{4}) = \frac{Z + D}{2}$$

with the even part $Z = (4^{-3} - 2^{-3})\zeta'(-3) + (4^{-3} \ln 4 - 2^{-3} \ln 2)\zeta(-3)$ and the odd part $D = -48\beta(4)/(64\pi^3)$ from Proposition 7.31.

Remark 7.36. $\psi^{(-3)}(\frac{1}{2})$ appears in Adamchik [18]; the flattened quarter- and third-values appear to be new — notably, Mathematica's `FunctionExpand` reduces `PolyGamma[-3, 1/2]` fully but leaves the jet $\zeta'(-2, \frac{1}{4})$ unevaluated inside `PolyGamma[-3, 1/4]`: the atom $\beta'(3)$ is exactly what is missing. The same atom resolves a naive integer-relation search over $\{\zeta(3)/\pi^2, \ln A, \log s\}$, which must fail for this value.

Corollary 7.37 (reflection and transport companions). *With the same atoms as Theorem 7.35,*

$$\begin{aligned} \psi^{(-3)}\left(\frac{2}{3}\right) &= \frac{13 \zeta(3)}{72\pi^2} - \frac{\sqrt{3} L'(3, \chi_{-3})}{8\pi^3} + \frac{2 \ln A}{3} + \frac{19 \ln 2\pi}{162} + \frac{\gamma}{162}, \\ \psi^{(-3)}\left(\frac{3}{4}\right) &= \frac{35 \zeta(3)}{256\pi^2} - \frac{\beta'(3)}{4\pi^3} + \frac{3 \ln A}{4} + \frac{19 \ln 2\pi}{128} + \frac{\gamma}{128}, \\ \psi^{(-3)}\left(\frac{1}{6}\right) &= \frac{\zeta(3)}{12\pi^2} + \frac{5\sqrt{3} L'(3, \chi_{-3})}{32\pi^3} + \frac{\ln A}{6} - \frac{\ln 2\pi}{1296} - \frac{\ln 2}{648} - \frac{5\gamma}{648}, \\ \psi^{(-3)}\left(\frac{5}{6}\right) &= \frac{\zeta(3)}{12\pi^2} - \frac{5\sqrt{3} L'(3, \chi_{-3})}{32\pi^3} + \frac{5 \ln A}{6} + \frac{235 \ln 2\pi}{1296} + \frac{\ln 2}{648} + \frac{5\gamma}{648}. \end{aligned}$$

Proof. The first two from Theorem 7.30 with $\text{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$ and $\text{Cl}_3(\pi/2) = -\frac{3}{32}\zeta(3)$; the sixths from the transport $\zeta(s, \frac{1}{6}) - \zeta(s, \frac{5}{6}) = (6^s + 3^s)L(s, \chi_{-3})$ differentiated at $s = -2$, the even part being $\zeta'(-2, \frac{1}{6}) + \zeta'(-2, \frac{5}{6}) = \frac{2}{3}\zeta'(-2)$ by inclusion–exclusion. The entire sixth family thus rides on the one atom $L'(3, \chi_{-3})$. $\square$

7.12.6 Denominators 5, 8, 12: quadratic and quartic atoms

The denominators 5, 8, 12 are the first where Proposition 7.33 forces genuinely new constants beyond the χ_{-3}/χ_{-4} families: at $k = 1$ each grid carries one new atom (the even real quadratic direction), and at $k = 2$, $q = 5$ the residual space is two-dimensional (a conjugate quartic pair).

Theorem 7.38 (the $k = 1$ grids). *Write $Z_1 = \zeta'(-1)$ and let χ_q be the even real quadratic primitive character mod $q \in \{5, 8, 12\}$ (Legendre mod 5; Kronecker $(8|\cdot)$, $(12|\cdot)$), with the exact values $L(-1, \chi_5) = -\frac{2}{5}$, $L(-1, \chi_8) = -1$, $L(-1, \chi_{12}) = -2$. Then for $\gcd(p, q) = 1$,*

$$\begin{aligned}\zeta'(-1, \frac{p}{5}) &= -\frac{Z_1}{5} + \frac{\text{Cl}_2(2\pi p/5)}{4\pi} + \chi_5(p) \left[\frac{L'(-1, \chi_5)}{20} - \frac{\ln 5}{50} \right] - \frac{\ln 5}{240}, \\ \zeta'(-1, \frac{p}{8}) &= -\frac{Z_1}{32} + \frac{\text{Cl}_2(\pi p/4)}{4\pi} + \chi_8(p) \left[\frac{L'(-1, \chi_8)}{32} - \frac{3 \ln 2}{32} \right] + \frac{\ln 2}{384}, \\ \zeta'(-1, \frac{p}{12}) &= \frac{Z_1}{24} + \frac{\text{Cl}_2(\pi p/6)}{4\pi} + \chi_{12}(p) \left[\frac{L'(-1, \chi_{12})}{48} - \frac{\ln 2}{12} - \frac{\ln 3}{24} \right] + \frac{\ln 3}{576},\end{aligned}$$

whence $\psi^{(-2)}(p/q) = \zeta'(-1, p/q) - Z_1 + Q_2(p/q)$ through the ladder; for instance

$$\psi^{(-2)}(\frac{1}{5}) = -\frac{6Z_1}{5} + \frac{\text{Cl}_2(2\pi/5)}{4\pi} + \frac{L'(-1, \chi_5)}{20} - \frac{29 \ln 5}{1200} + \frac{\ln 2\pi}{10} + \frac{2}{25}.$$

Proof. On each grid the two reflection rows (Theorem 7.29), the multiplication rows (7.20) for every divisor (non-coprime entries reduced through the sub-grids $q \in \{2, 3, 4, 6\}$), and the character row $\sum_p \chi_q(p) \zeta'(-1, \frac{p}{q}) = \frac{1}{q} [L'(-1, \chi_q) + \ln q L(-1, \chi_q)]$ form a system of full rank $\varphi(q) = 4$; solving it and reducing the composite-argument Clausen values ($\text{Cl}_2(\frac{3\pi}{4}) = \text{Cl}_2(\frac{\pi}{4}) - \frac{G}{2}$, $\text{Cl}_2(\frac{\pi}{6}) = \frac{2}{3}G + \frac{3}{8}\text{Cl}_2(\frac{2\pi}{3})$) gives the display. For the composite q the known rows are one short of independent — the single dependency is itself a Clausen duplication law in disguise (at $q = 8$: $\text{refl}(\frac{1}{8}) - \text{refl}(\frac{3}{8}) = m_2(\frac{1}{8}) - m_2(\frac{3}{8})$ is equivalent to $\text{Cl}_2(\frac{3\pi}{4}) = \text{Cl}_2(\frac{\pi}{4}) - \frac{G}{2}$) — and the quadratic character row supplies the missing dimension, exactly as Proposition 7.33 predicts ($r_{q,1} = 1$ for $q \in \{5, 8, 12\}$). $\square$

Theorem 7.39 (the quartic floor: $k = 2$, $q = 5$). *Let χ be the odd quartic character mod 5 with $\chi(2) = i$, for which $L(-2, \chi) = -\frac{4+2i}{5}$ exactly, and write $L'(-2, \chi) = R + iI$. Then*

$$\begin{aligned}\zeta'(-2, \frac{1}{5}), \zeta'(-2, \frac{4}{5}) &= -\frac{\text{Cl}_3(2\pi/5)}{4\pi^2} \pm \left[\frac{R}{50} - \frac{2 \ln 5}{125} \right], \\ \zeta'(-2, \frac{2}{5}), \zeta'(-2, \frac{3}{5}) &= -\frac{\text{Cl}_3(4\pi/5)}{4\pi^2} \pm \left[\frac{I}{50} - \frac{\ln 5}{125} \right],\end{aligned}$$

and, flattening through the ladder with $\text{Cl}_3(\frac{2\pi}{5}), \text{Cl}_3(\frac{4\pi}{5}) = -\frac{6}{25}\zeta(3) \pm \frac{\sqrt{5}}{4}L(3, \chi_5)$,

$$\begin{aligned}\psi^{(-3)}(\frac{1}{5}) &= \frac{31\zeta(3)}{200\pi^2} - \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} + \frac{R}{100} - \frac{\ln 5}{125} + \frac{\ln 2\pi}{100} + \frac{7}{1500} - \frac{Z_1}{5}, \\ \psi^{(-3)}(\frac{2}{5}) &= \frac{31\zeta(3)}{200\pi^2} + \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} + \frac{I}{100} - \frac{\ln 5}{250} + \frac{\ln 2\pi}{25} + \frac{41}{1500} - \frac{2Z_1}{5}, \\ \psi^{(-3)}(\frac{3}{5}) &= \frac{31\zeta(3)}{200\pi^2} + \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} - \frac{I}{100} + \frac{\ln 5}{250} + \frac{9\ln 2\pi}{100} + \frac{7}{125} - \frac{3Z_1}{5}, \\ \psi^{(-3)}(\frac{4}{5}) &= \frac{31\zeta(3)}{200\pi^2} - \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} - \frac{R}{100} + \frac{\ln 5}{125} + \frac{4\ln 2\pi}{25} + \frac{59}{750} - \frac{4Z_1}{5}.\end{aligned}$$

Remark 7.40. The remaining derivative coordinates in these formulas are the quartic pair $\{R, I\}$, matching $r_{5,2} = \varphi(5)/2 = 2$ of Proposition 7.33; the multiplication row $\sum_p \zeta'(-2, \frac{p}{5}) = \frac{6}{25}\zeta(3)/\pi^2$ is *dependent* on the reflections — it is the Clausen multiplication law $\text{Cl}_3(\frac{2\pi}{5}) + \text{Cl}_3(\frac{4\pi}{5}) = -\frac{12}{25}\zeta(3)$ in disguise. Through the conjugated harmonic-number bridge (Theorem 7.32 with $k = 2$, $H_2 = \frac{3}{2}$) the pair $\{R, I\}$ is interchangeable with $\{\text{Re}, \text{Im}\}$ of $L'(3, \chi)$.

Appendix: fully expanded values at the third and quarter points

The reduction recipes of Theorems 7.24 and 7.26, carried out to the end, give the following flattened closed forms. The Γ_2 values live in the atom set announced in Theorem 7.24; the Γ_3 values pick up, exactly as Theorem 7.26 predicts, the three further atoms γ_3, π^4 and $\gamma_4(\frac{1}{q}) - \gamma_4(1 - \frac{1}{q})$. The combinations $\gamma + \ln 6\pi$ and $\gamma + \ln 8\pi$ are the constants Λ_3, Λ_4 of Theorem 7.18. Each display has been verified at 90-digit working precision (residuals below 10^{-88}) against $\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1} [\zeta^{(n+1)}(0, p) - \zeta^{(n+1)}(0)]$ evaluated from Hurwitz-zeta derivative jets.

$$\begin{aligned}\Gamma_2(\frac{1}{3}) &= \frac{1}{2}\gamma_2 + \gamma\gamma_1 + \gamma_1 \ln 2\pi - \frac{1}{2}\gamma_1 \ln 3 \\ &\quad + \frac{4}{9}\gamma^3 + \frac{4}{3}\gamma^2 \ln 2\pi - \frac{1}{6}\gamma^2 \ln 3 + \frac{4}{3}\gamma \ln^2 2\pi \\ &\quad + \frac{7}{6}\gamma \ln 2\pi \ln 3 - \frac{1}{6}\gamma \ln^2 3 - \frac{4}{9}\zeta(3) - \frac{\pi^2}{12} \ln 2\pi \\ &\quad + \frac{\pi^2}{24} \ln 3 + \frac{4}{9} \ln^3 2\pi + \frac{4}{3} \ln^2 2\pi \ln 3 + \frac{7}{12} \ln 2\pi \ln^2 3 \\ &\quad - \frac{1}{18} \ln^3 3 \\ &\quad - \left((\gamma + \ln 6\pi)^2 - \frac{\pi^2}{12} \right) \ln \Gamma(\frac{1}{3}) \\ &\quad - \frac{\sqrt{3}}{6\pi} (\gamma + \ln 6\pi) (\gamma_2(\frac{1}{3}) - \gamma_2(\frac{2}{3})) \\ &\quad - \frac{\sqrt{3}}{18\pi} (\gamma_3(\frac{1}{3}) - \gamma_3(\frac{2}{3}))\end{aligned}$$

$$\begin{aligned}
\Gamma_2\left(\frac{1}{4}\right) &= \frac{1}{2} \gamma_2 + \gamma \gamma_1 + \frac{1}{2} \gamma_1 \ln 2 + \gamma_1 \ln \pi \\
&+ \frac{1}{2} \gamma^3 + 2 \gamma^2 \ln 2 + \frac{3}{2} \gamma^2 \ln \pi + 6 \gamma \ln^2 2 \\
&+ \frac{13}{2} \gamma \ln 2 \ln \pi + \frac{3}{2} \gamma \ln^2 \pi - \frac{1}{2} \zeta(3) - \frac{\pi^2}{24} \ln 2 \\
&- \frac{\pi^2}{12} \ln \pi + \frac{49}{6} \ln^3 2 + \frac{45}{4} \ln^2 2 \ln \pi + \frac{9}{2} \ln 2 \ln^2 \pi \\
&+ \frac{1}{2} \ln^3 \pi \\
&- \left((\gamma + \ln 8\pi)^2 - \frac{\pi^2}{12} \right) \ln \Gamma\left(\frac{1}{4}\right) \\
&- \frac{1}{4\pi} (\gamma + \ln 8\pi) (\gamma_2(\frac{1}{4}) - \gamma_2(\frac{3}{4})) \\
&- \frac{1}{12\pi} (\gamma_3(\frac{1}{4}) - \gamma_3(\frac{3}{4}))
\end{aligned}$$

$$\begin{aligned}
\Gamma_3\left(\frac{1}{3}\right) &= -\frac{1}{2} \gamma_3 - \frac{3}{2} \gamma \gamma_2 - \frac{3}{2} \gamma_2 \ln 2\pi + \frac{3}{4} \gamma_2 \ln 3 \\
&- \frac{3}{2} \gamma^2 \gamma_1 - 3 \gamma \gamma_1 \ln 2\pi + \frac{3}{2} \gamma \gamma_1 \ln 3 - \frac{3}{2} \gamma_1 \ln^2 2\pi \\
&+ \frac{3}{2} \gamma_1 \ln 2\pi \ln 3 + \frac{3}{4} \gamma_1 \ln^2 3 - \frac{\pi^2}{8} \gamma_1 - \frac{1}{2} \gamma^4 \\
&- 2 \gamma^3 \ln 2\pi + \frac{1}{4} \gamma^3 \ln 3 - 3 \gamma^2 \ln^2 2\pi - \frac{3}{2} \gamma^2 \ln 2\pi \ln 3 \\
&+ \frac{3}{8} \gamma^2 \ln^2 3 - \frac{\pi^2}{24} \gamma^2 - 2 \gamma \ln^3 2\pi - \frac{15}{4} \gamma \ln^2 2\pi \ln 3 \\
&- \frac{3}{2} \gamma \ln 2\pi \ln^2 3 + \frac{1}{4} \gamma \ln^3 3 + \frac{\pi^2}{6} \gamma \ln 2\pi - \frac{\pi^2}{48} \gamma \ln 3 \\
&+ \frac{5\pi^2}{24} \ln^2 2\pi + \frac{\pi^2}{24} \ln 2\pi \ln 3 - \frac{7\pi^2}{96} \ln^2 3 + \frac{\pi^4}{120} \\
&- \frac{1}{2} \ln^4 2\pi - 2 \ln^3 2\pi \ln 3 - \frac{15}{8} \ln^2 2\pi \ln^2 3 - \frac{1}{2} \ln 2\pi \ln^3 3 \\
&+ \frac{1}{16} \ln^4 3 \\
&+ \left((\gamma + \ln 6\pi)^3 - \frac{\pi^2}{4} (\gamma + \ln 6\pi) + 2 \zeta(3) \right) \ln \Gamma\left(\frac{1}{3}\right) \\
&+ \frac{\sqrt{3}}{48\pi} \left(12 (\gamma + \ln 6\pi)^2 - \pi^2 \right) (\gamma_2(\frac{1}{3}) - \gamma_2(\frac{2}{3})) \\
&+ \frac{\sqrt{3}}{6\pi} (\gamma + \ln 6\pi) (\gamma_3(\frac{1}{3}) - \gamma_3(\frac{2}{3})) \\
&+ \frac{\sqrt{3}}{24\pi} (\gamma_4(\frac{1}{3}) - \gamma_4(\frac{2}{3}))
\end{aligned}$$

$$\begin{aligned}
\Gamma_3\left(\frac{1}{4}\right) = & -\frac{1}{2}\gamma_3 - \frac{3}{2}\gamma\gamma_2 - \frac{3}{4}\gamma_2\ln 2 - \frac{3}{2}\gamma_2\ln\pi \\
& - \frac{3}{2}\gamma^2\gamma_1 - \frac{3}{2}\gamma\gamma_1\ln 2 - 3\gamma\gamma_1\ln\pi + \frac{9}{4}\gamma_1\ln^2 2 \\
& - \frac{3}{2}\gamma_1\ln 2\ln\pi - \frac{3}{2}\gamma_1\ln^2\pi - \frac{\pi^2}{8}\gamma_1 - \frac{9}{16}\gamma^4 \\
& - 3\gamma^3\ln 2 - \frac{9}{4}\gamma^3\ln\pi - \frac{45}{4}\gamma^2\ln^2 2 - \frac{51}{4}\gamma^2\ln 2\ln\pi \\
& - \frac{27}{8}\gamma^2\ln^2\pi - \frac{\pi^2}{32}\gamma^2 - 27\gamma\ln^3 2 - \frac{153}{4}\gamma\ln^2 2\ln\pi \\
& - \frac{33}{2}\gamma\ln 2\ln^2\pi - \frac{9}{4}\gamma\ln^3\pi + \frac{\pi^2}{4}\gamma\ln 2 + \frac{3\pi^2}{16}\gamma\ln\pi \\
& + \frac{3\pi^2}{8}\ln^2 2 + \frac{11\pi^2}{16}\ln 2\ln\pi + \frac{7\pi^2}{32}\ln^2\pi + \frac{29\pi^4}{3840} \\
& - \frac{207}{8}\ln^4 2 - \frac{179}{4}\ln^3 2\ln\pi - 27\ln^2 2\ln^2\pi - \frac{27}{4}\ln 2\ln^3\pi \\
& - \frac{9}{16}\ln^4\pi \\
& + \left((\gamma + \ln 8\pi)^3 - \frac{\pi^2}{4}(\gamma + \ln 8\pi) + 2\zeta(3)\right)\ln\Gamma\left(\frac{1}{4}\right) \\
& + \frac{1}{32\pi}\left(12(\gamma + \ln 8\pi)^2 - \pi^2\right)(\gamma_2(\tfrac{1}{4}) - \gamma_2(\tfrac{3}{4})) \\
& + \frac{1}{4\pi}(\gamma + \ln 8\pi)(\gamma_3(\tfrac{1}{4}) - \gamma_3(\tfrac{3}{4})) \\
& + \frac{1}{16\pi}(\gamma_4(\tfrac{1}{4}) - \gamma_4(\tfrac{3}{4}))
\end{aligned}$$

Chapter 8

The parameter-derivative tower

Differentiating in the Hurwitz parameter moves the spectral argument upward. The resulting coefficients are elementary symmetric functions of reciprocal integers, so harmonic numbers control every rung. The functional equation then pairs this tower with the integrated tower of the preceding chapter. Explicit character tables show how experiments can be replaced by uniform derivations.

8.1 The uniform master identity

Theorem 8.1 (Harmonic-number master identity). *For all $k \geq 1$ and $n \geq 0$,*

$$\gamma_n^{(k)}(a) = (-1)^{n+k} k! n! \sum_{i+j=n} e_i(k) \frac{\zeta^{(j)}(k+1, a)}{j!}, \quad (8.1)$$

where $e_i(k)$ is the i -th elementary symmetric function of $\{1, \frac{1}{2}, \dots, \frac{1}{k}\}$; in particular $e_0 = 1$, $e_1 = H_k$, $e_2 = \frac{1}{2}(H_k^2 - H_k^{(2)})$. The first two rows are

$$\gamma_1^{(k)}(a) = (-1)^{k+1} k! [\zeta'(k+1, a) + H_k \zeta(k+1, a)], \quad (8.2)$$

$$\gamma_2^{(k)}(a) = (-1)^k k! [\zeta''(k+1, a) + 2H_k \zeta'(k+1, a) + (H_k^2 - H_k^{(2)}) \zeta(k+1, a)]. \quad (8.3)$$

Proof. From $\partial_a \zeta(s, a) = -s \zeta(s+1, a)$, iteration gives $\partial_a^k \zeta(s, a) = (-1)^k (s)_k \zeta(s+k, a)$ with the rising factorial $(s)_k = s(s+1) \cdots (s+k-1)$. For $k \geq 1$ this annihilates the pole term in (7.1), so

$$\sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n^{(k)}(a) (s-1)^n = (-1)^k (s)_k \zeta(s+k, a).$$

Set $s = 1 + \varepsilon$. Then $(s)_k = (1 + \varepsilon)_k = \prod_{m=1}^k (m + \varepsilon) = k! \prod_{m=1}^k (1 + \varepsilon/m) = k! \sum_{i \geq 0} e_i(k) \varepsilon^i$, the generating identity for the elementary symmetric functions of $\{1/m\}$, while $\zeta(s+k, a) = \zeta(k+1+\varepsilon, a) = \sum_{j \geq 0} \zeta^{(j)}(k+1, a) \varepsilon^j / j!$ is analytic at $\varepsilon = 0$ because $k+1 \geq 2$. Comparing coefficients of ε^n yields (8.1); $n = 1, 2$ give (8.2)–(8.3). $\square$

Theorem 8.1 is the harmonic-number repackaging of Coffey's all-orders Stirling-number ladder [25]: the change of basis from the unsigned Stirling numbers $|s(k+1, i+1)|$ to the symmetric functions $e_i(k)$ turns the coefficients into $H_k, H_k^2 - H_k^{(2)}, \dots$. The specializations $\gamma_1' = \zeta'(2, a) + \zeta(2, a)$, $\gamma_1'' = -2\zeta'(3, a) - 3\zeta(3, a)$, $\gamma_2' = -[\zeta''(2, a) + 2\zeta'(2, a)]$ and $\gamma_2'' = 2\zeta''(3, a) + 6\zeta'(3, a) + 2\zeta(3, a)$ are the $s = 2, 3$ layers used below. Identity (8.1) was verified against numerical k -th parameter-derivatives of γ_1, γ_2 for $k = 1, \dots, 5$.

8.2 The uniform rational-argument landscape

Three structural facts hold uniformly in k ; all follow from $\sum_p \chi(p) \zeta(s, p/q) = q^s L(s, \chi)$ (and its multiplicative cousin) differentiated in s , fed through Theorem 8.1.

Proposition 8.2 (Character coordinate). *For a primitive nonprincipal Dirichlet character χ modulo $q > 1$ and all $k \geq 1$,*

$$\sum_{p=1}^{q-1} \chi(p) \gamma_1^{(k)}(p/q) = (-1)^{k+1} k! q^{k+1} [L'(k+1, \chi) + (H_k + \ln q) L(k+1, \chi)]. \quad (8.4)$$

Proposition 8.3 (Distribution row). *For $d \geq 2$ and all $k \geq 1$, with $\psi^{(k)}(a) = (-1)^{k+1} k! \zeta(k+1, a)$,*

$$\sum_{j=0}^{d-1} \gamma_1^{(k)}\left(\frac{a+j}{d}\right) = d^{k+1} \gamma_1^{(k)}(a) + d^{k+1} \ln d \psi^{(k)}(a). \quad (8.5)$$

Proof of (8.4) and (8.5). Apply $\sum_p \chi(p) (\cdot)$ (resp. $\sum_{j < d} (\cdot)$) to (8.2). For (8.4), use $\sum_p \chi(p) \zeta(s, p/q) = q^s L(s, \chi)$ and its s -derivative $q^s \ln q L(s, \chi) + q^s L'(s, \chi)$ at $s = k+1$; collect. For (8.5), use $\sum_{j < d} \zeta(s, (a+j)/d) = d^s \zeta(s, a)$ and its s -derivative, and recognize $(-1)^{k+1} k! \zeta(k+1, a) = \psi^{(k)}(a)$. $\square$

Observation 8.4 (Finite distribution-rank experiment). For the source's rational distribution matrices at $3 \leq q \leq 30$ and $k = 1, 2, 3$, exact row reduction gave residual dimension $(q-1) - \text{rank} = \varphi(q) - 1$.

This is an exact finite computation for those matrices, whose source scripts were not imported. A proof uniformly in q and k is not supplied. The character coordinates are exact for every k , but do not by themselves prove the distribution-row rank or numerical independence of the values. The identities above are the rigorous uniform statements.

8.3 Explicit tables for the second constant

We record the closed forms of $\gamma'_2 = -[\zeta''(2, a) + 2\zeta'(2, a)]$ (the $s = 2$ layer) and $\gamma''_2 = 2\zeta''(3, a) + 6\zeta'(3, a) + 2\zeta(3, a)$ (the $s = 3$ layer) at the elementary denominators. The new constants entering are the *second* s -derivatives $\zeta''(2), \zeta''(3), \beta''(s) = L''(s, \chi_{-4})$ and $L''(s, \chi_{-3})$; the values $L(3, \chi_{-4}) = \pi^3/32$ and $L(3, \chi_{-3}) = 4\pi^3/(81\sqrt{3})$ are elementary. As at the first-derivative layers, the odd-character block flips sign as a single unit under $p \mapsto q-p$ (including any elementary π -power value-part it carries). All entries are exact and were verified to $< 10^{-70}$.

Second constant, first parameter-derivative.

$$\begin{aligned} \gamma'_2(1) &= -\zeta''(2) - 2\zeta'(2), \\ \gamma'_2(\tfrac{1}{2}) &= -3\zeta''(2) - (6 + 8 \ln 2)\zeta'(2) - \tfrac{4}{3}\pi^2 \ln 2 - \tfrac{2}{3}\pi^2 \ln^2 2, \\ \gamma'_2(\tfrac{1}{3}) &= -4\zeta''(2) - (8 + 9 \ln 3)\zeta'(2) - \tfrac{3}{2}\pi^2 \ln 3 - \tfrac{3}{4}\pi^2 \ln^2 3 \\ &\quad - \left[\tfrac{9}{2}L''(2, \chi_{-3}) + 9(1 + \ln 3)L'(2, \chi_{-3}) + (9 \ln 3 + \tfrac{9}{2} \ln^2 3)L(2, \chi_{-3}) \right], \\ \gamma'_2(\tfrac{1}{4}) &= -6\zeta''(2) - (12 + 28 \ln 2)\zeta'(2) - \tfrac{14}{3}\pi^2 \ln 2 - 5\pi^2 \ln^2 2 \\ &\quad - \left[8\beta''(2) + 16(1 + 2 \ln 2)\beta'(2) + (32 \ln 2 + 32 \ln^2 2)G \right], \end{aligned}$$

with $\gamma'_2(2/3), \gamma'_2(3/4)$ obtained by negating the bracketed odd-character block, and the $q = 6$ entries given in the source report.

Second constant, second parameter-derivative.

$$\begin{aligned}
\gamma_2''(1) &= 2\zeta''(3) + 6\zeta'(3) + 2\zeta(3), \\
\gamma_2''(\tfrac{1}{2}) &= 14\zeta''(3) + (42 + 32\ln 2)\zeta'(3) + (14 + 48\ln 2 + 16\ln^2 2)\zeta(3), \\
\gamma_2''(\tfrac{1}{3}) &= 26\zeta''(3) + (78 + 54\ln 3)\zeta'(3) + (26 + 81\ln 3 + 27\ln^2 3)\zeta(3) \\
&\quad + \left[27L''(3, \chi_{-3}) + (81 + 54\ln 3)L'(3, \chi_{-3}) + (27 + 81\ln 3 + 27\ln^2 3)L(3, \chi_{-3}) \right], \\
\gamma_2''(\tfrac{1}{4}) &= 56\zeta''(3) + (168 + 240\ln 2)\zeta'(3) + (56 + 360\ln 2 + 248\ln^2 2)\zeta(3) \\
&\quad + \left[64\beta''(3) + (192 + 256\ln 2)\beta'(3) + (2 + 12\ln 2 + 8\ln^2 2)\pi^3 \right],
\end{aligned}$$

again with $\gamma_2''(2/3), \gamma_2''(3/4)$ the odd-block negations. The reducible fractions collapse correctly ($\frac{2}{4} = \frac{3}{6} = \frac{1}{2}, \frac{2}{6} = \frac{1}{3}$), an internal consistency check.

8.4 The derivative $\leftrightarrow$ negapolygamma duality

The functional equation pairs the layer $s = k + 1$ of $\gamma_1^{(k)}$ with the layer $s = -k$, which is exactly where the negative-order polygamma $\psi^{(-k+1)}$ lives (its ladder uses $\zeta'(-k, \cdot)$). The seam is a single uniform identity.

Theorem 8.5 (Harmonic-number bridge law). *Let χ be a primitive character mod q for which $L(-k, \chi) \neq 0$. Then*

$$\frac{L'(k+1, \chi)}{L(k+1, \chi)} + \overline{\left(\frac{L'(-k, \chi)}{L(-k, \chi)} \right)} = \gamma + \ln \frac{2\pi}{q} - H_k. \quad (8.6)$$

The same H_k that sets the master coefficient in Theorem 8.1 sets the bridge constant; the case $k = 1$ recovers the generalized-Glaisher constant $C_q = \gamma + \ln(2\pi/q) - 1$.

Proof of the bridge. For a primitive nonprincipal character of parity $\chi(-1) = (-1)^\epsilon$, the completed function is $\Lambda(s, \chi) = (q/\pi)^{(s+\epsilon)/2} \Gamma((s+\epsilon)/2) L(s, \chi)$. Its functional equation gives $\Lambda'/\Lambda(s, \chi) = -\Lambda'/\Lambda(1-s, \bar{\chi})$. At $s = k + 1$ with no trivial zero at $-k$, the two digamma terms, using reflection and recurrence, sum to $2(H_k - \gamma - \log 2)$ before their factor $1/2$ is applied. Collecting the conductor factor proves (8.6). The conductor-one principal character requires the meromorphic completed zeta function; the same nonzero-value calculation is valid for $k \geq 1$ where neither point is a pole. $\square$

Remark 8.6 (Trivial-zero bridge). For $k \geq 1$, $L(-k, \chi) = 0$ when $\epsilon \equiv k \pmod{2}$. At a simple trivial zero, cancellation of the pole in the completed gamma factor gives instead

$$\frac{L'(k+1, \chi)}{L(k+1, \chi)} + \frac{\overline{L''(-k, \chi)}}{2L'(-k, \chi)} = \gamma + \log(2\pi/q) - H_k.$$

The factor 2 in the denominator is essential. This follows by writing $L(-k+u) = L'(-k)u + \frac{1}{2}L''(-k)u^2 + O(u^3)$ in the same logarithmic functional equation and taking its finite part.

At $z = \frac{1}{4}$ the governing character is χ_{-4} (odd). For k even there is no trivial zero, and $\psi^{(-k+1)}(\frac{1}{4})$ carries the unreduced derivative coordinate $\beta'(k+1) = L'(k+1, \chi_{-4})$, so $\gamma_1^{(k)}(\frac{1}{4})$ is an exact finite combination of $\psi^{(-k+1)}(\frac{1}{4})$ and the remaining zeta-value and zeta-derivative terms. The representative instance $k = 2$ is, after eliminating $\beta'(3)$ entirely,

$$\begin{aligned}
\gamma_1''(\tfrac{1}{4}) &= -256\pi^3 \psi^{(-3)}(\tfrac{1}{4}) - 56\zeta'(3) - 84\zeta(3) - 120\ln 2 \zeta(3) + 35\pi \zeta(3) \\
&\quad - 3\pi^3 - 4\ln 2 \pi^3 + 64\ln A \pi^3 - 2\gamma \pi^3 + 2\ln(2\pi) \pi^3,
\end{aligned} \quad (8.7)$$

verified to $< 10^{-118}$. (The irrational coefficient $-256\pi^3$ on $\psi^{(-3)}(\frac{1}{4})$ is why a rational integer-relation search for this identity must take $\pi^3\psi^{(-3)}$, not $\psi^{(-3)}$, as the search atom.) For k odd the value-level duality *degenerates*: by Remark 8.6, $\psi^{(-(k+1))}(\frac{1}{4})$ then carries only the lower *value* atom (for $k = 1$, $\beta'(-1) = 2G/\pi$, a value rather than a derivative at the positive point; for $k = 3$, $\beta'(-3) \sim \beta(4)/\pi^3$), while the genuine derivative atom $\beta'(k+1)$ of $\gamma_1^{(k)}(\frac{1}{4})$ is tied instead to the second derivative $\beta''(-k)$, absent from $\psi^{(-(k+1))}$. Numerically the coefficient of $\beta'(k+1)$ in $\psi^{(-(k+1))}(\frac{1}{4})$ is zero to $\sim 10^{-163}$ for $k = 1, 3$. The clean odd- k value-level duality therefore lives at even-character points (e.g. $q = 5$), where the parity is reversed.

8.5 Experimental independence tests

The tables introduce $\zeta''(2), \zeta''(3), \beta''(2), \beta''(3)$ and $L''(2, \chi_{-3}), L''(3, \chi_{-3})$. Historical searches at 200 digits with planted controls recovered the known value reductions and found no accepted relation for the six derivative coordinates in the lower-order baskets. This supports using them as named coordinates in a reduction table. Their mutual independence, and the impossibility of reductions in larger baskets, remain open.

8.6 Elementary grids and the sixth-point completion

For $q = 3, 4, 6$ define

$$A_3(s) = \frac{3^s - 1}{2} \zeta(s), \quad A_4(s) = \frac{4^s - 2^s}{2} \zeta(s), \quad A_6(s) = \frac{(2^s - 1)(3^s - 1)}{2} \zeta(s),$$

$$B_3(s) = \frac{3^s}{2} L(s, \chi_{-3}), \quad B_4(s) = \frac{4^s}{2} \beta(s), \quad B_6(s) = \frac{6^s + 3^s}{2} L(s, \chi_{-3}).$$

Residue-class decomposition proves $\zeta(s, 1/q) = A_q(s) + B_q(s)$ and $\zeta(s, 1 - 1/q) = A_q(s) - B_q(s)$ for these three q . Consequently all four derivative tables, including the omitted sixth-point rows in the later article, follow from the differential operators

$$\begin{aligned} \gamma_1'(a) &= (\partial_s + 1)\zeta(s, a)|_{s=2}, & \gamma_1''(a) &= -(2\partial_s + 3)\zeta(s, a)|_{s=3}, \\ \gamma_2'(a) &= -(\partial_s^2 + 2\partial_s)\zeta(s, a)|_{s=2}, & \gamma_2''(a) &= (2\partial_s^2 + 6\partial_s + 2)\zeta(s, a)|_{s=3}. \end{aligned}$$

This supplies an exact symbolic recipe for every entry, without new fitted constants. Write $a = \log 2$, $b = \log 3$, and $L_j(s) = L^{(j)}(s, \chi_{-3})$. At the sixth points the first-constant rows are

$$\begin{aligned} \gamma_1'(1/6), \gamma_1'(5/6) &= 12\zeta'(2) + 2\pi^2 + \frac{8}{3}\pi^2 a + \frac{9}{4}\pi^2 b \\ &\quad \pm \left[\frac{45}{2} L_1(2) + \left(\frac{45}{2} + 18a + \frac{45}{2}b \right) L_0(2) \right], \\ \gamma_1''(1/6), \gamma_1''(5/6) &= -182\zeta'(3) - (273 + 208a + 189b)\zeta(3) \\ &\quad \mp \left[243 L_1(3) + \left(\frac{729}{2} + 216a + 243b \right) L_0(3) \right]. \end{aligned}$$

The second-constant rows are

$$\begin{aligned}
\gamma'_2(1/6), \gamma'_2(5/6) &= S_2 \mp O_2, \\
S_2 &= -12\zeta''(2) - (24 + 32a + 27b)\zeta'(2) \\
&\quad - \pi^2 \left(\frac{16}{3}a + \frac{9}{2}b + \frac{8}{3}a^2 + \frac{9}{4}b^2 + 6ab \right), \\
O_2 &= \frac{45}{2}L_2(2) + (45 + 36a + 45b)L_1(2) \\
&\quad + \left(36a + 45b + 18a^2 + \frac{45}{2}b^2 + 36ab \right) L_0(2), \\
\gamma''_2(1/6), \gamma''_2(5/6) &= S_3 \pm O_3, \\
S_3 &= 182\zeta''(3) + (546 + 416a + 378b)\zeta'(3) \\
&\quad + (182 + 624a + 567b + 208a^2 + 189b^2 + 432ab)\zeta(3), \\
O_3 &= 243L_2(3) + (729 + 432a + 486b)L_1(3) \\
&\quad + (243 + 648a + 729b + 216a^2 + 243b^2 + 432ab)L_0(3).
\end{aligned}$$

The upper sign is the 1/6 entry. These formulas consolidate the tables from the first-derivative, second-derivative and general-order reports.

Chapter 9

Herglotz arithmetic and quadratic regulators

The Herglotz function joins the digamma series to finite dilogarithm sums. At rational arguments this supplies exact evaluations; at quadratic units it connects continued-fraction cycles with real-quadratic Kronecker limits. The same experimental strategy thus leads from a finite cyclotomic sum to regulator candidates in class fields.

9.1 The function and the exact dilogarithm sum

For $x > 0$ define

$$F(x) = \sum_{n \geq 1} \frac{\psi(nx) - \log(nx)}{n}, \quad J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t} dt.$$

The series is locally uniformly convergent on the positive line and extends analytically to the cut plane $\mathbb{C} \setminus (-\infty, 0]$. Radchenko–Zagier [35] relate the two functions by

$$J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t} dt, \quad J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}, \quad (9.1)$$

which relates the companion J to the Herglotz function F . For the rational-argument results the engine is Radchenko–Zagier’s Theorem 3, an *exact, finite* evaluation obtained from the second integral representation of F via the distribution relations:

$$F\left(\frac{P}{Q}\right) - F(1) = \sum_{\alpha^P=1} \left(\sum_{\substack{\beta^Q=1 \\ \beta \neq 1}} f(\alpha, \beta) - \sum_{\substack{\beta^P=1 \\ \beta \neq 1}} f(\alpha, \beta) \right), \quad f(\alpha, \beta) = \text{Li}_2\left(\frac{\beta}{\beta-1}\right) - \text{Li}_2\left(\frac{\alpha-\beta}{1-\beta}\right). \quad (9.2)$$

The finite sum is an exact analytic identity, with compatible principal branches at its arguments. Floating-point evaluation still has rounding and cancellation errors. It is a useful independent comparator for the defining integral, not a zero-error numerical oracle.

9.2 Quadratic-unit values: Tables 1 & 2, and two errata

For real quadratic units the relevant object is the companion $\mathcal{J}(x) = J(x) - \frac{\log^2 2}{2} + \frac{\pi^2}{24} \left(x - \frac{1}{x}\right)$, and the arithmetic atoms are $S_{p,q} = \log \varepsilon_p \log \varepsilon_q$ where ε_d is the fundamental unit of $\mathbb{Q}(\sqrt{d})$ for a fundamental

discriminant $d > 1$ and $\varepsilon_1 := 2$. Tables 1 and 2 of the source paper express, for the 45 known one-class-per-genus cases, the quantities $4(\mathcal{J}(n + \sqrt{n^2 - 1}) - n\frac{\pi^2}{24})$ and $2(\mathcal{J}(n + \sqrt{n^2 + 1}) - n\frac{\pi^2}{24})$ as integer combinations of the $S_{p,q}$.

We re-implemented $\mathcal{J}$ and $S_{p,q}$ in PARI/GP (`intnum` integrals, `bnfinit` regulators) and evaluated both sides of all 45 rows at 200-digit precision. All 16 rows of Table 2 and 27 of the 29 rows of Table 1 agree to better than 10^{-80} . The two failures are $O(1)$:

$$n = 43: \text{LHS} - \text{RHS} = -13.0160\dots, \quad n = 61: \text{LHS} - \text{RHS} = -2.9011\dots$$

Each resolves to a single exact correction, confirmed against the high-DPI scan to be a typo in the printed paper rather than an OCR artifact.

Proposition 9.1 (Two corrections to Table 1).

- $n = 43$ ($n^2 - 1 = 2^3 \cdot 3 \cdot 7 \cdot 11$): the printed right-hand side overshoots by exactly $S_{33,56}$ ($(\text{LHS} - \text{RHS})/S_{33,56} = -1.000\dots$ to 110 digits); the spurious $+S_{33,56}$ term should be deleted.
- $n = 61$ ($n^2 - 1 = 2^3 \cdot 3 \cdot 5 \cdot 31$): the printed $S_{60,1032}$ should read $S_{60,248}$. The printed subscript $1032 = 2^3 \cdot 3 \cdot 43$ carries the prime 43, foreign to $n^2 - 1 = 3720$, whereas $60 \cdot 248 = 14880 = 4 \cdot 3720$ is the correct genus splitting.

A brute-force search over all single edits (delete a term, change a coefficient in $[-4, 4]$, or replace a subscript by any admissible fundamental-discriminant pair) returns these as the unique fixes. With both applied, all 45 rows verify to better than 10^{-80} .

The quadratic-unit tables concern specified arithmetic families. They do not imply that every even-trace unit has a pure-logarithm evaluation: outside genus characters the Kronecker-limit formula introduces additional class-group coordinates. Those coordinates are developed below.

9.3 Rational arguments: a family of theorems

The paper writes out only two rational-argument cases: $F(n)$ (pure logarithms, its eq. 23) and the lone fractional example $F(2/5)$ (its eq. 24). We now give the general laws.

9.3.1 Two reductions on the line $\text{Re} = \frac{1}{2}$

Everything rests on one observation: for a unit $\beta = e^{i\theta}$,

$$\frac{\beta}{\beta - 1} \text{ has real part } \frac{1}{2}, \quad \arg \frac{\beta}{\beta - 1} = \frac{\theta}{2} - \frac{\pi}{2}, \quad \left| \frac{\beta}{\beta - 1} \right| = \frac{1}{2 \sin(\theta/2)}. \quad (9.3)$$

By the reflection $\text{Li}_2(z) + \text{Li}_2(1 - z) = \frac{\pi^2}{6} - \log z \log(1 - z)$ with $1 - z = \bar{z}$ on $\text{Re } z = \frac{1}{2}$,

$$2 \text{Re Li}_2\left(\frac{1}{2} + iy\right) = \frac{\pi^2}{6} - \log^2 |z| - (\arg z)^2. \quad (9.4)$$

The second reduction handles the $\alpha = -1$ part of (9.2): $\frac{-1-\beta}{1-\beta} = -i \cot \frac{\theta}{2}$ is purely imaginary, and for real y

$$\text{Re Li}_2(iy) = \frac{1}{4} \text{Li}_2(-y^2). \quad (9.5)$$

9.3.2 $F(1/q)$: pure logarithms

Theorem 9.2. *For all $q \geq 2$, with $m = \lfloor (q-1)/2 \rfloor$,*

$$F\left(\frac{1}{q}\right) - F(1) = -\frac{(q-1)(3q-2)}{24q}\pi^2 - \sum_{k=1}^m \log^2\left(2 \sin \frac{\pi k}{q}\right) - [q \text{ even}] \frac{1}{2} \log^2 2.$$

Proof. With $P = 1$ only $\alpha = 1$ occurs and the inner β^P -sum is empty, so (9.2) reads $F(1/q) - F(1) = \sum_{\beta^q=1, \beta \neq 1} (\text{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6})$. Pairing β with $\bar{\beta}$ and applying (9.4) gives the π^2 coefficient $-\left[\frac{q-1}{12} + \sum_{k=1}^m \left(\frac{k}{q} - \frac{1}{2}\right)^2\right] = -\frac{(q-1)(3q-2)}{24q}$ and the log-sine sum. For even q the self-conjugate root $\beta = -1$ contributes $\text{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2} \log^2 2$. $\square$

9.3.3 $F(2/q)$: the general theorem

Theorem 9.3. *For every $q \geq 3$,*

$$F\left(\frac{2}{q}\right) - F(1) = -\frac{(q-1)(q-2)}{12q}\pi^2 + \log^2 2 - 2 \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \log^2\left(2 \sin \frac{\pi k}{q}\right) - \frac{1}{2} \sum_{k=1}^{\lfloor (q-1)/2 \rfloor} \text{Li}_2\left(-\cot^2 \frac{\pi k}{q}\right),$$

where for even q the two sums run to $k = \frac{q}{2} - 1$ and a further $-\log^2 2$ is added.

Proof (odd q). Here $\alpha \in \{1, -1\}$ and the inner β^P -sum is the single term $\beta = -1$. For the first part, $f(1, \beta) + f(-1, \beta) = 2 \text{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6} - \text{Li}_2(\frac{-1-\beta}{1-\beta})$. Summing over $\beta \neq 1$ and using (9.4) for the first pieces and (9.5) (with $\frac{-1-\beta}{1-\beta} = -i \cot \frac{\pi k}{q}$, pairing $k \leftrightarrow q-k$) for the last gives $2S_1 - (q-1)\frac{\pi^2}{6} - \frac{1}{2} \sum_k \text{Li}_2(-\cot^2 \frac{\pi k}{q})$, where $S_1 = \sum_{k=1}^{(q-1)/2} [\frac{\pi^2}{6} - \log^2(2 \sin \frac{\pi k}{q}) - \pi^2(\frac{k}{q} - \frac{1}{2})^2]$. The boundary gives $-\sum_{\alpha} f(\alpha, -1) = +\log^2 2$ from $\text{Li}_2(\frac{1}{2})$. Collecting, the $\frac{\pi^2}{6}$ terms cancel and the π^2 coefficient is $-2 \sum_{k=1}^{(q-1)/2} (\frac{k}{q} - \frac{1}{2})^2$, which equals $-\frac{(q-1)(q-2)}{12q}$ by an elementary identity (verified symbolically). The even- q case differs only in that $\beta = -1$ is now a genuine q -th root, contributing the extra $-\log^2 2$. $\square$

The algebraic dilogarithm arguments make the field structure explicit. At $q = 5$ the symmetric sum can be reduced further as described below. Unsuccessful searches at larger conductors do not prove that no other closed form exists; the finite-sum theorem remains valid at every q .

9.3.4 The $F(n/(n+1))$ family and the $F(p/2)$ reciprocity

Theorem 9.4. *For all $n \geq 2$, with $S(q) = \sum_{k=1}^{q-1} \log^2(2 \sin \frac{\pi k}{q})$,*

$$F\left(\frac{n}{n+1}\right) - F(1) = -\frac{3n^2 + 3n + 2}{24n(n+1)}\pi^2 + \text{Li}_2\left(\frac{n}{n+1}\right) + \frac{1}{2} \log^2 \frac{n+1}{n} - \frac{1}{2} S(n+1) + \frac{1}{2} S(n),$$

the single dilogarithm $\text{Li}_2(n/(n+1))$ appearing with coefficient exactly $+1$ (the source paper's “bilinear-log” Example 2, made explicit). The π^2 coefficient equals $-\frac{1}{8} - \frac{1}{12n(n+1)}$.

Proof. The two functional equations of F are

$$F(x) - F(x+1) - F(x/(x+1)) = -F(1) + \text{Li}_2(1/(x+1)),$$

$$F(x) + F(1/x) - 2F(1) = \frac{1}{2} \log^2 x - \frac{\pi^2}{6x}(x-1)^2.$$

The second equation and Theorem 9.2 give

$$F(n) - F(1) = -\frac{(n-1)(n-2)}{24n}\pi^2 + \frac{1}{2}\log^2 n + \frac{1}{2}S(n).$$

Subtract the cases n and $n+1$ in the first equation, and apply Euler reflection to $\text{Li}_2(1/(n+1))$. Collecting the logarithms and rational π^2 coefficients yields the displayed formula. $\square$

Theorem 9.5 (Reciprocity). *For all odd $p \geq 3$,*

$$F\left(\frac{p}{2}\right) + F\left(\frac{2}{p}\right) - 2F(1) = -\frac{(p-2)^2}{12p}\pi^2 + \frac{1}{2}\log^2 \frac{p}{2}.$$

The $\text{Li}_2(-\cot^2 \frac{\pi k}{p})$ terms of $F(p/2)$ are exactly minus those of $F(2/p)$ and cancel in the sum; combined with Theorem 9.3 this yields $F(p/2)$ in closed form.

Proof. Substitute $x = p/2$ into the second functional equation above. $\square$

9.3.5 The derivative law

Theorem 9.6 (Radchenko–Zagier Proposition 4, $P = 2$). *For all $q \geq 3$, with $T(q) := \frac{2}{q}F'(2/q) - (1 + \log 2)$,*

$$T(q) = \frac{q^2 - 4}{24q}\pi^2 + W(q), \quad W(q) = \sum_{k=1}^{q-1} C(2k/q) \text{Cl}_2\left(\frac{2\pi k}{q}\right),$$

where $C(t) = \cot(\pi t)$ for $t \notin \mathbb{Z}$ and $C(t) = 0$ for $t \in \mathbb{Z}$, with an extra $-\log 2$ for even q . This assigned value at integers is part of the formula; a literal cotangent there would be undefined. The sum provides a Clausen representation, without an independence assertion.

9.3.6 Rational-argument samples and the companion integral

The source’s proposed criterion $\text{rad}(q) \mid 30$ is false as a necessary condition for a reduction to rational dilogarithms: Theorem 9.2 gives a pure-logarithm answer for $F(1/7)$. Numerator, denominator and the available functional equations must all be considered. No replacement universal classification is asserted here. The following experimentally recovered samples are useful:

$$\begin{aligned} F(3/4) - F(1) &= -\frac{19}{144}\pi^2 - \frac{3}{4}\log^2 2 - 2\log 2 \log 3 + \frac{7}{4}\log^2 3 + 2\text{Li}_2(1/3), \\ F(4/5) - F(1) &= \frac{3}{80}\pi^2 + \frac{11}{4}\log^2 2 - \frac{1}{2}\log^2 5 - \text{Li}_2(1/5) - \log^2(2\sin(\pi/5)) - \log^2(2\sin(2\pi/5)), \\ J(2/3) &= -\frac{\pi^2}{144} + \frac{3}{4}\log^2 2, \\ J(2/5) &= \frac{11\pi^2}{240} + \frac{3}{4}\log^2 2 - 2\log^2 \phi. \end{aligned}$$

The last two values follow by combining the F evaluations through (9.1). Their existence does not prove that these are the only q for which $J(2/q)$ simplifies.

9.4 Bridges to the golden ladders and the Clausen lattice

The new constants are not isolated; they touch two corpora we have already mapped.

The golden bridge ($q = 5$). The arguments $-\cot^2 \frac{\pi}{5} = -\frac{5+2\sqrt{5}}{5}$ and $-\cot^2 \frac{2\pi}{5} = -\frac{5-2\sqrt{5}}{5}$ lie in $\mathbb{Q}(\sqrt{5})$, the field of the golden polylogarithm ladders. Comparing Theorem 9.3 ($q = 5$) with eq. 24 gives the functional equation

$$\mathrm{Li}_2\left(\frac{4}{5}\right) + \mathrm{Li}_2\left(-\cot^2 \frac{\pi}{5}\right) + \mathrm{Li}_2\left(-\cot^2 \frac{2\pi}{5}\right) = 2\log^2 2 - 2\log 2 \log \frac{2}{5} - 6(\log^2(2 \sin \frac{\pi}{5}) + \log^2(2 \sin \frac{2\pi}{5})),$$

hence the golden-sum closed form (with $\varphi = \frac{1+\sqrt{5}}{2}$)

$$\mathrm{Li}_2\left(-\cot^2 \frac{\pi}{5}\right) + \mathrm{Li}_2\left(-\cot^2 \frac{2\pi}{5}\right) = \mathrm{Li}_2\left(\frac{1}{5}\right) - \frac{\pi^2}{6} - 3\log^2 \varphi + \frac{1}{4}\log^2 5.$$

The symmetric sum has the displayed reduction. The antisymmetric combination $X = \frac{1}{2}[\mathrm{Li}_2(-\cot^2(\pi/5)) - \mathrm{Li}_2(-\cot^2(2\pi/5))]$ did not reduce in the source's tested basket. Its independence is open. A formal symbol with nonzero Bloch boundary is not subject to a vanishing-rank conclusion for the Bloch group.

The Clausen bridge. The coefficients of $W(q)$ are real algebraic numbers contained in $\mathbb{Q}(\zeta_{\mathrm{lcm}(4,q)})^+$; at odd conductors the factor i implicit in the cotangent can require an extension of $\mathbb{Q}(\zeta_q)^+$. This is an algebraic coefficient issue, not a proof of numerical period independence. Applying the distribution identities of Chapter 2 gives

$$W(8) = 2\mathrm{Cl}_2(\pi/4) - 2\mathrm{Cl}_2(3\pi/4) = G, \quad W(12) = \frac{11\sqrt{3}}{9}\mathrm{Cl}_2(\pi/3).$$

For the second formula, pair k with $12 - k$:

$$W(12) = 2\sqrt{3}[\mathrm{Cl}_2(\pi/6) - \mathrm{Cl}_2(5\pi/6)] + \frac{2}{\sqrt{3}}[\mathrm{Cl}_2(\pi/3) - \mathrm{Cl}_2(2\pi/3)].$$

Use $\mathrm{Cl}_2(\pi/6) - \mathrm{Cl}_2(5\pi/6) = V/2$ and $\mathrm{Cl}_2(2\pi/3) = 2V/3$. This proves the original report's coefficient $22\sqrt{3}/9$ multiplying the first difference. It is also checked against direct derivative quadrature in the fresh verification battery.

9.5 Quadratic units and Kronecker-limit coordinates

Define the normalized functions

$$\mathcal{F}(x) = F(x) - F(1) + \frac{\pi^2}{12}(x - 2 + x^{-1}) - \frac{1}{4}\log^2 x, \quad \mathcal{J}(x) = J(x) - \frac{1}{2}\log^2 2 + \frac{\pi^2}{24}(x - x^{-1}).$$

For a real quadratic order of discriminant D , a narrow ideal class $\mathcal{B}$ has a partial zeta function. Its Kronecker constant is

$$\varrho(\mathcal{B}) = \lim_{s \rightarrow 1} \left[D^{s/2} \zeta(\mathcal{B}, s) - \frac{\log \varepsilon_D}{s-1} \right],$$

where $\varepsilon_D > 1$ is the least norm-one unit. In the reduced-form cycle representation, $w_j > 1 > w'_j > 0$, Radchenko–Zagier give

$$\varrho(\mathcal{B}) = 2\gamma \log \varepsilon_D - \ell(\mathcal{B}) \frac{\pi^2}{6} + \sum_{j \bmod \ell} [\mathcal{F}(w_j) - \mathcal{F}(w'_j) + L(w'_j/w_j)], \quad (9.6)$$

where L is the unnormalized Rogers dilogarithm. Class characters turn the ϱ coordinates into Hecke L -values. Their Theorem 5 expresses $\mathcal{J}$ at specified even-trace units in these coordinates. The source's independent cycle engine reported checks at $n = 2, 4, 6, 8$ with residuals about 5×10^{-38} ; the engine was not imported into this checkout.

9.6 Regulator candidates and field degrees

The published Stark examples and the source’s follow-up fits are conjectural or numerical evaluations. A cubic polynomial over $\mathbb{Q}$ defines a degree-three field E . It cannot itself be a Hilbert or ring class field containing a quadratic field K . In the examples at issue, its degree-six splitting field can be a cyclic cubic extension of K ; E is a cubic subfield. The distinction corrects a repeated field-degree error in the report.

For $u = 16 + \sqrt{257}$ the source reports

$$\mathcal{J}(u) \stackrel{\text{num}}{=} 4\zeta(2) + \log 2 \log u + 2 \operatorname{reg}(E), \quad E = \mathbb{Q}[x]/(x^3 - 2x^2 - 3x + 1). \quad (9.7)$$

The determinant of the two selected unit-logarithm columns is signed; with the source’s ordering it is $-\operatorname{reg}(E)$. Vertical bars around a matrix denote its determinant, not an absolute value. Replacing this negative determinant by its absolute value without changing its external coefficient reverses the regulator term.

The other published example concerns $u = 10 + 3\sqrt{11}$ and the order of discriminant 396. Its unit-logarithm expression is a Stark minor rather than a full field regulator. We retain that structural distinction; the unimported algebraic-unit definitions do not provide a locally replayable formula for the minor.

n	D_0	cubic field polynomial	c_1	c_2	c_3
6	37	$x^3 - x^2 - 3x + 1$	$3/2$	$2/3$	$2/3$
10	101	$x^3 - x^2 - 5x - 1$	$5/2$	$2/3$	$2/3$
14	197	$x^3 - x^2 - 7x - 3$	$7/2$	$2/3$	$2/3$
16	257	$x^3 - x^2 - 4x + 3$	4	1	2
26	677	$x^3 - x^2 - 11x - 7$	$13/2$	$2/3$	$2/3$

Table 9.1: Historical numerical candidates $\mathcal{J}(n + \sqrt{n^2 + 1}) = c_1\zeta(2) + c_2 \log 2 \log u + c_3 \operatorname{reg}(E)$. The source reports residuals below 1.2×10^{-109} . These are not unconditional theorems.

The source reports that $n = 6, 10, 14, 16, 26$ are the narrow-class-number-three cases in its scan of $2 \leq n \leq 5000$. This is a bounded census, not a proof that the family is finite. Negative fits at class numbers five and seven likewise do not establish a universal “single regulator if and only if cyclic cubic” theorem.

There is a precise conditional route to a leading- L interpretation. Suppose a non-Galois totally real cubic field E has quadratic resolvent K , and its normal closure supplies a nontrivial order-three class character χ of K . Artin induction gives $\zeta_E(s) = \zeta(s)L_K(s, \chi)$. The analytic class-number formula then gives

$$L_K(s, \chi) = h_E \operatorname{reg}(E)s^2 + O(s^3),$$

because $\zeta_E(s) = -h_E \operatorname{reg}(E)s^2/2 + O(s^3)$ and $\zeta(0) = -1/2$. Thus the coefficient equals $\operatorname{reg}(E)$ only after $h_E = 1$ and the field-character identification have been established. This argument does not convert the numerical Herglotz fits into proofs of Stark’s conjecture.

Chapter 10

Experimental discovery and the route to proof

The mathematical progression of the manuscript suggests a practical method. One begins with a value that is easy to compute but hard to recognize, finds an exact transformation that organizes its arguments, and then asks which coordinates remain after known laws are imposed. Numerical experiments are especially effective at proposing those coordinates. Their usefulness increases when the proposed identity is subsequently derived, and when a negative result is recorded with enough detail to explain its limits.

10.1 Constructing a meaningful integer-relation experiment

An experiment consists of a target, a finite ordered basket of constants, their coefficient field, a working precision, a coefficient bound and an acceptance rule. Weight and character parity give strong guidance for the basket. For instance a rational-angle Fourier transform can require $\sqrt{q}$ -weighted atoms; searching only with rational coefficients in the unweighted values misses that structure. The $\pi^3\psi^{(-3)}(1/4)$ coordinate in Chapter 8 is another example: omitting the π^3 factor changes the question.

Before searching, remove known basket dependencies. A search containing both $\log \Gamma(1/2)$ and $\frac{1}{2} \log \pi$ may return their tautology instead of a relation involving the target. Reflecting arguments, distributing roots of unity and replacing elementary odd beta values reduce this problem before numerical linear algebra starts.

Working precision must exceed the cancellation loss as well as the digits needed to discriminate the proposed coefficient height. The heuristic $10^{P/d}$, for P available digits and d basket coordinates, estimates a scale at which accidental short numerical relations can appear. It is neither a theorem about a specific basket nor a rigorous exclusion bound. Coefficient sizes, conditioning and the number of searches all matter. The earlier false negative for S_5 illustrates this: a 51-coordinate basket at 150 digits obscured the height-3840 relation that a smaller, properly structured basket exposed.

10.2 Controls and independent recomputation

A planted positive control checks that the search can recover a known relation of comparable scale. A perturbation of one coefficient tests the resolution of the residual check. A negative control and an auxiliary “canary” constant can expose numerical coincidences. These controls are diagnostics.

In particular no independence theorem is known between the Champernowne constant and the periods in these baskets, and a zero canary coefficient cannot certify an identity.

A successful candidate should be frozen as exact rational or algebraic coefficients, then reevaluated at higher precision using an independent representation. For a nested sum that may be an iterated integral; for a CM row sum it may be a modular Fourier series; for Herglotz values it may be a defining integral compared with the finite dilogarithm sum. Agreement between two engines using the same incorrect formula is less informative than agreement between different mathematical representations. Error-controlled interval evaluation can establish a finite residual bound, but even an arbitrarily small nonzero numerical bound alone does not establish equality of general transcendental constants.

10.3 What exact linear algebra proves

An exact relation matrix proves an upper bound on the number of necessary coordinates in its presented quotient. Its row-combination matrix is a replayable finite certificate relative to the input identities. A free column is independent in that formal quotient, not automatically among the evaluated numerical periods. Additional functional equations or regularization can reduce the quotient further.

Likewise a motivic dimension theorem concerns motivic objects. To infer numerical independence one needs a period-injectivity statement, generally conjectural. A dimension for a chosen word-length grading need not be the dimension of the classical depth filtration. The Gaussian triples are a natural test case for this distinction.

Certificates require both exact atom cancellation and sound law instantiation. Substitution strings must be parsed within a restricted grammar; branches and domains must be established mathematically. A numerical branch guard is useful for rejecting errors but leaves a separate domain obligation. These issues motivate a future formal development rather than a current claim of proof-assistant verification.

10.4 Computational conventions

The manuscript uses decreasing nested indices. In the tested Wolfram Language 15 interface,

$$\text{Li}_{a,b}(z_1, z_2) = \text{MultiplePolyLog}[\{\mathbf{a}, \mathbf{b}\}, \{\mathbf{z1}, \mathbf{z2}\}], \quad \zeta(a, b) = \text{MultipleZeta}[\{\mathbf{a}, \mathbf{b}\}].$$

Finite checks such as $\zeta(2, 1, 1) = \zeta(4)$ and $H(0, 1; z) = \text{Li}_2(z)$ calibrate the interface. A divergent call does not resolve ordering ambiguities on its own. The same calibration is needed for signed harmonic indices and generalized-polylogarithm letter order.

Negative-order polygamma means the integral based at zero: $\psi^{(-1)} = \log \Gamma$ and $\psi^{(-n)}(z) = \int_0^z \psi^{(-(n-1))}(t) dt$ for $n \geq 2$. Balanced conventions can differ by explicit polynomials. In Hurwitz jets, $\zeta^{(j)}$ always differentiates the spectral variable; the superscript (k) on γ_n differentiates its parameter. The CM hexagonal point $e^{i\pi/3}$ and the Eisenstein cube root $e^{2\pi i/3}$ are different arguments even when a local section denotes each by ρ .

10.5 The surviving research programme

Several concrete proof problems now fit into one framework. The odd-weight S_p directions need a precise choice of comparison algebra and a rigorous reduction or obstruction; the even-weight

direction has the generating-integral proof of Chapter 4. The Gaussian and Eisenstein mixed points need a full regularized relation computation with an explicit matrix and certified reductions. The Stieltjes distribution ranks need a uniform proof of the divisor-row structure. Numerical independence of character derivative coordinates remains a separate arithmetic problem.

The gamma programme can transport its exact row certificates across the DFT and prove branch validity for each law instance. Its finite-matrix termination should remain separate from Rohrlich's conjecture. The CM programme can replace fitted coefficients by modular equations and class-field computations, while taking account of periods under Galois conjugation. The Herglotz programme needs complete explicit class-character and unit definitions for the regulator candidates before they become locally replayable, followed by the relevant Stark argument.

The cubic log-gamma moment and the opaque individual CM components are good discovery targets because their defining integrals are accessible, but the naive single-value baskets miss their likely depth or modular structure. A richer representation is often more productive than a larger unstructured search. The scientific value of the experiments is that they tell us which representation to seek next.

Appendix A

Literature problems and source provenance

The two source catalogs organize questions that motivated or could extend the programme. Their categories and answer status describe the June 2026 survey, not a current audit of the websites. A catalog entry is a research lead, not an accepted identity. Closely overlapping questions are retained under a single thread identifier. The external reports numbered one through four contributed leads to these catalogs; their unverified prose and claims are not used as mathematical authorities.

One erroneous external formula is corrected here with a proof. Euler reflection at $1/3$, together with Landen at $-1/3$ and duplication, gives

$$6 \operatorname{Li}_2(1/3) - \operatorname{Li}_2(1/9) = \frac{\pi^2}{3} - \log^2 3.$$

For completeness, put $A = \operatorname{Li}_2(1/3)$ and $B = \operatorname{Li}_2(-1/3)$. The five-term relation at $x = y = 1/2$ and Landen at $-1/3$ give

$$2 \operatorname{Li}_2(1/2) = \operatorname{Li}_2(1/4) + 2A + \log^2(2/3), \quad \operatorname{Li}_2(1/4) = -B - \frac{1}{2} \log^2(4/3).$$

With $2 \operatorname{Li}_2(1/2) = \pi^2/6 - \log^2 2$, collecting the logarithms gives $2A - B = \pi^2/6 - \frac{1}{2} \log^2 3$. Duplication says $\operatorname{Li}_2(1/9) = 2A + 2B$, so subtracting it from $6A$ proves the corrected formula. This derivation fixes the factor-two error in the external report without relying on numerical recognition.

A.1 Polylogarithm questions

Thread	Research question	Historical category
MSE 1424600	Real part of $\operatorname{Li}_2(1/2 + i/6)$	Seed question
MSE 1430169	Dilogarithm values at algebraic complex arguments	Seed question
MSE 1373123	An identity for six rational tetralogarithm values	Seed question
MSE 945972	Relations among rational polylogarithm values	Seed question
MSE 1579355	Trilogarithms and the imaginary golden ratio	Seed question
MSE 918680	Closed Form for the Imaginary Part of $\operatorname{Li}_3((1+i)/2)$	Primary answered matches on Math Stack Exchange
MSE 3290798	A conjectured value for $\operatorname{Re} \operatorname{Li}_4(1+i)$	Primary answered matches on Math Stack Exchange

Thread	Research question	Historical category
MSE 3447529	On the relationship between $\operatorname{Re} \operatorname{Li}_n(1+i)$ and $\operatorname{Li}_n(1/2)$ when $n \geq 5$	Primary answered matches on Math Stack Exchange
MSE 676124	Proving a $\operatorname{Li}_3(-1/3) - 2 \operatorname{Li}_3(1/3)$ identity	Primary answered matches on Math Stack Exchange
MSE 1450917	A special value polylogarithm identity involving $\operatorname{Li}_3(-1/2)$, $\operatorname{Li}_3(-1/3)$, $\operatorname{Li}_3(2/3)$, $\operatorname{Li}_2(-1/3)$, $\operatorname{Li}_2(2/3)$	Primary answered matches on Math Stack Exchange
MSE 935366	Simplification of an expression containing $\operatorname{Li}_3(x)$ terms	Primary answered matches on Math Stack Exchange
MSE 5092144	How to prove this $\zeta(3)$ form with Li_3 and Li_2	Primary answered matches on Math Stack Exchange
MSE 463682	Closed form of $\operatorname{Li}_2(\phi)$ and $\operatorname{Li}_2(\phi-1)$	Primary answered matches on Math Stack Exchange
MSE 1410276	Closed-form of $\operatorname{Li}_2(1 + i \sqrt{3})$	Primary answered matches on Math Stack Exchange
MSE 967398	Extract real and imaginary parts of $\operatorname{Li}_2(i(2 + \sqrt{3}))$	Primary answered matches on Math Stack Exchange
MSE 1602647	Extract imaginary part of $\operatorname{Li}_3(2/3 - i \sqrt{2}/3)$ in closed form	Primary answered matches on Math Stack Exchange
MSE 980179	Closed-form of a special value dilogarithm identity	Primary answered matches on Math Stack Exchange
MSE 989704	Closed-forms of real parts of special value dilogarithm identities from inverse tangent integral function	Primary answered matches on Math Stack Exchange
MSE 1436493	Dilogarithm identity containing the tribonacci constant	Primary answered matches on Math Stack Exchange
MSE 481682	Polylogarithm ladders for the tribonacci and n-nacci constants	Primary answered matches on Math Stack Exchange
MSE 1064100	Why does the tribonacci constant have a trilogarithm ladder?	Primary answered matches on Math Stack Exchange
MSE 1959503	Proof of a dilogarithm identity	Primary answered matches on Math Stack Exchange
MSE 932932	Known exact values of the Li_3 function	Primary answered matches on Math Stack Exchange
MSE 1411312	Closed form of a series (dilogarithm)	Primary answered matches on Math Stack Exchange
MSE 2389425	A very odd-looking statement about $\zeta(3)$ and $\operatorname{Li}_2(1/\phi^2)$	Primary answered matches on Math Stack Exchange
MO 492746	Proving identity involving dilogarithms and $\pi/9$	Primary answered matches on MathOverflow
MO 85986	A dilogarithm identity: known or new?	Primary answered matches on MathOverflow
MO 144322	The relationship between the dilogarithm and the golden ratio	Primary answered matches on MathOverflow
MO 261408	Several conjectured identities for polylogarithms	Primary answered matches on MathOverflow
MO 450252	An identity involving polylogarithms	Primary answered matches on MathOverflow
MO 511718	Two-term dilogarithmic identity simplification	Primary answered matches on MathOverflow
MO 511765	Daniel Shanks "simplest cubic fields" and dilogarithm identities	Primary answered matches on MathOverflow
MO 131111	Is this combination of generalized polygamma and dilogarithm actually zero?	Primary answered matches on MathOverflow

Thread	Research question	Historical category
MSE 5080786	Ramanujan type formulas for polylogarithm	Open prompts with usable identity examples but no meaningful answers
MSE 5023055	How to find closed form for a complex expression involving special functions and polylogarithms?	Open prompts with usable identity examples but no meaningful answers
MO 173492	Dilogarithm of $-1/2$?	Open prompts with usable identity examples but no meaningful answers
MO 141185	A second polylogarithm ladder for the tribonacci and n-nacci constants	Open prompts with usable identity examples but no meaningful answers
MO 490573	The imaginary part of two-term complex dilogarithms	Open prompts with usable identity examples but no meaningful answers
MO 496682	Deriving a 4-term functional dilogarithm identity by Kirillov	Open prompts with usable identity examples but no meaningful answers
MSE 3938336	Evaluate a series equal to $\text{Li}_3((3-\sqrt{5})/2)$	Duplicate-like or low-new-information threads
MSE 5110303	Does the polylogarithm $\text{Li}_k(x)$ solve a first or second order ODE/ADE?	Rejected or parked: no answer and no special-value identity example
MSE 5127863	Completing proof of formula for a hard polylogarithmic integral	Rejected or parked: no answer and no special-value identity example
MSE 4986171	Understanding polylogarithm branch points, monodromy and multivaluedness	Rejected or parked: no answer and no special-value identity example
MO 85774	Is the quantum dilogarithm related in any way to cohomology of quantum groups?	Rejected or parked: no answer and no special-value identity example
MO 160646	Inverse of polylogarithm	Rejected or parked: no answer and no special-value identity example
MO 333610	Limiting property of polylogarithm ratio	Rejected or parked: no answer and no special-value identity example
MO 279588	Approximations of polylogarithm and Lerch transcendent?	Rejected or parked: no answer and no special-value identity example
MO 218158	Root polylogarithm dominance questions	Rejected or parked: no answer and no special-value identity example
MSE 1210644	Proving a $\sum \cos(n)/n^4$ evaluation	Near misses and background threads
MSE 376180	Integral identity involving $\text{Li}_2(-e^{ix})$	Near misses and background threads
MSE 4284361	Closed form evaluation of a trigonometric integral in terms of polylogarithms	Near misses and background threads
MSE 5125161	Verify a polylogarithmic evaluation of a 5D integral over a positive orthant	Near misses and background threads

Thread	Research question	Historical category
MO 347747	Simplify the difference of two dilogarithms	Near misses and background threads
MO 2402	Abel's equation for the dilog	Near misses and background threads
MO 25428	What is special about polylogarithms that leads to so many interesting identities and applications?	Near misses and background threads
MO 242920	Polylogarithm sheaves	Near misses and background threads
MO 309945	A remarkable almost-identity	Near misses and background threads

A.2 Polygamma and Hurwitz questions

Thread	Research question	Historical category
MSE 897967	Trigamma identity $4 \psi_1(1/5) + \psi_1(2/5) - \psi_1(1/10) = 4 \pi^2 / (\phi \sqrt{5})$	Primary answered matches on Math Stack Exchange
MSE 4290297	New trigamma identity for $\Psi_1(3/20) + 6 \Psi_1(1/5) + 10 \Psi_1(2/5) - \Psi_1(1/20)$	Primary answered matches on Math Stack Exchange
MSE 953976	Prove $\psi^{(1)}(5/6) - \psi^{(1)}(1/6) = 5(\psi^{(1)}(2/3) - \psi^{(1)}(1/3))$	Primary answered matches on Math Stack Exchange
MSE 2802872	Trigamma identity $\psi_1(11/12) - \psi_1(5/12) = 4 \sqrt{3} \pi^2 - 80G$	Primary answered matches on Math Stack Exchange
MSE 3063097	Alternative proof for $\zeta(2, 1/4) = \psi^{(1)}(1/4) = \pi^2 + 8G$	Primary answered matches on Math Stack Exchange
MSE 2952474	Is it possible to simplify $\psi^{(2)}(1/8)$ or $\psi^{(2)}(p/q)$?	Primary answered matches on Math Stack Exchange
MSE 4040435	General summation formula for $\psi^{(s)}(1/2)$, including negative orders	Primary answered matches on Math Stack Exchange
MSE 4784920	Closed form for $\psi^{(-2)}(1/4)$	Primary answered matches on Math Stack Exchange
MSE 893486	Identity connecting polylogarithm and polygamma functions of arguments $1/4$ and $3/4$	Primary answered matches on Math Stack Exchange
MSE 4504218	Identity relating the polygamma function and the Lerch transcendent	Primary answered matches on Math Stack Exchange
MSE 3653746	Decomposition of $\psi^{(n)}(1)$ in terms of $\psi^{(n)}(k)$	Primary answered matches on Math Stack Exchange
MSE 1154373	An identity on Polygamma	Primary answered matches on Math Stack Exchange
MSE 597867	Multiplication formula for the Hurwitz zeta function	Primary answered matches on Math Stack Exchange
MSE 1335002	An infinite series in polygamma function	Primary answered matches on Math Stack Exchange
MSE 4441095	Closed form for a derivative limit involving $\psi^{(0)}$ and higher polygamma values at 1	Primary answered matches on Math Stack Exchange
MO 312479	π in terms of polygamma	Primary answered matches on MathOverflow
MO 312550	$\psi(2, 1/6)$, $\psi(4, 1/6)$ in terms of zeta and π only	Primary answered matches on MathOverflow

Thread	Research question	Historical category
MO 42696	Connection between Bernoulli polynomials and polygamma function	Primary answered matches on MathOverflow
MO 347520	Reference request: sums of rational functions and polygamma functions	Primary answered matches on MathOverflow
MO 251727	Closed form for sum involving digamma?	Primary answered matches on MathOverflow
MSE 1909595	Interesting connection between polylogarithms and polygamma functions	Open or no-meaningful-answer leads
MSE 4319642	Simplification of a difficult identity involving the digamma function	Open or no-meaningful-answer leads
MSE 3555737	Sum involving 2nd antiderivative of the digamma function	Open or no-meaningful-answer leads
MSE 7183	Connection between Bernoulli polynomials and polygamma function	Open or no-meaningful-answer leads
MSE 521203	Recurrence relation for polygamma reflection polynomials	Open or no-meaningful-answer leads
MSE 4845364	Closed form for $\psi^{(1/k)}(1)$, where k is an integer	Open or no-meaningful-answer leads
MO 152299	Recurrence formula for digamma function with rational number	Open or no-meaningful-answer leads
MO 372726	Can a digamma generating-function expression be simplified?	Open or no-meaningful-answer leads
MO 455136	Negativity of a determinant involving trigamma, tetragamma, and pentagamma	Open or no-meaningful-answer leads
MSE 1427206	Conjecture for an integral involving $\psi^{(1)}(1/3)$	Adjacent answered leads
MSE 1346396	Logarithmic Integral I	Adjacent answered leads
MSE 3075324	Improper integral with trigamma identity in the solution path	Adjacent answered leads
MSE 4634827	Closed form for a fractional-part integral	Adjacent answered leads
MSE 2683542	Meaning and definition of $\psi^{(-2)}(x)$	Adjacent answered leads
MSE 5080998	Integral of a trigamma difference at conjugate half-line arguments	Adjacent answered leads
MO 220759	Closed form for $\sum 1/(1+n+\dots+n^a)$	Adjacent answered leads
MO 378665	Closed form of $\sum_{r \geq 2} \zeta(r)/r^2$	Adjacent answered leads

A.3 The provenance boundary

The original drafts retain their snapshot identifiers. The accompanying source inventory records the SHA-256 digest of all thirty-nine textual sources. The editorial ledger maps each source to the merged mathematical development, records which results later drafts supersede, and identifies claims that have been withdrawn or weakened. The old PDFs are source evidence and are superseded as reading artifacts by this manuscript.

The original identity stores, reduction tools, notebook parsers, certificate checker and many high-precision search scripts were not included in the import. A historical receipt in a source report is therefore not a fresh replay here. The new verification scripts exercise representative corrected formulas from independent definitions. Their receipt specifies engine, precision, tolerance and actual residuals; it does not assert a complete replay of every historical experiment.

Bibliography

- [1] L. Lewin, *Polylogarithms and Associated Functions*, North-Holland, 1981; and (ed.) *Structural Properties of Polylogarithms*, AMS, 1991. (The trigamma Clausen-DFT law; cited via MathWorld, *Trigamma Function*, eq. (14).)
- [2] Wolfram Research, *The Wolfram Functions Site*, functions.wolfram.com, PolyLog identity 10.08.03.0007.01. (The forward cyclotomic formula.)
- [3] E. E. Kummer, *Beitrag zur Theorie der Function $\Gamma(x)$* , J. reine angew. Math. **35** (1847), 1–4. (The Fourier series of $\log \Gamma$.)
- [4] C. J. Malmsten, *De integralibus quibusdam definitis, seriebusque infinitis*, J. reine angew. Math. **38** (1849), 1–39 (dated 1846). (The log-log integrals and the reflection identity for γ_1 .)
- [5] I. Vardi, *Integrals, an introduction to analytic number theory*, Amer. Math. Monthly **95** (1988), 308–315.
- [6] I. V. Blagouchine, *A theorem for the closed-form evaluation of the first generalized Stieltjes constant at rational arguments and some related summations*, J. Number Theory **148** (2015), 537–592. (Vendored in the repository at src/PolyLog/docs/pdfs/1401.3724v3/.)
- [7] O. Espinosa and V. Moll, *On some integrals involving the Hurwitz zeta function, Part 1*, Ramanujan J. **6** (2002), 159–188. (The second log-gamma moment.)
- [8] E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis*, 4th ed., Cambridge University Press, 1927. (Binet’s formulas; the Hurwitz-series ladder.)
- [9] C. Deninger, *On the analogue of the formula of Chowla and Selberg for real quadratic fields*, J. reine angew. Math. **351** (1984), 171–191. (The $\zeta''(0, a)$ layer.)
- [10] N. Koblitz and A. Ogus, *Algebraicity of some products of values of the Γ function*, appendix (pp. 343–346) to P. Deligne, *Valeurs de fonctions L et périodes d’intégrales*, in *Automorphic Forms, Representations and L -Functions* (Corvallis, 1977), Proc. Sympos. Pure Math. **33**, part 2, Amer. Math. Soc., 1979, 313–346.
- [11] C. F. Gauss, *Disquisitiones generales circa seriem infinitam*, Comment. Soc. Regiae Sci. Gottingensis Rec. **2** (1813), 3–46; Werke **3**, 265–327. (The multiplication theorem.)
- [12] V. Reshetnikov, *FunctionTransformations* (Wolfram Language package), 2016, <https://github.com/VladimirReshetnikov/FunctionTransformations>. (The 437-rule polylog and 670-rule Γ reduction tables anchoring the corpus.)
- [13] A. Selberg and S. Chowla, *On Epstein’s zeta-function*, J. reine angew. Math. **227** (1967), 86–110.

- [14] R. M. Damerell, *L-functions of elliptic curves with complex multiplication, I*, Acta Arith. **17** (1970), 287–301.
- [15] A. Hurwitz, *Über die Entwicklungskoeffizienten der lemniskatischen Funktionen*, Math. Ann. **51** (1899), 196–226.
- [16] D. A. Cox, *Primes of the Form $x^2 + ny^2$: Fermat, Class Field Theory, and Complex Multiplication*, Wiley, 1989. (Genus theory; class invariants; the $3 \mid D$ obstruction for γ_2 .)
- [17] E. W. Weisstein, *Digamma Function*, MathWorld — A Wolfram Web Resource. (Source of the FoxTrot identity.)
- [18] V. S. Adamchik, *Polygamma functions of negative order*, J. Comput. Appl. Math. **100** (1998), 191–199.
- [19] T. M. Apostol, *Formulas for higher derivatives of the Riemann zeta function*, Math. Comp. **44** (1985), 223–232.
- [20] T. M. Apostol, *Introduction to Analytic Number Theory*, Springer, 1976.
- [21] B. C. Berndt, *Ramanujan’s Notebooks, Part I*, Springer, 1985.
- [22] I. V. Blagouchine, *Rediscovery of Malmsten’s integrals, their evaluation by contour integration methods and some related results*, Ramanujan J. **35** (2014), 21–110; Erratum–Addendum: **42** (2017), 777–781.
- [23] K. Chakraborty, S. Kanemitsu and T. Kuzumaki, *Finite expressions for higher derivatives of the Dirichlet L-function and the Deninger R-function*, Hardy–Ramanujan J. **32** (2009), 38–53.
- [24] B. K. Choudhury, *The Riemann zeta-function and its derivatives*, Proc. R. Soc. Lond. A **450** (1995), 477–499.
- [25] M. W. Coffey, *Series representations for the Stieltjes constants*, Rocky Mountain J. Math. **44** (2014), 443–477 (arXiv:0905.1111).
- [26] D. F. Connon, *On some new formulae involving the Stieltjes constants*, arXiv:1902.00510 (2019).
- [27] O. Espinosa and V. H. Moll, *A generalized polygamma function*, Integral Transforms Spec. Funct. **15** (2004), 101–115.
- [28] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 1*, Ramanujan J. **6** (2002), 159–188.
- [29] O. Espinosa and V. H. Moll, *On some integrals involving the Hurwitz zeta function: Part 2*, Ramanujan J. **6** (2002), 449–468.
- [30] O. Kouba, *Generalized Stieltjes constants and integrals involving the log–log function: Kummer’s theorem in action*, J. Classical Anal. **9** (2016), no. 2, 79–88 (arXiv:1702.05439).
- [31] J. Miller and V. S. Adamchik, *Derivatives of the Hurwitz zeta function for rational arguments*, J. Comput. Appl. Math. **100** (1998), 201–206.
- [32] D. H. Bailey and J. M. Borwein, *Computation and structure of character polylogarithms with applications to character Mordell–Tornheim–Witten sums*, Math. Comp. **85** (2016), 295–324.

- [33] E. Panzer, *The parity theorem for multiple polylogarithms*, J. Number Theory **172** (2017), 93–113. <https://arxiv.org/abs/1512.04482>.
- [34] T. Rivoal and J. Roques, *Hadamard products of algebraic functions*, Section 3 (gamma relations and Rohrlich’s conjecture). <https://rivoal.perso.math.cnrs.fr/articles/algcoef.pdf>.
- [35] D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*. <https://arxiv.org/abs/2012.15805>. Author version: <https://people.mpim-bonn.mpg.de/zagier/files/preprints/HerglotzFunction.pdf>.
- [36] H. Cohen, L. Lewin and D. Zagier, *A Sixteenth-order Polylogarithm Ladder*, Experimental Mathematics **1** (1992), no. 1. <https://people.mpim-bonn.mpg.de/zagier/files/exp-math-1/fulltext.pdf>. (The higher-order ladders in this paper are explicitly conjectural numerical identities.)
- [37] C. Hakimoglu-Brown, *Analytic Dilogarithm Identities* (2025). <https://arxiv.org/abs/2506.10206>.
- [38] E. Remiddi and J. A. M. Vermaseren, *Harmonic polylogarithms*, Int. J. Mod. Phys. A **15** (2000), 725–754. <https://arxiv.org/abs/hep-ph/9905237>.
- [39] F. Brown, *Mixed Tate motives over $\mathbb{Z}$* , Annals of Mathematics **175** (2012), 949–976. <https://arxiv.org/abs/1102.1312>.